

When Redevelopment Transforms the Housing Stock: Carrot & Stick Policy and the Affordability Puzzle^{*}

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Abstract

Housing supply in mature cities evolves primarily through the replacement of existing structures. In this setting, policies affect supply and rents by changing individual developers' redevelopment decisions rather than simply shifting an aggregate supply curve. We analyze this redevelopment margin by developing a forward-looking spatial model of real estate development in which developers choose whether, when, and how intensively to redevelop individual parcels. Calibrated to New York City, the model shows that tax incentives raise expected supply by accelerating development, whereas density bonuses raise it by increasing building height, although they often postpone redevelopment. Because new development replaces lower-rent units with higher-rent ones, we find that policies incentivizing construction raise average rents. Under a representative upzoning counterfactual in which expected housing supply increases by 1.8 percent, average rent rises by approximately 1.3 percent. We also show that affordability mandates primarily crowd out market-rate construction and generate costs of approximately \$60,000 per affordable unit. The incidence of those costs depends critically on the incentive paired with the mandate.

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Housing affordability pressures persist across cities even where housing production has remained strong. In New York City, roughly two-thirds of residential building permits filed between 2023 and 2025 are for parcels that already carried a structure, except on Staten Island, where vacant land remains. Buildings over a century old are redeveloped at two and a half times the rate of buildings under forty and rent for roughly 14 percent less. Housing production in mature markets therefore works by replacing the oldest and cheapest stock with higher-rent construction. The supply that policymakers seek to encourage is thus the very mechanism by which the cheapest units disappear.

Yet quantitative urban models, the dominant tools for studying urban housing markets, are not built to capture this mechanism. These models typically resolve household and firm location choices finely, but treat housing supply through an aggregate elasticity, with no individual parcel choice of redevelopment timing and density (Ahlfeldt et al., 2015; Redding, 2023; Tsivanidis, 2026).¹ These models abstract from the replacement mechanism, because housing created by expanding the housing stock cannot be distinguished from housing created by replacing existing structures.² They therefore do not explain why policies that stimulate construction simultaneously reduce the stock of inexpensive housing. We address this gap by modeling housing supply as the aggregation of parcel-level redevelopment decisions. We refer to these decisions as the *redevelopment margin*.

We argue that in urban housing markets where redevelopment is the dominant source of new housing, the relevant object for analyzing housing supply is the parcel-level redevelopment decision. In our model, developers decide whether, when, and how intensively to redevelop individual parcels—including development on empty parcels—by comparing the value of immediate redevelopment with the option value of waiting (McDonald and Siegel, 1986; Paddock et al., 1988; Capozza and Li, 1994).³ The stochastic and parcel-specific rent process each developer

¹Recent applications of quantitative urban models particularly focus on the importance of including dynamics to understand long-term urban outcomes (Favilukis et al., 2023; Peng, 2023; Greaney et al., 2025; Rollet, 2025), the endogenous quality of the supply (Ma, 2026; Koster and Loumeau, 2026), endogenous urban amenities (Almagro and Domínguez-Iino, 2025), and the distributional consequences of housing supply (Giacoletti et al., 2025; Hong and Macek, 2025; Rollet and Weiwu, 2025; Borck et al., 2026). Our framework speaks to these features.

²A related literature, that follows the monocentric urban modeling tradition, analyzes the renewal and upgrading of the built environment (Braid, 2001; Brueckner and Rosenthal, 2009). Henderson et al. (2021), for instance, includes durable structures and dynamic redevelopment decisions and highlights the determinants and consequences of the informal-to-formal housing replacement in emerging countries, an analog of the redevelopment margin we study here. Our model goes further by treating redevelopment as a parcel-specific decision and quantifying changes in the rent distribution across units.

³Two contemporaneous papers likewise emphasize parcel-level characteristics in explaining redevelopment patterns (Diamond et al., 2025; Rollet, 2025). Our framework differs in how it models the developer’s decision, separating for each parcel the value of the existing structure, the cost of redevelopment over the construction period, and the value of the completed building under demand uncertainty. This structure allows us to study how alternative policy instruments affect redevelopment timing, the resulting transformation of the housing stock, and rents through the redevelopment margin.

takes as given must be consistent with the collective redevelopment decisions of all developers. This fixed point defines a spatial equilibrium, one in which the market clears through redevelopment, where the timing, intensity, and location of redevelopment, and hence the housing supply, emerge endogenously. In this framework, housing policies incentivizing development (*Carrots*), such as density bonuses and tax exemptions, and affordability mandates (*Sticks*), affect supply not by shifting an exogenously specified aggregate housing curve, but by changing individual redevelopment decisions through instruments. Because these decisions are made parcel by parcel, housing policy changes not only the quantity of housing but also its composition, because redeveloped units enter the housing stock without the depreciation carried by the units they replace.

We calibrate the model to New York City’s housing market, using parcel-level data on density and building age, rents from the American Community Survey, and neighborhood rent dynamics from CoStar. Rental income is estimated to fall with building age (Rosenthal, 2014) and calibrated to rise with height (Danton and Himbert, 2018; Liu et al., 2018; Ahlfeldt and Barr, 2022). Demolition costs exhibit increasing returns to scale, and construction costs are convex in height (Ahlfeldt and McMillen, 2018; Eriksen and Orlando, 2022; Ahlfeldt et al., 2026). Housing supply from redevelopment feeds back into rents through two channels. A citywide supply effect lowers rent growth everywhere (Mast, 2023; Bratu et al., 2023). A local spillover effect, which we estimate from rental data, lowers rent in the vicinity of new construction (Li, 2022; Asquith et al., 2023). The calibrated model reproduces key features of New York City’s housing market, matching observed cross-neighborhood property values and predicting out-of-sample construction patterns.

Our study delivers three central findings. First, supply-side policy instruments that deliver similar aggregate supply do so through different channels. Tax incentives raise expected supply by accelerating development, whereas density bonuses raise it by increasing building height, although they often postpone redevelopment. Parcels near their redevelopment thresholds, meaning older structures and underdeveloped parcels, respond more strongly to fiscal incentives, while high-rent locations respond more strongly to density bonuses. Because parcels differ in these characteristics, policies generate heterogeneous effects across space.

Second, because housing production in mature cities occurs primarily through replacement rather than expansion, policies that stimulate construction simultaneously affect the quantity and composition of the housing stock. We refer to this compositional shift, and its consequences for the distribution of rents, as the *replacement effect*. Within our calibration, policies that increase housing production raise rather than lower average rents, because the replacement effect dominates the supply effect. Under a representative upzoning counterfactual, construction rises by 1.76 percent and average rents rise by 1.31 percent.⁴ The replacement effect raises rent per unit

⁴The rents we report are not adjusted for housing quality. They are the rents that prevail across the stock of units

by approximately \$60 per month, more than offsetting the approximately \$24 decline generated by additional supply. This replacement effect dominates the supply effect for empirically plausible demand elasticities. More generally, the mechanism is strongest in mature cities, where new construction largely requires replacing existing structures, and weaker in expanding cities, where development occurs primarily on vacant or underutilized land, where there is little existing stock to replace.

Third, affordability mandates produce affordable housing, but primarily by crowding out market-rate construction. A 30 percent mandate, for instance, yields roughly 42,000 affordable units but reduces market-rate construction by about 58,000 units. Because redevelopment with a mandate is profitable only where rents are already high, these units expand the middle of the rent distribution while leaving the bottom quintile largely unchanged. Mandated affordable units are also costly, and the policy instrument paired with the mandate determines who bears that cost. Holding total housing supply fixed, each affordable unit substituted for a market-rate unit costs about \$59,900 under density incentive zoning and \$61,200 under fiscal incentive zoning. Using density bonuses to compensate for the mandate splits the burden evenly between landowners and the city, whereas tax exemptions place it almost entirely on the public sector.

Beyond providing a quantitative framework for policy evaluation, the redevelopment margin also reshapes our understanding of housing supply in urban settings. A large literature examines the economic forces governing urban housing production (Glaeser and Gyourko, 2018; Baum-Snow and Duranton, 2025), emphasizing the roles of market structure and productivity (D’Amico et al., 2024; Goolsbee and Syverson, 2025), construction technology (Ahlfeldt and McMillen, 2018; Henderson et al., 2021; Ahlfeldt and Barr, 2022; Ahlfeldt et al., 2026), and physical and regulatory constraints determining housing supply elasticity (Saiz, 2010; Albouy and Ehrlich, 2018; Brueckner and Singh, 2020; Gyourko et al., 2021; Büchler and v. Ehrlich, 2023; Baum-Snow and Han, 2024). By modeling redevelopment decisions at the parcel level, our study contributes to this literature by making explicit how the current built environment and policy instruments jointly determine the timing and intensity of redevelopment.

Our study is closely related to work that models forward-looking developers in dynamic settings (Cunningham, 2006; Murphy, 2018; Peng, 2023; Hsiao, 2025; Rollet, 2025), where land embodies an option to redevelop. Research on investment under uncertainty has shown how the option value of waiting shapes the timing of investment (Myers, 1977; McDonald and Siegel, 1986), and real estate investment timing in particular (Arnott and Lewis, 1979; Titman, 1985; Quigg, 1993; Grenadier, 1996; Capozza and Li, 1994; Lebreton and Liu, 2023; Lange and Teulings, 2024). These

actually occupied, the rents a tenant pays, a developer underwrites, and a policymaker measures. Controlling for quality, housing rents fall when supply expands.

models highlight the individual developer’s investment decision, but generally take the price or rent process governing that decision as exogenous rather than as an equilibrium outcome. We extend this class of models by embedding parcel-level decisions in a spatial equilibrium. Because the rent process each developer takes as given is itself an equilibrium outcome of the redevelopment the policy induces, the model can trace how a policy affects individual parcels and citywide outcomes, allowing for counterfactual policy evaluation.

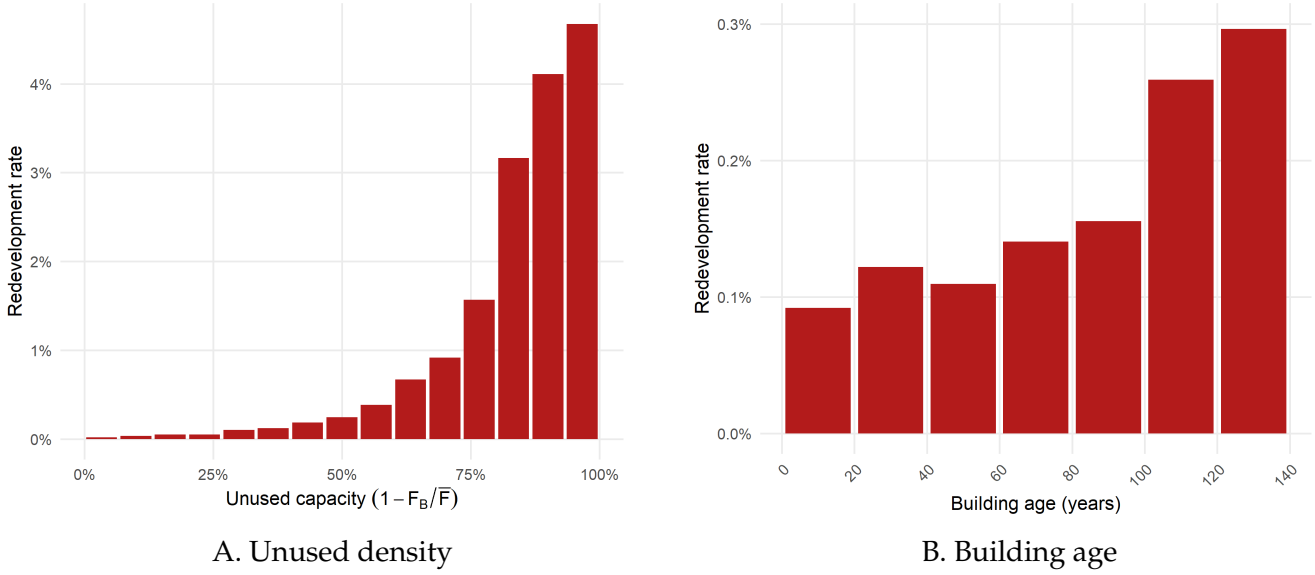
A large literature evaluates how individual regulatory and fiscal instruments affect housing markets (Quigley and Rosenthal, 2005; Gyourko and Molloy, 2015), including tax incentives (Baum-Snow and Marion, 2009; Eriksen and Rosenthal, 2010; Chen et al., 2023; Corinth and Feldman, 2024; Soltas, 2024b,a; Cook et al., 2026), upzoning (Greenaway-McGrevy and Phillips, 2023; Büchler and Lutz, 2024; Liao, 2025; Anagol et al., 2026), and affordable housing mandates and inclusionary policies (Schuetz et al., 2011; Kim, 2020; Favilukis et al., 2023; Krimmel and Wang, 2026). This literature has generated important insights into the effects of individual policy instruments, evaluated largely in isolation. As a result, few frameworks compare alternative instruments within a single structural urban model. We provide a unified structural framework that compares policies and quantifies how alternative instruments affect housing production, rents, affordability, and the incidence of policy costs.

The paper proceeds as follows. Section 1 empirically documents the central role of the parcel-level characteristics in redevelopment decisions. Section 2 develops the forward-looking spatial model of real estate development. Section 3 illustrates how housing policies change a representative parcel’s property value and redevelopment probability. Section 4 describes the empirical setting and calibration, and Section 5 validates the model against observed property values and out-of-sample redevelopment patterns. Section 6 uses the calibrated framework to evaluate the effects of affordability mandates, density bonuses, and tax exemptions on housing production and rents. Section 7 describes the incidence of each policy and quantifies the cost of mandated affordable units. Section 8 discusses the conditions under which the redevelopment margin matters the most. Section 9 concludes.

1 Development Patterns in New York City: Motivating Facts

To document what determines housing development decisions, we collected residential permits for new construction from the New York City housing database (New York City Department of City Planning, 2026a). We retain all permits filed by private for-profit entities between 2023 and 2025 and match them to parcel characteristics from the Primary Land Use Tax Lot Output (PLUTO) dataset (New York City Department of City Planning, 2026b). For each parcel with a

Figure 1: Residential Permits and Built Environment



Notes: These bar graphs show the share of parcels filing a private for-profit residential building permit. Panel A splits parcels based on their unused density $(1 - F_{B,i}/\bar{F}_i)$, where $F_{B,i}$ is the current built floor-area ratio and $\bar{F}_i$ is the maximum allowable floor-area ratio under existing zoning. Panel B splits parcels by the age of their building, defined as the time since the last recorded renovation or construction. Parcel characteristics are taken from the 2023 PLUTO matched to permit filings from the NYC Department of City Planning Housing Database from 2023 to 2025.

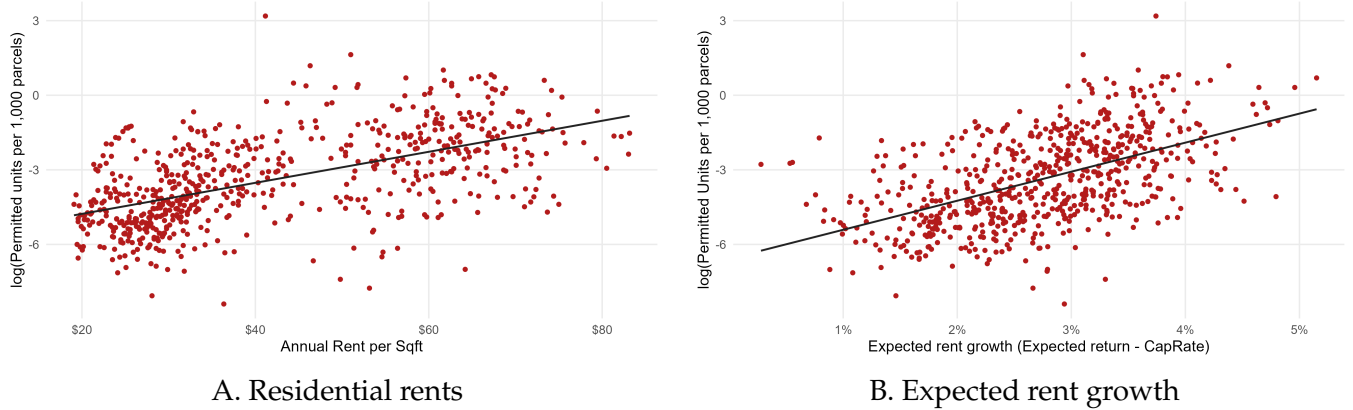
permit, we determine whether it had residential or commercial floorspace before filing. In every borough except Staten Island, roughly two-thirds of permits were filed on parcels that already carried a structure. In Staten Island, where more vacant land remains, the share falls to 34%.⁵ New housing production in New York therefore predominantly takes the form of redevelopment rather than construction on vacant land.

Figure 1 further highlights the relation between parcel characteristics and the share of parcels for which a permit was filed. Panel A shows that redevelopment concentrates on parcels with substantial unused density, defined as the share of allowable built floor-space that remains unused. Parcels built at or near their zoning maximum rarely redevelop, while 4.0% of the parcels with more than 75% of their allowable density unused filed for a construction permit. This pattern reflects that parcels with room to build taller assets have greater incentives to redevelop. Panel B shows that older buildings are also more likely to be redeveloped. We observe that buildings older than 100 years redevelop at rates roughly 2.5 times those of buildings below 40 years. Thus, more depreciated structures, which provide lower rental incomes, bring parcels closer to their redevelopment threshold.⁶

⁵Borough-level shares are reported in Table C1 in the Appendix. Because a parcel's use history in PLUTO is imperfectly recorded, this figure is best read as a lower bound.

⁶We provide empirical foundations for this observation by estimating age elasticity of housing rents and prices

Figure 2: Residential Permits and Rent Dynamics



Notes: These scatter plots show the log permitted housing units per parcel (y-axis) against annual rent per square foot (Panel A), and expected rent growth (Panel B). Expected rent growth is defined as the total return on multifamily real estate minus the capitalization rate. The data comprises 37 CoStar-defined submarkets from 2004-2023 merged with private for-profit residential permits from NYC Department of City Planning Housing Database.

We next examine how rent dynamics influence development decisions. We collected annual rent per square foot and expected rent growth at the neighborhood level from CoStar market data, covering 37 New York City submarkets from 2004 to 2023. We merge these measures with a panel of private for-profit residential permit filings and show their relations with permitted units per parcel in Figure 2. Panel A shows that neighborhoods with higher rent levels attract more development activity. A 10% increase in rent per square foot is associated with a 7.1% increase in permitted units, consistent with higher neighborhood rents raising the profitability of redevelopment. Panel B shows that expected rent growth is also a strong predictor of permitting activity. A one percentage point increase in expected rent growth is associated with a 39% increase in permitted units, reflecting the forward-looking nature of development decisions.

Together, these empirical patterns identify the key economic determinants of the redevelopment margin examined in this paper. They show that redevelopment decisions depend on redevelopment opportunities, reflected in parcel characteristics such as unused development capacity, building age, current rents, and expected rent growth. The forward-looking spatial model of real estate development developed in the following section formalizes these determinants.

of approximately -0.09 in Appendix C.1. The magnitude of this elasticity determines how depreciation lowers the option value of waiting in the structural model.

2 A Forward-looking Spatial Model of Real Estate Development

We develop a forward-looking model of urban development in which developers decide both when to develop a parcel and how intensively to build. Each developer takes the rent process as given, but that process is itself policy-dependent in the aggregate. We develop the model in two stages. In Section 2.1, we first solve an individual developer's redevelopment decision as a real-options problem. Because redevelopment requires irreversible demolition and construction expenditures while future rents remain uncertain, a developer does not redevelop as soon as redevelopment becomes marginally profitable. Instead, investment is undertaken only after expected rents exceed a threshold that compensates for the value of waiting. Policy reforms therefore change redevelopment incentives by shifting this threshold rather than acting directly on housing supply, implying that redevelopment responds to policy through discrete threshold decisions rather than smooth adjustments. These redevelopment decisions determine the expected housing supply produced on each parcel.

In Section 2.2, we then embed these parcel-level decisions in a spatial equilibrium by specifying how redevelopment feeds back into the rent process developers expect. Citywide redevelopment increases aggregate housing supply and lowers expected rent growth throughout the city. Nearby redevelopment also affects local rent growth, but its net effect reflects opposing competition and amenity forces, making the sign and magnitude of the local elasticity an empirical question. Together, these elasticities determine the rent process developers face, while developers' redevelopment decisions determine the housing supply that feeds back into rents. The model therefore closes as a fixed point in which developers' expectations about rents are consistent with the redevelopment decisions those expectations induce.

2.1 The Real Estate Developer Decision Rule

We consider a city composed of parcels $i \in I$, each owned by a price-taking developer. Parcels of size L_i may contain an existing building characterized by its built density $F_{B,i}$ and its rental depreciation index $z_{B,i}$. The location of parcel i determines the market rent per unit of built density, denoted R_{it} , which reflects both local and citywide housing market conditions. Developers form expectations about future rents when making redevelopment decisions. In our setting, the government regulates redevelopment through a policy environment that may change at time zero from a baseline regime $\mathcal{P}_0$ to an alternative regime $\mathcal{P}$. Policy affects redevelopment incentives directly by affecting individual development decisions, and indirectly by altering future housing supply and, thus, future rent dynamics across the city.

2.1.1 Decision rule under status quo regulation

To characterize the redevelopment decision, we first analyze the problem of a single parcel and suppress the parcel index i for notational simplicity. The market rent per unit of space, R_t , is our fundamental state variable. Under the risk-neutral measure $\mathbb{Q}$, rent evolves according to a geometric Brownian motion:

$$dR_t = \delta R_t dt + \sigma R_t dW_t \quad (1)$$

where δ is the risk-neutral expected growth rate of rent used for valuation, σ is the volatility, and W_t is a standard Brownian motion on the probability space $(\Omega, \mathcal{F}, \mathbb{Q})$. Within this first layer of the model, δ is fixed from the perspective of an individual developer's own redevelopment decision. σ remains fixed throughout the model. This process is log-normal, implying that both expected growth and volatility scale proportionally with the current rent level. Throughout, we maintain the standard assumption $\delta < r$, which simply requires the property's rent-to-price ratio (or cap rate) to be positive ($c > 0$), since $\delta_i = r - c$ under standard asset-pricing conditions.⁷

Valuation relies on standard assumptions that ensure the existence of a risk-neutral measure. Under these conditions, the stochastic process for rents can serve as the fundamental state variable for arbitrage-free pricing, and development options can be valued as contingent claims on future rents.⁸ Denoting R as the market rent per unit of space at $t = 0$, we can rewrite the potential rent at $t \geq 0$ as $R_t = Re^{(\delta - \sigma^2/2)t + \sigma W_t} = Re^{\sigma Z_t}$, where $Z_t = bt + W_t$, and $b = 1/\sigma(\delta - \sigma^2/2)$.

To obtain the rental income of a parcel, we introduce two elements. First, we define $\Gamma(F, z, \alpha)$ as the function that scales the potential rent per unit of density R_t . This function depends on the characteristics of the built property and yields the stochastic rental income of a developed property. In particular, the rent increases with the built density F , decreases with the accumulated rental depreciation index ($z \in [0, 1]$), and decreases with the portion of built space allocated to affordable housing (α).⁹ Because market rent R_t is defined on a per-unit basis of newly built property, $\Gamma(F = 1, z = 1, \alpha = 0) = 1$ by definition. It corresponds to a property with a built density of 1 ($F = 1$), with no operational efficiency losses due to rental depreciation ($z = 1$), and no affordable housing ($\alpha = 0$) that would alter the rental incomes. We revert to the full parametrization of $\Gamma(F, z, \alpha)$ in Section 4.

⁷This condition ensures the infinite-horizon rental-income integral $\int_0^\infty e^{ru} \Gamma(\cdot) R_u du$ converges, and it is equivalent to the condition under which the second-order condition for the optimal expansion threshold (Proposition 2) holds.

⁸In practice, rents are not continuously traded, and markets are incomplete. The assumption that rents are spanned by traded assets should be interpreted as an analytical convenience supported by the existence of correlated securities (e.g., REIT shares, lease-backed instruments) that allows developers to partially hedge rent risk. We discuss these assumptions in Appendix A.1.

⁹Density F should be interpreted as effective built volume per unit of land rather than purely floor-area density, capturing building height that affects productive floor space. In this theoretical section, we assume parcels are of unit size one ($L = 1$). In the empirical application, parcel size scales both rents and costs proportionally.

Second, we assume the government imposes a tax on developed properties proportional to the rental incomes, reducing the developer's effective payout by ψ . Consequently, the value of the developed parcel absent the option to redevelop is

$$\mathbf{E}_{\mathbb{Q}} \left\{ \int_0^{\infty} e^{-ru} \underbrace{(1 - \psi)}_{\text{Net of tax}} \underbrace{\Gamma(F_B, z_B, \alpha_B) R_u}_{\text{Gross rental income}} du \right\},$$

where the subscript B describes the characteristics of the current building.

Private developers have the option to replace the current building with a new one constructed at density $F \leq \bar{F}_0$, where $\bar{F}_0$ denotes the maximum allowable built density under the regulatory environment $\mathcal{P}_0$.¹⁰ Exercising this option requires paying an instantaneous demolition cost $\gamma(F_B)$ to remove the existing structure, in addition to incurring an annual construction cost during the construction period of d years, denoted as $\kappa(F, \alpha)$. Construction costs increase with the density of the new construction (F) and vary with the portion of built-up area dedicated to affordable housing (α).

Developers exercise the option to redevelop under $\mathcal{P}_0$ only if favorable market conditions prevail. Because developers already own the parcel and the existing structure, the redevelopment decision corresponds to exercising an embedded option to replace the current building. We denote by R^e the predetermined rent level (the expansion threshold) at which developers initiate construction at density F , by $T_{R^e}(R) = \inf\{t \geq 0 : R_t = R^e\}$ the first time this threshold is reached, and by $H_{R^e}(R) = T_{R^e}(R) + d$ the completion time under deterministic time-to-build d . Consequently, the value of the parcel under $\mathcal{P}_0$ is

$$\begin{aligned} \pi_0(R, F_B, z_B, \alpha_B, F; \mathcal{P}_0, \Theta) = & \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_0^{T_{R^e}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right\}}_{\text{Value of existing property}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ e^{-rT_{R^e}} \gamma(F_B) \right\}}_{\text{Demolition costs}} \\ & - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{T_{R^e}}^{H_{R^e}} e^{-ru} \kappa(F, \alpha) du \right\}}_{\text{Construction costs}} + \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{R^e}}^{\infty} e^{-ru} (1 - \psi) \Gamma(F, 1, \alpha_B) R_u du \right\}}_{\text{Value of new property}} \quad (2) \end{aligned}$$

where Θ is the vector of parameters. The value of a parcel thus consists of four discounted terms: net-of-tax rents from the current property until demolition, demolition costs at the stopping time, expenditures during construction, and the net-of-tax rental income from the completed asset.

¹⁰In Appendix E, we extend the model by adding the option for developers to renovate their properties in addition to the option to redevelop and show how the baseline outcomes and the counterfactual results differ under this extended model.

2.1.2 Adding policy instruments into the development decision rule

We summarize the policy environment by the vector $\mathcal{P} = [\alpha, \phi, \tau]$. The parameter α denotes a mandate requiring that a share of the project be allocated to affordable housing units, ϕ represents additional allowable density, and τ denotes a temporary tax exemption following redevelopment. The status quo corresponds to $\mathcal{P}_0 = [\alpha = 0, \phi = 0, \tau = 0]$.¹¹ The three policy instruments affect redevelopment decisions through different channels. The mandate α reduces the market rental income generated by the property and alters construction costs through the functions $\Gamma(\cdot)$ and $\kappa(\cdot)$, respectively. The density bonus ϕ relaxes the density constraint by allowing developers to choose $F \leq \bar{F}$, where $\bar{F} = \bar{F}_0(1 + \phi)$. Finally, the tax exemption τ postpones property taxation for τ years after redevelopment is completed.

We can thus rewrite the value of a parcel under the new regulatory environment as:

$$\begin{aligned} \pi(R, F_B, z_B, \alpha_B, F; \mathcal{P}, \Theta) = & \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_0^{T_{Re}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right\}}_{\text{Value of existing property}} \\ & - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ e^{-rT_{Re}} \gamma(F_B) \right\}}_{\text{Demolition costs}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{T_{Re}}^{H_{Re}} e^{-ru} \kappa(F, \alpha) du \right\}}_{\text{Construction costs}} \\ & + \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{Re}}^{\infty} e^{-ru} \Gamma(F, 1, \alpha) R_u du \right\}}_{\text{Value of new property gross of tax}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{Re} + \tau}^{\infty} e^{-ru} \psi \Gamma(F, 1, \alpha) R_u du \right\}}_{\text{Property taxes after } \tau \text{ years of exemption}} \quad (3) \end{aligned}$$

Similar to Equation (2), the property value consists of the sum of present values with an additional final term that captures the impact of postponing property taxes for τ years after the development has been completed. Developers maximize their parcel values, yielding a closed-form solution for the optimal expansion threshold and parcel value, and a numerical solution for the optimal density.

Parcel value. We solve the model analytically by first characterizing the five integrals that define parcel value conditional on a given expansion threshold and density.

Proposition 1 *In the presence of the expansion threshold R^e and given a construction density $F \leq \bar{F}$, the*

¹¹Some parcels may already contain units subject to rent stabilization, rent control, or existing affordability requirements. Accordingly, the policy parameter α should be interpreted as an incremental mandate beyond existing regulations. If the current share of regulated or affordable units on a parcel (denoted α_B) already exceeds the policy requirement, the mandate does not bind, and the existing constraint remains in place.

property value $\pi(\cdot)$ satisfies

$$\begin{aligned} \pi(R, R^e, F, F_B, z_B, \alpha_B; \mathcal{P}, \Theta) = & \underbrace{\Gamma(F_B, z_B, \alpha_B) \frac{1-\psi}{r-\delta} \left[R - R^e \left(\frac{R^e}{R} \right)^\xi \right]}_{\text{Value of existing property}} - \underbrace{\gamma(F_B) \left(\frac{R^e}{R} \right)^\xi}_{\text{Demolition costs}} \\ & - \underbrace{\frac{\kappa(F, \alpha)}{r} (1 - e^{-rd}) \left(\frac{R^e}{R} \right)^\xi}_{\text{Construction costs}} + \underbrace{\Gamma(F, 1, \alpha) \frac{R^e}{r-\delta} e^{-(r-\delta)d} \left(\frac{R^e}{R} \right)^\xi}_{\text{Value of new property gross of tax}} \\ & - \underbrace{\psi \Gamma(F, 1, \alpha) \frac{R^e}{r-\delta} e^{-(r-\delta)(d+\tau)} \left(\frac{R^e}{R} \right)^\xi}_{\text{Property taxes after } \tau \text{ years of exemption}}, \quad (4) \end{aligned}$$

with $b = (1/\sigma)(\delta - \sigma^2/2)$, $\lambda = \sqrt{2r + b^2}$, and $\xi = (b - \lambda)/\sigma$.

The proof is in Appendix A.2.

Probability of development. Conditional on current market conditions and a given regulatory regime, developers optimally initiate construction when rent reaches a critical expansion threshold R^{e*} . At this threshold, the expected value of redevelopment exactly equals the continuation value of the existing structure, accounting for construction costs, policy distortions, and the stochastic timing of redevelopment.

Proposition 2 *Given a construction density $F \leq \bar{F}$ and a regulatory regime $\mathcal{P}$, suppose $\vartheta > 0$, where*

$$\vartheta = \Gamma(F, 1, \alpha) \frac{1}{r-\delta} (e^{-(r-\delta)d} - \psi e^{-(r-\delta)(d+\tau)}) - \Gamma(F_B, z_B, \alpha_B) \frac{1-\psi}{r-\delta}.$$

Then the optimal expansion threshold is

$$R^{e*} = \frac{\xi}{1+\xi} \frac{\gamma(F_B) + \frac{\kappa(F, \alpha)}{r} (1 - e^{-rd})}{\vartheta}. \quad (5)$$

If instead $\vartheta \leq 0$, redevelopment is never optimal and the parcel is never redeveloped at density F .

The proof is in Appendix A.3.

Proposition 2 characterizes the rent level at which redevelopment becomes optimal. Because market rents evolve stochastically, a higher expansion threshold does not simply delay development deterministically. Instead, it lowers the probability that redevelopment occurs within any finite horizon. Conditional on the current rent level R , the timing of development is governed by the stopping time $T_{R^{e*}}(R) = \inf\{t \geq 0 \mid R_t = R^{e*}\}$, the first time the rent process reaches the

optimal expansion threshold.

This formulation allows us to characterize the redevelopment completion in probabilistic terms. Policies that raise construction or regulatory costs increase R^{e*} and therefore reduce the likelihood that development is initiated by a given date, even holding fixed the underlying rent dynamics.

Proposition 3 *Given the expansion threshold R^{e*} and a construction density $F \leq \bar{F}$, the probability that development is completed by horizon h satisfies*

$$\mathbf{P}_{\mathbb{P}}(H_{R^{e*}} \leq h) = \mathbf{P}_{\mathbb{P}}(T_{R^{e*}} \leq h - d), \quad (6)$$

where $H_{R^{e*}} \equiv T_{R^{e*}} + d$. The probability is computed under the physical measure $\mathbb{P}$, under which rent evolves as $dR_t = gR_t dt + \sigma R_t dW_t^{\mathbb{P}}$. Here g denotes the expected rent growth rate under the physical measure.

For a parcel whose current rent lies below its redevelopment threshold ($R < R^{e*}$), let $a^* = \ln(R^{e*}/R)/\sigma > 0$ and $m = (g - \sigma^2/2)/\sigma$. Then for $h > d$ the redevelopment probability is

$$\mathbf{P}_{\mathbb{P}}(T_{R^{e*}} \leq h - d) = \mathcal{N}\left(\frac{m(h - d) - a^*}{\sqrt{h - d}}\right) + e^{2ma^*} \mathcal{N}\left(-\frac{m(h - d) + a^*}{\sqrt{h - d}}\right), \quad (7)$$

while $\mathbf{P}_{\mathbb{P}}(H_{R^{e*}} \leq h) = 0$ for $h \leq d$. If instead $R \geq R^{e*}$, redevelopment is initiated immediately, so $\mathbf{P}_{\mathbb{P}}(H_{R^{e*}} \leq h) = 1$ for $h > d$ and 0 otherwise.

The proof is in Appendix A.4.

Optimal density. The preceding analytical results characterize optimal redevelopment timing conditional on a chosen construction density $F \leq \bar{F}$. We now turn to the developer's choice of density. For a fixed expansion threshold R^e , the tradeoff between higher post-redevelopment rental income and higher construction costs admits a closed-form solution for the optimal density (see Appendix A.5). Higher density increases post-redevelopment rental income but also raises construction costs, so the optimal choice of F^* balances these forces. Once the expansion threshold is chosen optimally as a function of density, the joint problem of selecting both redevelopment timing and density no longer admits a closed-form solution.

As a result, we determine the developer's optimal density numerically. Formally, we solve

$$\begin{aligned} F^* &= \arg \max_F \pi(R, R^{e*}(F), F, F_B, z_B, \alpha_B) \\ \text{s.t. } & 0 \leq F \leq \bar{F}, \end{aligned} \quad (8)$$

where $R^{e*}(F)$ is given by Proposition 2. This numerical optimization yields the optimal density F^* , the associated expansion threshold $R^{e*}(F^*)$, and the associated probability of development $\mathbf{P}^*$. Together, these objects characterize the developer's decision rule, incorporating both the optimal timing and intensity of redevelopment, conditional on replacement construction.¹²

Given this optimal decision rule, for any horizon h and any policy $\mathcal{P}$, the expected floor-space supply on a parcel of size L is

$$s(h; \mathcal{P}) = \underbrace{(1 - \mathbf{P}^*(h; \mathcal{P}))F_B L}_{\text{Supply with no development}} + \underbrace{\mathbf{P}^*(h; \mathcal{P})F^*(\mathcal{P}; \Theta)L}_{\text{Supply with development}}. \quad (9)$$

This expression makes the redevelopment margin explicit. With probability $(1 - \mathbf{P}^*)$, the parcel keeps the existing structure. With probability $\mathbf{P}^*$, the parcel is redeveloped at density F^* .

2.2 Rent Determination and Market Clearing

The parcel-level decision rule described above characterizes redevelopment incentives for a given rent process. To evaluate housing policies, however, these parcel-level decisions must be aggregated into a citywide equilibrium because future expected supply must change the rent process facing every developer. The pricing drift δ , fixed within layer one for an individual developer's valuation problem, and the physical growth rate g , fixed within layer one for the redevelopment probability in Proposition 3, are both policy-dependent once we account for how redevelopment across the city feeds back into future rents. To capture this feedback, we embed the parcel-level model in a citywide environment in which redevelopment decisions determine future housing supply and, in turn, expected rental growth.

In equilibrium, the pricing drift δ_i and the physical rent growth rate g_i depend on the policy environment and the evaluation horizon h . We denote these policy-adjusted parameters by $\delta_i(h; \mathcal{P})$ and $g_i(h; \mathcal{P})$.¹³ We allow rents to respond to both *citywide* and *local* changes in housing stock.

¹²In this framework, developers choose *when* to redevelop and *how much* to build, but not the quality of construction conditional on redevelopment. New construction is assumed to enter without accumulated rental depreciation, reflecting the fact that redevelopment typically replaces aging structures with buildings that must meet current building codes, safety standards, and regulatory requirements. To the extent that developers in high-demand locations could choose to upscale quality, allowing for such choices would tend to amplify replacement-driven rent increases.

¹³Because the redevelopment probability in Proposition 3 depends on the physical drift only through the constant m , the closed-form expression in Equation (7) continues to hold verbatim once g is replaced by the policy-adjusted growth rate $g_i(h; \mathcal{P})$. Within each policy counterfactual and horizon h , $g_i(h; \mathcal{P})$ is constant over $[0, h - d]$, so the first-passage law is unchanged and only the value of m shifts.

Negative supply-induced adjustments. Let $S(h; \mathcal{P}) = \sum_i s_i(h; \mathcal{P})$ denote the expected citywide housing stock at horizon h under policy $\mathcal{P}$, and let $S(h; \mathcal{P}_0)$ denote the corresponding expectation under the status-quo. We measure the policy-induced change in expected total housing stock as the log difference:

$$\Delta S(h; \mathcal{P}) = \ln S(h; \mathcal{P}) - \ln S(h; \mathcal{P}_0).$$

We assume that rental growth responds to this aggregate supply shock with an elasticity of rents with respect to housing supply $\zeta_C < 0$, so that larger citywide increases in housing supply reduce expected rent growth.

Local spillover adjustment. Let $N_i(h; \mathcal{P})$ denote the expected housing supply in the vicinity of parcel i under policy $\mathcal{P}$, and define the associated policy-induced change as the log difference

$$\Delta N_i(h; \mathcal{P}) = \ln N_i(h; \mathcal{P}) - \ln N_i(h; \mathcal{P}_0).$$

The parameter $\zeta_L \leq 0$ captures how an increase in neighboring housing units affects local rent growth. This local channel reflects opposing forces: an amenity effect from an improved surrounding environment, and a local competition effect from added nearby stock. Its sign is thus not restricted a priori.

Combining the citywide supply and local spillover channels yields a common adjustment summarizing the effect of policy-induced construction on rent growth. For parcel i , the supply-induced adjustment is

$$\Delta \delta_i(h; \mathcal{P}) = \frac{1}{h} \left(\zeta_C \Delta S(h; \mathcal{P}) + \zeta_L \Delta N_i(h; \mathcal{P}) \right).$$

This term captures the effect of policy-driven supply changes on rent dynamics.¹⁴

Mapping supply adjustments into rents. We incorporate this supply adjustment into the two drift parameters governing rent dynamics. The *risk-neutral drift* governs valuation under the pricing measure $\mathbb{Q}$. For parcel i , the baseline pricing drift implied by the cap rate c_i is $\delta'_i \equiv r - c_i$. Rent growth, however, evolves under the physical measure $\mathbb{P}$. We denote the baseline physical rent growth rate by g'_i , which reflects the expected rent growth consistent with the total return on real estate net of the cap rate in the status-quo.

¹⁴Under the risk neutral measure, the expected potential rent satisfies $\mathbb{E}_{\mathbb{Q}}[R_h] = Re^{\delta h}$. At horizon h under policy $\mathcal{P}$, an exogenous supply shock $\Delta S(h; \mathcal{P})$ scales expected rents by $e^{\zeta_C \Delta S(h; \mathcal{P})}$ and implies $Re^{(\delta + \Delta \delta)h} = Re^{\delta h + \zeta_C \Delta S(h; \mathcal{P})}$, which yields this drift adjustment.

For a given policy $\mathcal{P}$ and evaluation horizon h , the policy-adjusted parameters are

$$\delta_i(h; \mathcal{P}) = \delta'_i + \Delta\delta_i(h; \mathcal{P}), \quad g_i(h; \mathcal{P}) = g'_i + \Delta g_i(h; \mathcal{P}).$$

Within each policy counterfactual, both parameters remain constant over time in the geometric Brownian motion dynamics. The dependence on h reflects the annualization of the horizon-specific supply adjustment rather than a time-varying drift process.

Fixed point and convergence. In equilibrium, policy-induced supply adjustments must be consistent with developers' redevelopment responses. This is how the market clears: the policy-induced changes in citywide and local supply determine the drift adjustment $\Delta\delta_i(h; \mathcal{P})$, which updates both the pricing drift $\delta_i(h; \mathcal{P})$ and the physical rent growth rate $g_i(h; \mathcal{P})$. These updated parameters affect redevelopment probabilities and densities and, therefore, expected supply. The resulting equilibrium corresponds to a fixed point in which supply adjustments and redevelopment responses are mutually consistent. Appendix A.6 describes the numerical procedure used to compute this fixed point.

2.3 Expected Rental Incomes

Using the completion probability from Proposition 3 and the optimal density F^* , we derive the expected parcel-level rental income at horizon h incorporating the redevelopment option and spatial adjustments. These adjustments account for the effects of citywide supply changes and neighborhood spillovers on expected rent levels.

Corollary 1 *For parcel i , the expected rental income at horizon h is*

$$\begin{aligned} \text{RentInc}_i(h, \mathcal{P}) = & \underbrace{\left[1 - \widehat{\mathbf{P}}_i^*(H_{Re^*} \leq h; \mathcal{P})\right]}_{\text{Rent-weighted prob. of no development}} \times \underbrace{\Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}) \times \mathbf{E}_{\mathbb{P}}\{R_{ih}\}}_{\text{Rental income of existing property}} \\ & + \underbrace{\widehat{\mathbf{P}}_i^*(H_{Re^*} \leq h; \mathcal{P})}_{\text{Rent-weighted prob. of development}} \times \underbrace{\Gamma(F_i^*, 1, \alpha) \times \mathbf{E}_{\mathbb{P}}\{R_{ih}\}}_{\text{Rental income of new property}}, \end{aligned} \quad (10)$$

where the spatially-adjusted physical expectation of rents is

$$\mathbf{E}_{\mathbb{P}}\{R_{ih}\} = R_i \exp(g_i(h; \mathcal{P}) h), \quad (11)$$

which reduces to $R_i e^{g_i^* h}$ under the status quo, and $\widehat{\mathbf{P}}_i^*$ is the rent-weighted completion probability.¹⁵

¹⁵The physical completion probability $\mathbf{P}_i^*$ of Proposition 3 (drift $m_i(h; \mathcal{P})$) governs the redevelopment event and enters expected floor-space supply in Equation (9). The rent-weighted probability $\widehat{\mathbf{P}}_i^*$, with drift $m_i(h; \mathcal{P}) + \sigma$, enters

For a parcel with $R_i < R_i^{e*}$, $\hat{P}_i^*$ is given by the closed form in Proposition 3 (Equation (7)) with the physical drift $m_i(h; \mathcal{P})$ replaced everywhere by $m_i(h; \mathcal{P}) + \sigma$; the domain condition ($h > d$, $\hat{P}^* = 0$ for $h \leq d$) carries over unchanged. For $R_i \geq R_i^{e*}$, $\hat{P}_i^* = 1$ for $h > d$ and 0 for $h \leq d$, mirroring the immediate-development branch of Proposition 3. This substitution reflects that expected rental income weights the redevelopment event by the contemporaneous rent level.

The proof is in Appendix A.7.

Expression (10) shows that expected rental income is a weighted average of the income generated by the existing structure and the potentially redeveloped property. Spatial effects enter through two channels. First, the drift adjustment $\Delta\delta_i(h; \mathcal{P})$ modifies the pricing drift $\delta_i(h; \mathcal{P})$ and therefore affects the completion probability and the optimal density $F_i^*(\mathcal{P})$. The weights in (10) therefore incorporate spatial feedback induced by policy. Second, spatial supply responses also affect the level of expected rents through the physical growth rate $g_i(h; \mathcal{P})$ in (11). These adjustments apply even when redevelopment does not occur, so spatial feedback affects both new and existing structures.

3 Impact of Public Policies on Parcel-level Outcomes

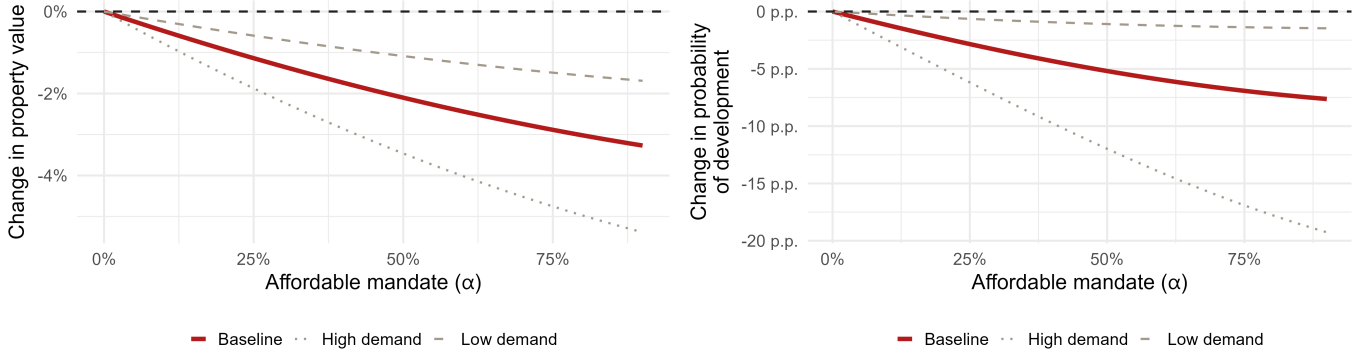
Before examining the calibrated spatial equilibrium, we first isolate the parcel-level mechanisms through which alternative policy instruments affect redevelopment decisions. The objective is to show that affordability mandates, density bonuses, and tax exemptions operate through fundamentally different margins of the redevelopment decision. We therefore focus on two outcomes: parcel value and redevelopment probability. Appendix B develops these effects further, reports comparative statics for the remaining parameters, shows why parcel heterogeneity matters for redevelopment, analyzes the trade-off between mandate and incentive at the parcel level, and numerically describes how exogenous supply propagates through the spatial spillovers.

Figure 3 shows how each policy instrument changes property values (left panels) and the ten-year redevelopment probability (right panels). We use the parametrization from Section 4, that allows density to raise both revenues and costs, and also allows the affordability mandate α to reduce rents and construction costs. We study a representative parcel near its redevelopment threshold and also show how these effects vary with initial demand, by moving potential rent R above or below the baseline.

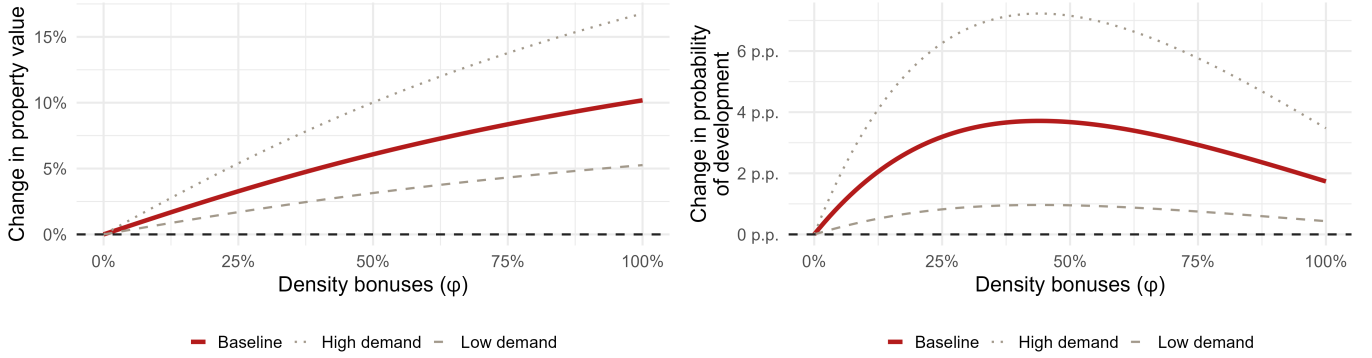
Panel A of Figure 3 shows that a higher affordable share requirement (α) lowers both property values and redevelopment probabilities. The mandate lowers property values by shrinking

only the rental-income decomposition, because the redevelopment event and the rent level are combined.

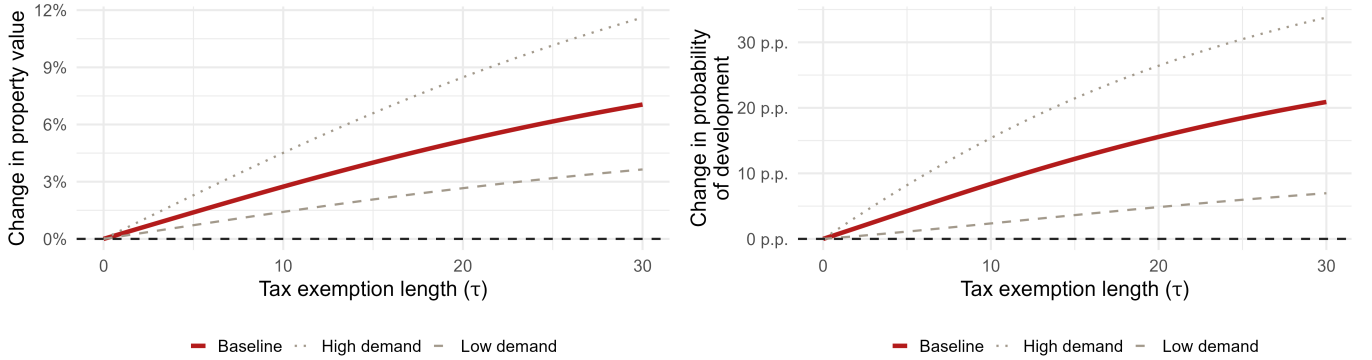
Figure 3: Effects of Policy Instruments on Property Value and Development Probability



A. Mandatory share for affordable housing (α)



B. Density bonuses (ϕ)



C. Length of tax exemptions (τ)

Notes: These figures show the effects of policy instruments on property value (left panels) and the probability of development within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, aged 150, with a rental growth of 4%, a volatility of 15%, and an annual potential rent \$50 per square foot. All other parameters and elasticities are reported in Table 1.

the redevelopment option, as the value of the existing structure does not change. Parcels with higher demand lose more value and face larger reductions in redevelopment probability, so de-

developers delay replacement more often on those parcels. The effect is also stronger for parcels closer to the redevelopment threshold, where the mandate binds most tightly.

Panel B of Figure 3 shows that a higher density bonus (ϕ) raises property values by making redevelopment more valuable. Because developers retain the option to build at higher density, property value always rises. However, the effect on redevelopment probability is ambiguous. The probability can rise or fall with ϕ , depending on the parcel characteristics and local housing demand. When ϕ is large, developers often delay construction, waiting until rents are high enough to justify a taller project. The delay is largest for parcels near the redevelopment threshold. On those parcels, extra density raises the value of waiting and pushes replacement further into the future.

Panel C of Figure 3 shows that a tax exemption raises property values. It also accelerates redevelopment by lowering the rent level that triggers replacement. The value gain comes entirely from the redevelopment option because the exemption applies only to new buildings. The effect on development probability is strongest for parcels close to their redevelopment threshold.

Taken together, the parcel-level analysis demonstrates that affordability mandates, density bonuses, and tax exemptions alter redevelopment through fundamentally different economic margins, even before citywide equilibrium effects are considered. These parcel-level mechanisms provide the economic foundation for understanding the calibrated spatial equilibrium analyzed in the following sections, where each policy generates distinct housing production, rent, and incidence outcomes across the city.

4 Model Quantification

We calibrate the model to New York City’s housing market and focus on affordable housing mandates as the central policy. First, we specify the functional forms $\Gamma(\cdot)$, $\gamma(\cdot)$, and $\kappa(\cdot)$ that govern the instantaneous return and costs to redevelopment. Second, we calibrate the parameter vector Θ using microdata, estimates from the literature, and selected empirical moments. Table 1 reports the resulting parameter values. Third, we construct the parcel-level dataset. We summarize its sources and summary statistics in Table 2.

4.1 Parametrization

4.1.1 The rental income scaling function

The underlying potential rent per square foot R is scaled by $\Gamma(\cdot)$, which maps parcel characteristics into annual rental income. We parametrize this scaling function as

$$\Gamma(F, z, \alpha, L) = LF^{(1+\epsilon_F)}z(1 - \alpha\epsilon_\alpha). \quad (12)$$

Here L is the parcel size, F is the floor-area ratio, $z \in [0, 1]$ is the rental depreciation index, and α is the share of affordable units in the building.¹⁶ The parameter $\epsilon_F > 0$ is the elasticity of rental incomes with respect to building height, while $\epsilon_\alpha > 0$ captures the rental discount associated with subsidized units. We set ϵ_F at 0.07 using estimates from [Danton and Himbert \(2018\)](#), [Ahlfeldt and Barr \(2022\)](#), and [Ahlfeldt et al. \(2026\)](#). The positive value of ϵ_F reflects the capitalization of height-related amenities, such as improved views and reduced street-level noise, into market rents.

The rental discount parameter ϵ_α could, in principle, be an element of the policy vector $\mathcal{P}$. To limit the counterfactual space, we, however, vary the share of affordable units α , while holding ϵ_α fixed. We follow [Favilukis et al. \(2023\)](#), who report that rent-stabilized units in New York City rent at an average discount of 26.6% relative to market-rate units, while rent-controlled units rent at an average discount of 77.7%. To span this range, we consider $\epsilon_\alpha \in \{0.25, 0.50, 0.75\}$. Our baseline counterfactuals set $\epsilon_\alpha = 0.25$. Section 6.1.3 examines how results vary with the rental discount, with particular attention to developers' voluntary adoption of policy $\mathcal{P}$.

4.1.2 The demolition cost function

We parametrize demolition costs as

$$\gamma(F_B, L) = k_d LF_B^{\eta_D}, \quad (13)$$

where η_D governs how demolition costs scale with existing built density F_B , and $k_d > 0$ is a cost shifter. We calibrate η_D using information from developer interviews. They reported that fixed cost components (those that do not scale with structure size) account for approximately 50–60% of total demolition costs. To match this feature, we choose η_D so that the average ratio of the cost of demolishing a building with $F_B = 0.1$ to the cost of demolishing the full structure equals 55%.

¹⁶The theoretical model abstracts from the size of the parcel because the optimal decision rule is unaffected by it (i.e., the expansion threshold R^{e*} is homogeneous of degree zero in L). Because it affects aggregate outcomes, such as total housing production, we reintroduce it here explicitly.

Table 1: Model Elasticities and Parameters

Parameters	Description	Value	Methods & main sources
<u>The Rental Scaling function</u>			
ε_F	Density elasticity of rent	0.07	Calibration – Literature
ε_z	Age elasticity of rent	-0.089	Estimation – Table C2
ε_α	Affordable units rental discount	{0.25; 0.50; 0.75}	Calibration – Literature
<u>The demolition cost function</u>			
η_D	Demolition cost elasticity of density	0.296	Target: Fixed cost demolition (55%)
k_d	Baseline demolition cost per density unit	72.10	Target: Share of development cost (15%)
<u>The construction cost function</u>			
η_F	Density elasticity of construction cost	0.60	Calibration – Literature
η_α	Affordable housing construction cost discount	0.20	Estimation - RCLCO report
k_c	Baseline construction cost per density unit	425.98	Target: Land-value ratio (23.68%)
<u>Citywide demand and local spillover parameters</u>			
ζ_C	Citywide supply elasticity of rent	-0.66	Calibration – Literature & CEX
ζ_L	Local supply elasticity of rent	-0.01	Estimation – Section C.2
	Distance for local supply spillovers	500 feet	
<u>Others</u>			
ψ	Real estate tax rate (% of rental income)	20.15	Estimation - DoF & Costar
d	Construction periods (years)	3.08	Estimation – DCP
r	Risk free rate (%)	1.90	Estimation – FED

Formally,

$$\hat{\eta}_D : \frac{1}{n_B} \sum_{i: F_{B,i} > 0} \frac{\gamma(0.10, L_i; \eta_D)}{\gamma(F_{B,i}, L_i; \eta_D)} = 0.55,$$

where $n_B \equiv \#\{i : F_{B,i} > 0\}$. This condition yields $\eta_D = 0.296$, implying substantial returns to scale in the demolition industry. Developers also reported that demolition costs typically account for 13–20% of total redevelopment costs when an existing structure is removed. This range pins down the relationship between the demolition cost shifter k_d and the construction cost shifter k_c . We determine these parameters jointly as described in Section 4.3.

4.1.3 The construction cost function

We parametrize annual construction costs as

$$\kappa(F, \alpha, d, L) = \frac{1}{d} k_c L F^{(1+\eta_F)} (1 - \alpha \eta_\alpha), \quad (14)$$

where k_c is the baseline construction cost per square foot, $\eta_F > 0$ governs how costs scale with building height, and η_α captures cost differences between market-rate and affordable units. We set $\eta_F = 0.60$ to reflect the convex relationship between height and construction costs. This value is consistent with estimates in Ahlfeldt and McMillen (2018), Ahlfeldt and Barr (2022), and Ahlfeldt

et al. (2026), which report elasticities in the range of 0.55–0.61 for tall residential buildings.

The parameter η_α captures systematic differences in per-square-foot construction expenditures between market-rate and subsidized units. We calibrate η_α using data from the National Apartment Association’s Income and Expenses Report, as summarized in the RCLCO report “[The Affordable Housing Asset Class](#)”. This report documents capital expenditures per square foot for market-rate and subsidized units from 2012 to 2021. On average, subsidized units receive 19.9% lower capital expenditure per square foot than market-rate units. We treat this differential as a reasonable proxy for construction cost differences and set $\eta_\alpha = 0.20$.

4.2 Citywide Demand and Local Spillover Parameters

The spatial equilibrium model developed in Section 2 holds because of two elasticities that link redevelopment to rent dynamics. The first is the citywide inverse demand elasticity, ζ_C , which governs how aggregate housing supply affects citywide rent growth. Because this relationship governs market-wide equilibrium, we calibrate it using evidence on long-run housing demand from the empirical literature. The second is the local spillover elasticity, ζ_L , which governs how nearby redevelopment affects neighborhood rent growth. Unlike ζ_C , this relationship cannot be inferred from demand theory because nearby redevelopment simultaneously generates opposing competition and amenity effects. We therefore estimate ζ_L directly using an event-study analysis of New York City building completions.

4.2.1 Citywide inverse demand elasticity (ζ_C).

We parametrize ζ_C as the inverse of the long-run price elasticity of housing demand. In an open-city model, this elasticity decomposes into an intensive margin, the price elasticity of floorspace demand, and an extensive margin, the labor supply elasticity scaled by the housing expenditure share.¹⁷ This decomposition holds for any preferences and implies that:

$$\zeta_C = - \left(\underbrace{\text{Housing.Share} \times \text{Labor.Supply.Elasticity}}_{\text{Migration component}} + \underbrace{\text{Floorspace.Demand.Elasticity}}_{\text{Intensive margin component}} \right)^{-1} \quad (15)$$

We use the shelter expenditure share for the New York metropolitan area, 27.1%, from the

¹⁷We derive this decomposition by assuming that aggregate floorspace demand is the demand per household times the population, and that there are idiosyncratic location preferences drawn from a Fréchet distribution with shape parameter ϱ , such that $Pop \propto V(R, w)^\varrho$, where w is income. By Roy’s identity, it follows that the migration component of the long-run demand elasticity simplifies to $\text{Housing.Share} \times \partial \ln Pop / \partial \ln w$, where the last derivative is the labor supply elasticity.

Consumer Expenditure Surveys (U.S. Bureau of Labor Statistics, 2025). For the labor supply elasticity, we use 3, which lies within the range [2.1, 4.0] of estimates for the United States (Diamond, 2016; Bryan and Morten, 2019; Hornbeck and Moretti, 2024). We set the floorspace demand elasticity to 0.70, the preferred estimate of Albouy et al. (2016), which falls between the 0.61 of Finlay and Williams (2025) for the United States and the 0.90 implied for France by Combes et al. (2019), as reported in Robert-Nicoud et al. (2026).

Our preferred parametrization is therefore $\zeta_C = -(0.271 \times 3 + 0.70)^{-1} = -0.66$. Varying the three inputs over their plausible ranges (Floorspace Demand Elasticity $\in [0.6, 0.9]$, Labor Supply Elasticity $\in [2, 4]$, and a housing share between 25% and 35%) yields values between -0.91 and -0.43 , and we report robustness of our main results across this interval.

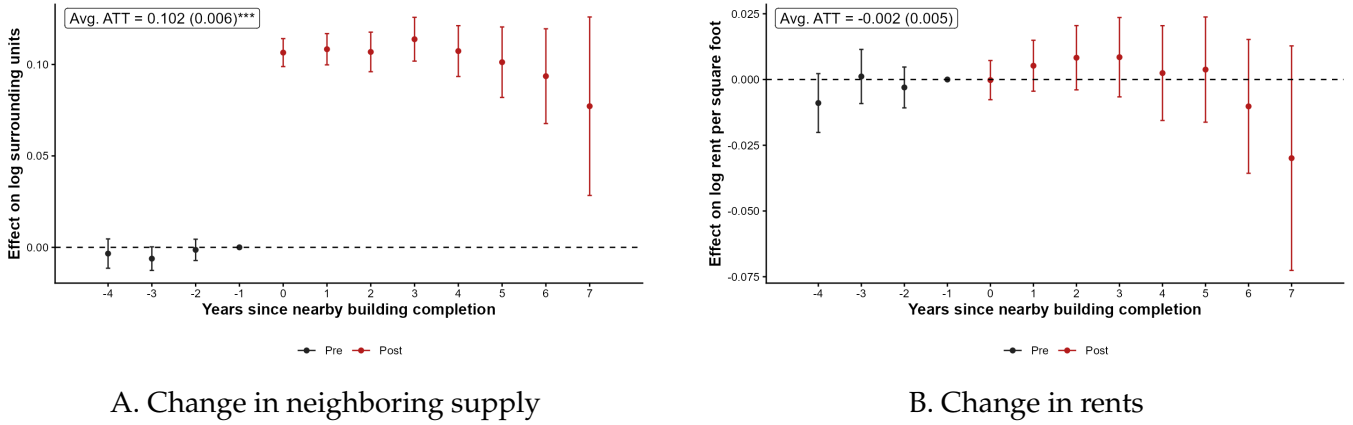
4.2.2 Local spillover parameter (ζ_L).

In our model, ζ_L captures how much rents respond to a nearby increase in supply, beyond the citywide effect. The sign of this elasticity is an empirical question because new units are close substitutes for the incumbent stock and compete with it, pushing nearby rents down, while they may also raise local amenities (Autor et al., 2014; Diamond, 2016; Gechter and Tsivanidis, 2023; Almagro and Domínguez-Iino, 2025), pushing nearby rents up. We estimate it for New York City by comparing buildings exposed to nearby new construction with otherwise similar buildings not yet exposed. We combine building rental incomes from the Notice of Property Value (NOPV) with residential building completions (New York City Department of City Planning, 2026a).

We use the staggered difference-in-differences estimator of Callaway and Sant’Anna (2021) to trace dynamic treatment effects by length of exposure. Appendix C.2 provides more details regarding the data, method, and robustness. A building is treated in the year a new residential building of at least 50 units is completed within 500 feet of it. To avoid comparing exposed buildings to the entire stock of buildings in the city, our preferred sample is composed of treated and buildings near parcels with a filed-but-uncompleted residential project. As such, identification comes from the timing of nearby completion rather than from distance to redevelopment.

Figure 4 reports the dynamic effects on neighboring supply (Panel A) and on rental incomes (Panel B). A nearby completion raises the surrounding housing stock by roughly 10% (an overall ATT of 0.102, significant at the 1% level) and holds it at that level. This result confirms that our methodology captures a substantial and durable local supply shock. Rents, by contrast, do not respond significantly. The post-completion effect is statistically non-significant and economically negligible throughout, with at most a slight downward drift at the longest horizons. The ratio of the two responses implies $\zeta_L = -0.01$. The near-zero estimate thus reflects that the effects of increased local housing competition, which lowers rents, and of neighborhood amenity improve-

Figure 4: Local effects of a nearby building completion on supply and rents



Notes: These Figures show the [Callaway and Sant’Anna \(2021\)](#) event-study estimates for buildings exposed to a new residential building of more than 50 units within their 500-foot radius. Panel A shows the estimates for the log number of residential units within 500 feet (the 2013 PLUTO count rolled forward with net completions from the DCP Housing Database). Panel B shows the estimates for the building’s log rent per square foot (gross income divided by floor area). Event time zero is the completion year of the nearby building. The sample comprises residential buildings of at least eleven units, excluding public housing, buildings with a commercial floor-area share above 10%, the building’s own lot, and buildings exposed to a second nearby completion during the period. The control group is restricted to not-yet-treated buildings and to buildings with a filed-but-uncompleted project nearby. Bars denote 95% confidence bands from a multiplier bootstrap clustered at the building level.

ments, which raise them, offset each other ($\text{LocalEffect} = \text{CompetitionEffect} + \text{AmenityEffect} \approx 0$).¹⁸ Appendix C.2 shows this near-zero effect is robust across radii, samples, and specifications. We therefore calibrate $\zeta_L = -0.01$ and assess robustness to $\zeta_L \in \{-0.10, -0.05, 0, +0.05, +0.10\}$ in explaining patterns of gentrification in Section 6.2.3.

4.3 Calibration of Other Parameters

Construction length (d) We calibrate the construction period using the Housing Database Project-Level Files ([New York City Department of City Planning, 2026a](#)). We define construction duration as the interval between permit issuance and building completion. During this period, developers incur construction costs but earn no rental income. We restrict the sample to multi-family buildings developed by private for-profit firms. We exclude projects completed in less than three months or more than ten years. We set $d = 3.08$ years, which matches the average time between permit issuance and completion.

¹⁸Our near-zero estimate is smaller than recent local estimates. [Li \(2022\)](#) finds that a 10% increase in nearby supply lowers local rents by about 1% within 500 feet, and [Rollet \(2025\)](#) finds a larger negative effect at a comparable scale. [Anenberg and Kung \(2020\)](#), [Asquith et al. \(2023\)](#), [Mense \(2025\)](#), and [Pennington \(2025\)](#) report negative but limited effects. These studies imply that local competition modestly dominates amenity gains. Our results are consistent with those but not statistically distinguishable from zero.

Risk-free rate (r) We calibrate the risk-free rate r using the average one-year yield on U.S. Treasury securities reported by the Board of Governors of the Federal Reserve System ([FRED](#)). The average yield over the period from January 2000 to December 2023 equals 1.90%. We therefore set $r = 0.019$. This rate serves as the discount rate in the risk-neutral valuation.

Property tax rate (ψ) We construct an effective property tax rate on residential properties by combining multiple datasets. We first link assessment data for multi-family properties from PLUTO with statutory tax rates ([NYC Department of Finance](#)) to compute parcel-level tax liabilities since 2000. We then merge these tax liabilities with 36,000 multi-family transactions from CoStar and compute tax rates by dividing tax bills by corresponding sale prices. Since 2000, the median effective tax rate equals 1.1%, with an interquartile range of 0.57–1.83%. The model requires a tax rate on rental flows. We therefore convert the asset-based tax rate into an income-based rate by dividing by the sample’s average capitalization rate of 5.34%. This mapping translates observed property tax liabilities into the effective tax wedge on rental income that enters developers’ decisions.

Baseline construction and demolition costs (k_c and k_d) Three parameters govern the monetary level of the model: the market rent of a non-depreciated square foot, R , the demolition cost, k_d , and the construction cost, k_c . All three quantities are expressed per square foot. Property values (Proposition 1) are homogeneous of degree one in R , k_d , and k_c , and development probabilities (Proposition 3) are homogeneous of degree zero. Only their relative magnitudes therefore matter for counterfactual analyses. We take the distribution of R_i across parcels as given and calibrate k_d and k_c to match empirical moments. We treat these parameters as constant, since labor, material, and regulatory costs are broadly similar within the city ([Glaeser and Gyourko, 2018](#)).

To pin down k_c , we use PLUTO data on the ratio of land to property values. For each parcel, we compute the model-implied land-value ratio $LandRatio_i = \pi_i(F_{B,i} = 0; \Theta) / \pi_i(F_{B,i} = F_{B,i}; \Theta)$, which compares the value of the parcel with no structure to its value at the observed density $F_{B,i}$. We interpret this ratio as the redevelopment option value when the existing structure is removed. Lower construction cost k_c increases redevelopment incentives and raises land values relative to current property values. We therefore choose k_c so that the average model-implied $LandRatio_i$ matches the observed mean of 23.7%.

To identify k_d , we use evidence that demolition costs account for 13–20% of total development costs when an existing structure is removed. We target the midpoint of this range by imposing

$$\frac{1}{n_B} \sum_{i:F_{B,i}>0} \frac{\gamma(F_{B,i}, L_i)}{d \cdot \kappa_i(F_i^*, \alpha_{B,i}, d, L_i) + \gamma(F_{B,i}, L_i)} = 0.15$$

Table 2: Summary Statistics of Lot-level Data

Statistic	Source	Mean	St. Dev.	Min	Median	Max
Current FAR – $F_{B,i}$	PLUTO	1.05	1.17	0.00	0.73	88.55
Maximum FAR – $\bar{F}_i$	PLUTO	1.43	1.43	0.50	0.90	88.55
Lot size (th. sqft) – L_i	PLUTO	3.72	6.20	1.25	2.50	250.00
Age (years)	PLUTO	72.94	36.63	0	84	302
Rental depreciation index (%) – $z_{B,i}$	Estimated	70.14	6.84	60.13	67.34	100.00
Share of aff. units (%) – $\alpha_{B,i}$	DHCR & HPD & DOF	0.40	3.96	0	0	50
Annual Rent per sqft (\$)	ACS + PLUTO	24.07	7.98	2.48	23.11	99.24
Annual Potential Rent per sqft (\$) – R_i	Estimated	34.85	10.87	3.37	34.01	124.65
Expected growth (%) – g_i	Costar	2.25	0.45	1.34	2.32	4.10
Caprate (%) – $r - \delta_i$	Costar	5.34	0.34	4.22	5.21	6.17
Volatility (%) – σ_i	Costar	8.21	0.39	7.46	8.26	11.08

Notes: This table presents summary statistics for the 765,305 parcels included in the baseline analysis. The majority of parcel-level data is sourced from the Primary Land Use Tax Lot Output (PLUTO) as of 2023. Age measures the number of years since the last recorded renovation or construction. Rental depreciation index is the rental discount associated with property aging, with 1 representing a newly constructed building. Annual rent per square foot is derived from the 2023 ACS 5-Year data at the census block-group or tract level. Potential rent reflects market-rate rents for one non-depreciated square foot at FAR=1, obtained by adjusting observed rent for the typical building within each geography. Cap rates, rental returns, and volatility are sourced from CoStar, following their definitions for New York City residential markets, as detailed in Appendix C.

in the model. This condition matches the share of the total development costs attributable to demolition. Higher demolition costs k_d increase the share of total redevelopment costs attributable to demolition. Matching the land-value ratio and demolition share targets jointly yields calibrated values of $k_c = 425.98$ and $k_d = 72.10$, very close to the 2024 estimate of \$564 from [New York Building Congress \(2025\)](#). Because the mean value of R_i is \$34.85. The cost of constructing one unit of density is approximately twelve times the annual rental income.

4.4 Parcel-level Data

Each parcel i is characterized by its current density $F_{B,i}$, maximum allowable density $\bar{F}_i$, rental depreciation index $z_{B,i}$, share of current affordable units $\alpha_{B,i}$, and location. These variables determine the value of the existing structure and the constraints on redevelopment. Location determines the potential rent per square foot R_i , and the expected return and volatility of rental income. Together, these variables pin down the evolution of rents and the payoffs to redevelopment. We restrict the sample to parcels zoned or partially zoned for residential use. We exclude parcels designated exclusively for public or other non-residential purposes. The final sample contains 765,305 parcels. Appendix C describes the data construction and cleaning procedure, and includes additional summary statistics of key parcel-level variables.

Built and maximum allowed density ($F_{B,i}$ and $\bar{F}_i$). We measure current density $F_{B,i}$ and maximum allowable density $\bar{F}_i$ using the built Floor-Area-Ratio (FAR) and zoning FAR from PLUTO (New York City Department of City Planning, 2026b). We also record the lot size L_i , so that the total built space on parcel i equals $s_i = F_{B,i}L_i$. In the baseline calibration, the average built density $F_{B,i}$ equals 1.05. The average building size is approximately 4,680 square feet. The average maximum allowable density $\bar{F}_i$ equals 1.43. A total of 36.7% of parcels are built to their maximum allowable density. As Figure C4 shows, unused capacity is spatially heterogeneous, with greater scope for densification in areas such as the Southern Bronx.

Rental depreciation index ($z_{B,i}$). We model depreciation as a reduction in rental income as the structure ages. Specifically, we set $z_i = \text{Age}_i^{\epsilon_z}$, $\epsilon_z < 0$ where Age_i is the number of years since construction or last recorded renovation. Older buildings, therefore, generate lower effective rental income. This functional form implies faster depreciation for younger buildings. Lopez and Yoshida (2022) provides evidence consistent with this pattern. Because New York City includes many types of rent regulations that might affect the relation between age and rental incomes, we estimate this elasticity rather than calibrate it from the literature using other empirical settings. We follow Rosenthal (2014) and regress transaction prices on building age in a hedonic pricing framework, as described in Appendix C.1. The estimated elasticity ranges from -0.096 to -0.089 . As a robustness check, we also perform the same analysis using the rental incomes for the buildings included in the NOPV data and obtain an elasticity ranging from -0.088 to -0.084 , consistent with our transaction-based estimate. These estimates imply an average annual decline of approximately 2% over the first 10 years, and slower depreciation thereafter. We set ϵ_z to -0.088 , yielding a mean $z_{B,i}$ of 0.70 and a standard deviation of 0.07. Lower values of $z_{B,i}$, concentrated mostly in Queens and Brooklyn (Panel B of Figure C4), reduce the value of the existing structure and increase the likelihood of redevelopment by lowering the expansion threshold R_i^{e*} .

Current share of affordable units ($\alpha_{B,i}$). We construct $\alpha_{B,i}$ from multiple data sources, because no single dataset reports building-level affordability shares. We first use publications from the New York State Division of Housing and Community Renewal (DHCR), which we complement with the exemption field from the NOPV, to identify buildings that receive a 421-a or a J-51 tax exemption. These programs provide property tax benefits in exchange for affordable units. We identify 5,142 parcels with a 421-a exemption and 2,898 parcels with a J-51 exemption. We assign $\alpha_{B,i} = 30\%$ for 421-a properties and $\alpha_{B,i} = 50\%$ for J-51 properties, to reflect the stronger affordability requirement under J-51. We then incorporate buildings that the Department of Housing Preservation & Development reports as participating in Mandatory Inclusionary Housing (MIH) or Voluntary Inclusionary Housing (VIH) programs. These programs also require the provision of affordable units. We identify 498 MIH/VIH buildings and assign $\alpha_{B,i} = 30\%$ for those parcels.

Because VIH allows FAR bonuses to be transferred to parcels other than those hosting affordable units, we assign a positive $\alpha_{B,i}$ only to the generating sites.

Potential rent per square foot (R_i). We estimate R_i using block-group-level median rents from the 2023 American Community Survey (ACS) 5-year data. Appendix C.4 provides details and shows that our rent measure correlates strongly with other available sources. We first convert reported rents to a per-square-foot measure by normalizing ACS rents using the median residential unit size in the block group of i (UnitSize_i), computed from parcel-level data in PLUTO. The average annual rent per square foot in the sample equals \$24.07. The variable R_i however represents the potential rent of *market-rate*, non-depreciated space at $\text{FAR} = 1$. We, therefore, adjust observed rent per square foot in geography j to remove the effects of depreciation, built density, and existing affordability requirements. For instance, low observed rent per square foot in a block-group may reflect low potential rent R_i or a high prevalence of obsolete structures. Specifically, we define

$$R_{i(j)} = \frac{\text{Rent}_{i(j)}}{\Gamma\left(\tilde{F}_{B,i(j)}, \tilde{z}_{B,i(j)}, \tilde{\alpha}_{B,i(j)}\right)},$$

where $\text{Rent}_{i(j)}$ denotes the average observed rent per square foot in geography j . The terms $\tilde{F}_{B,i(j)}$, $\tilde{z}_{B,i(j)}$, and $\tilde{\alpha}_{B,i(j)}$ denote the weighted median built density, rental depreciation index, and affordable unit share across parcels in geography j , respectively. We weight each parcel by built square footage $F_{B,i}L_i$ so that the characteristics reflect the stock of space occupied by renters. The average annual potential rent in the sample equals \$35.46 per square foot.

Rental return and volatility (δ_i , g_i and σ_i). We calibrate the baseline pricing drift δ'_i using observed capitalization rates (c_i) defined as the rent-to-price ratio. We set $\delta'_i = r - c_i$ where r is the risk-free rate. This specification treats δ'_i as a reduced-form, risk-neutral measure of expected rental growth that incorporates both fundamentals and the required risk premia. We obtain capitalization rates from CoStar market data and assign them to each parcel based on its location in one of the 37 CoStar-defined New York submarkets. Appendix C.5 describes the assignment procedure and reports summary statistics. We calibrate the baseline physical rent growth rate g'_i using Costar data with $g'_i = \text{TotalReturn}_i - c_i$. The average implied rent growth rate across submarkets equals 2.25%. This measure captures developers' expectations about future rental growth under the status-quo policy. We estimate volatility σ_i using asset and rental returns for multifamily properties over the past 20 years in each submarket. We compute standard deviations of these return series. The average annual volatility across submarkets equals 8.21%.

5 Baseline Model Outcomes and Validation

Before using the calibrated model to evaluate housing policies, we establish that it produces sensible baseline outcomes, that its central mechanism is robust to changes in the model’s underlying assumptions, and that it credibly describes New York City’s housing market. We report the model’s baseline equilibrium under the no-policy environment $\mathcal{P}_0 = [\alpha = 0, \phi = 0, \tau = 0]$, which serves as the reference point for all counterfactuals. At the parcel-level, we recover three objects: $\mathbf{P}_{ih}^*(\mathcal{P}_0)$, $F_i^*(\mathcal{P}_0)$, $\pi_i(\mathcal{P}_0)$. The redevelopment probability $\mathbf{P}_{ih}^*(\mathcal{P}_0)$ describes the probability that parcel i is redeveloped within horizon h . The optimal density $F_i^*(\mathcal{P}_0)$ determines how intensively the parcel is rebuilt once redevelopment occurs. The parcel value $\pi_i(\mathcal{P}_0)$ combines the continuation value of the existing structure and the option value of redevelopment: $\pi_i(\mathcal{P}_0) = \pi_i^{\text{existing}}(\mathcal{P}_0) + \pi_i^{\text{option}}(\mathcal{P}_0)$. Together, these objects characterize the margin through which policy affects supply and rents.

For each parcel i , we compute from these objects the expected housing supply $s_{ih}(\mathcal{P}_0)$ and the expected rental income $\text{RentInc}_{ih}(\mathcal{P}_0)$ following Equations (9) and (10), respectively. To ease the interpretation of our quantity results, we report outcomes in housing units rather than square footage by scaling the expected supply $s_{ih}(\mathcal{P}_0)$ by block-group median dwelling size. This transformation aids interpretation but leaves the redevelopment decisions unchanged. Rents are reported in dollars per month per unit. Appendix C.6 describes the aggregation in detail.

5.1 Baseline Results

Housing production and expected rents. Table 3, row 1, reports the outcomes under $\mathcal{P}_0$. We expect 167,758 additional housing units over ten years, in line with New York City’s historical net production of 10,000–28,000 units per year since 2010 (NYC housing production). After accounting for demolitions, we obtain a net supply addition of 126,215 units. We find that construction concentrates in locations with high potential rents R_i and on parcels with obsolete and underutilized structures. Parcels in these locations are more likely to reach the redevelopment threshold R_i^{e*} and therefore exhibit higher redevelopment probabilities $\mathbf{P}_{ih}(\mathcal{P}_0)$. Panel A of Figure C6 shows this spatial pattern.

Expected average rent rises from \$2,124 today to \$2,714 in 10-years. This increase follows directly from Equation (10). The first term captures non-redeveloped properties for which depreciation lowers $\Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}, L_i)$ over time and reduces rents on the old structure. The second term captures redeveloped properties with replacement raising rents because $\Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}, L_i) < \Gamma(F_i^*, 1, \alpha_{B,i}, L_i)$. The growth term g_i' increases $\mathbb{E}_{\mathbb{P}}\{R_{ih}\}$ for both states.¹⁹

¹⁹In the baseline scenario, δ_i' and g_i' do not vary with future expected supply, $\Delta S(\mathcal{P}_0, t)$, because the calibrated cap

Table 3: Baseline Calibration Outcomes and Robustness to Parameter Changes

	UnitCons (# of Unit)	UnitNet (# of Unit)	Rent (\$ per month)	$PV = \sum_i \pi_i$ (\$ Bn.)	Option (% of PV)
	(1)	(2)	(3)	(4)	(5)
<i>Baseline</i>	167,758	126,215	2,714	1,557.34	0.82
Varying key underlying model's elasticities					
<i>Shorter duration time ($d = d/2$)</i>	272,562	181,888	2,769	1,562.04	1.12
<i>Higher cost to height ($\eta_F = 2\eta_F$)</i>	190,297	134,729	2,697	1,563.99	1.24
<i>Higher return to height ($\epsilon_F = 2\epsilon_F$)</i>	196,253	144,285	2,941	1,685.67	0.79
<i>Higher depreciation ($\epsilon_z = 2\epsilon_z$)</i>	308,265	176,334	2,073	1,132.30	1.50
Other model benchmarks					
<i>No endogenous density choice ($F_i^* = \bar{F}_i$)</i>	171,221	131,343	2,725	1,553.13	0.55
<i>Adding an option to renovate</i>	170,356	126,166	2,715	1,557.34	0.82
<i>No development ($k_c = \infty$)[†]</i>	0	0	2,688	1,614.24	0
<i>No maximum allowable density ($\bar{F}_i = \infty$)[†]</i>	304,428	236,214	2,787	1,504.18	0.89

Notes: This table reports the aggregate outcomes of our baseline calibration with a horizon of 10 years and additional sensitivity analyses performed by varying underlying model parameters. Column (1) shows the expected number of housing units completed at the 10-year horizon, Column (2) shows expected net unit addition, and Column (3) shows expected monthly rent. Column (4) shows the sum of the parcel present values, and Column (5) shows the average value of the redevelopment option. The first row reports results under the baseline calibration described in Table 1, and subsequent rows report the same outcomes under alternative parameter assumptions. [†]Except for the last two model specifications, the rent drift is held fixed at its baseline value in these exercises and does not adjust with excess supply.

Asset valuation, and option value. Our calibration implies a total residential asset value of \$1.56 trillion, close to the Department of Finance's reported valuation of \$1.83 trillion for both commercial and residential buildings ([Tax Lot Assessment Portal](#)). At the parcel level, values satisfy $\pi_i(\mathcal{P}_0) = \pi_i^{\text{existing}}(\mathcal{P}_0) + \pi_i^{\text{option}}(\mathcal{P}_0)$. The redevelopment option accounts for 0.82% of total residential asset value. Thus, in a mature city like New York, most of the parcel's value comes from the existing structure rather than from growth options.

Parcel-level heterogeneity. Redevelopment varies across parcels through the optimal redevelopment probability P_i^* and density F_i^* . Figure C7 and Table C6 show these patterns across built space and potential rent. Older buildings redevelop more often. Depreciation lowers the existing payoff $\Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i)$, which reduces the continuation value π_i^{existing} . This lowers the expansion threshold R_i^{e*} and increases the redevelopment probability. Parcels with greater unused capacity ($\bar{F}_i - F_{B,i}$) also redevelop more often. Higher feasible density raises the post-redevelopment payoff $\Gamma(F_i^*, 1, \alpha_{B,i}, L_i)$, increases the option value π_i^{option} , and lowers R_i^{e*} . High existing density has the opposite effect. It delays redevelopment. It raises the opportunity cost of demolition

rate already incorporates developers' expectations under the status quo.

through the value of current cash flows, $\pi_i^{\text{existing}} \propto (1-\psi)/(r-\delta_i) \Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i) R_i$. A higher continuation value increases R_i^{e*} and reduces P_i^* . High-rent locations redevelop more often because higher potential rent increases the likelihood that $R_{ih} \geq R_i^{e*}$ and raises P_i^* . The model thus generates a rich cross-section, that aligns with the motivating evidence outlined in Section 1.

5.2 Robustness to Underlying Parameters

The remaining panels of Table 3 assess how changes in parameters affect aggregate outcomes. In these scenarios, we hold developers' expectations about future aggregate and local supply fixed and thus do not update $\mathbb{E}_{\mathbb{P}}\{R_{ih}\}$ through equilibrium feedback. We do so to isolate the direct effect of parameter assumptions on the redevelopment decisions and, thus, aggregate outcomes. The objective is to illustrate how baseline outcomes vary with underlying parameters, not to evaluate a counterfactual in which a given parameter shifts exogenously.

5.2.1 Sensitivity to parameter assumptions

Shorter construction durations d increase redevelopment. A lower d raises the option value π_i^{option} through discounting which lowers the rent threshold R_i^{e*} , which increases P_i^* . Redevelopment occurs earlier and more often.

Higher curvature of construction costs with respect to height raises marginal costs $\kappa(F, \alpha, L)$. This change reduces the optimal density F_i^* . At the same time, developers respond to this lower optimal density by lowering the rent threshold. Consequently, they anticipate redeveloping their parcels sooner. The timing component dominates the scale component. Total expected supply, therefore, increases despite lower density per project. These results underscore the central role of construction technology in shaping urban outcomes (Ahlfeldt et al., 2026).

Higher returns to height increase $\Gamma(F, z, \alpha, L)$ which raises the payoff to redevelopment if $F_i^* > F_{B,i}$. For most parcels, this shift reduces rent thresholds and increases the redevelopment probability. Faster depreciation works through the existing assets. It lowers $\Gamma(F, z, \alpha, L)$, reduces the continuation value π_i^{existing} , and again lowers the replacement threshold. Both changes make redevelopment more attractive, either by improving the upside or by eroding the value of the existing asset.

5.2.2 Alternative models

The final panel of Table 3 reports baseline outcomes under alternative model assumptions. The first row imposes a density constraint $F_i^* = \bar{F}_i$ upon redevelopment (i.e., developers do not choose the optimal density). Aggregate outcomes remain close to the baseline, although the sum

of present values declines. This comparison shows that optimizing F_i^* matters mainly for surplus, not for citywide volumes. In New York City, zoning is often binding, so developers already build close to the regulatory limit (Brueckner et al., 2024). Fixing $F_i^* = \bar{F}_i$ therefore trims gains from optimization but does not greatly alter output.

The second row shows the results of a model in which we not only permit redevelopment but also the renovation of the current asset. We detail this model extension and present our main counterfactual policy results under this extended model in Appendix E. Renovation is the preferred option on many parcels, yet it has limited aggregate implications for housing supply because this option is preferred for parcels unlikely to redevelop. Aggregate housing supply remains essentially unchanged. Total expected production (renovation and reconstruction) equals 170,356 units, and 126,166 units net of demolitions. The larger difference between the two numbers indicates that, under this model, a greater share of the housing stock consists of renovated units. The expected citywide rent is marginally higher with renovation, reflecting quality improvements in the existing stock rather than a contraction of housing supply.

The third row of Table 3 shuts down redevelopment by setting the cost of construction prohibitively high ($k_c = \infty$). We use this comparison to isolate the contribution of the replacement effect relative to the supply effect, and thus solve for the spatial equilibrium with $\zeta_C + \zeta_L < 0$. In this case, the expected rent is 0.95% lower than in the baseline despite the supply being held fixed. Without redevelopment, parcels remain at $F_{B,i}$. With redevelopment, some parcels move to F_i^* . Because $\Gamma(F_i^*, 1, \alpha_{B,i}, L_i) > \Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i)$, new construction raises average rents by replacing older, lower-rent buildings with newer and taller ones. In the absence of redevelopment, rents fall because the stock deteriorates.

The final row of Table 3 removes the density constraint. Construction increases by a factor of approximately two. This result reflects a higher redevelopment payoff, since unconstrained density raises $\Gamma(F^*, 1, \alpha_{B,i}, L_i)$. We include the negative supply effect ($\zeta_C + \zeta_L < 0$) to avoid unrealistic infinite construction and ensure equilibrium. Even in this case, average rents rise to \$2,787. The supply response remains too small to offset the replacement effect. Land value in this scenario declines by 3.2%, highlighting the difficulty of convincing landowners to support such policy.

Taken together, the results of Table 3 show that the main driver of outcomes is the redevelopment margin. Density and redevelopment constraints matter, but they operate largely through shifting the rent threshold R_i^{e*} and the redevelopment probability $\mathbf{P}_i^*$, rather than through aggregate changes in supply.

5.3 Model Validation

The preceding exercises demonstrate that the model’s qualitative behavior is internally consistent and robust to changes in key assumptions. Internal robustness alone, however, does not establish that the calibrated model credibly describes New York City’s housing market. We therefore turn to external validation by comparing the model’s predictions with independent observations on property values and subsequent construction patterns.

Does the model predict real estate values? We compare predicted asset values to observed transaction prices to validate the model. For each community district j , we compute the average model-implied value per housing unit:

$$\widehat{PricePerUnit}_j = \frac{1}{n_j} \sum_{i \in j} \frac{\pi_i(\mathcal{P}_0; \Theta)}{F_{B,i} L_i / \text{UnitSize}_i}.$$

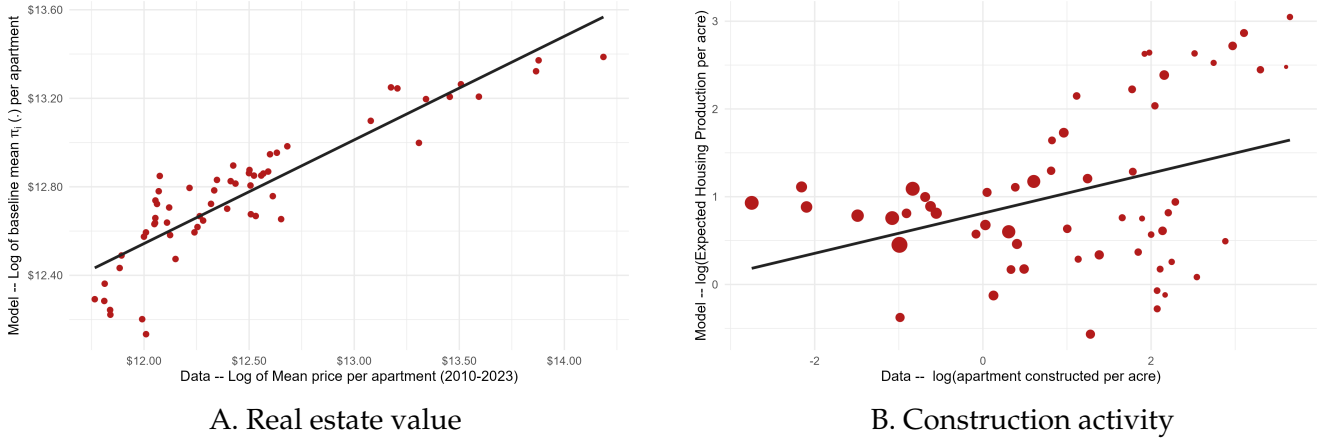
The denominator converts parcel value into a per-unit measure using built density and local unit size UnitSize_i . Panel A of Figure 5 compares the log of $\widehat{PricePerUnit}_j$ to observed log prices per unit from CoStar transactions from 2010 to 2023. The correlation equals 0.89, highlighting that the model captures cross-neighborhood variation in property values. This fit is central to our contribution because developers’ redevelopment decisions rely on maximizing π_i . By matching cross-sectional variations in values, we show that the model ranks parcels correctly by their incentives to redevelop across policy environments.

Does the model predict real estate construction activity? We compare model-implied net construction to observed net construction to validate the model’s predictions for redevelopment activity. The key object is the expected number of units constructed that depends on both the development probability $\mathbf{P}_i^*(\mathcal{P}_0)$ and the optimal density $F_i^*(\mathcal{P}_0)$. We measure observed construction using data on net housing units built by the private sector in each community district between 2016 and 2025 (New York City Department of City Planning, 2026a). We recalibrate parcel characteristics to their 2015 values using the procedure described in Section 4. For each district j , the model-implied number of housing units constructed is

$$\widehat{\text{UnitNet}}_j(\mathcal{P}_0, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}_0) \frac{L_i(F_i^*(\mathcal{P}_0) - F_{B,i})}{\text{UnitSize}_i},$$

expressed in housing units. Citywide, our model predicts net construction of 211,521 housing units. There were 199,992 housing units net of demolition during this period. Panel B of Figure 5 compares $\widehat{\text{UnitNet}}_j(\mathcal{P}, h)$ to observed construction, with both measures scaled by district acreage. The model matches the cross-sectional distribution of construction across neighborhoods. The

Figure 5: Model Validation



Notes: Panel A shows the relation between the log mean price per housing unit of Multi-Family buildings, using transaction from CoStar from 2010 to 2023, and the log of the model-implied mean price per unit in each community district. Panel B plots the log number of housing units constructed in each community district between 2016 and 2025, scaled by community district acreage, against the corresponding model-implied construction outcomes based on the 2015 calibration.

model assigns redevelopment to the correct locations by capturing variations in $\mathbf{P}_{ih}^*(\mathcal{P}_0) F_i^*(\mathcal{P}_0)$ across parcels. We provide this model fit by borough in Figure C8 of the Appendix. It shows an even greater fit for all boroughs except for the Bronx, where a large share of housing production during the 2016–2025 period was associated with programs such as the Harlem River Waterfront District. Excluding the Bronx community districts shows a 60.3% correlation between the predicted and realized housing construction measures.

Taken together, the results of this section show that the calibrated model credibly represents New York City’s redevelopment process. The baseline outcomes reproduce the qualitative features of the city’s housing market, the robustness exercises show that the paper’s central mechanism holds across a wide range of assumptions, and the external validation demonstrates that the resulting outcomes closely match observed property values and construction patterns. This validation matters beyond demonstrating aggregate fit. Because the counterfactuals shift redevelopment thresholds across heterogeneous parcels, the model must first capture redevelopment incentives across the city before it can credibly evaluate the effects of alternative housing policies.

6 Policy Counterfactuals

This section applies the spatial equilibrium developed in Section 2 and calibrated in Section 4 to evaluate how alternative housing policies affect housing production and rents. Because each policy alters parcel-level redevelopment incentives in different ways, it alters when, how inten-

sively, and where redevelopment occurs. The resulting citywide outcomes illustrate the economic implications of the redevelopment margin, showing how parcel-level redevelopment decisions aggregate into housing production, rent dynamics, and the trade-offs among alternative policy instruments.

Our counterfactual analyses are constructed such that, at $t = 0$, the city moves from the baseline environment $\mathcal{P}_0$ to a counterfactual regime $\mathcal{P} = [\alpha \geq 0, \phi \geq 0, \tau \geq 0]$, where α denotes the affordable housing requirement, ϕ the density bonus, and τ the duration of the post-redevelopment property tax exemption. A policy change affects outcomes by changing developers' optimization problems. It shifts the value of redevelopment relative to the continuation value of the existing structure, alters the optimal redevelopment threshold R^{e*} , and also changes the optimal redevelopment density F^* . Each developer's choice, in turn, changes expectations about future rental growth on nearby and citywide parcels.

We present all counterfactual results relative to the baseline. For example, the ten-year change in expected net housing production under $\mathcal{P}$ is

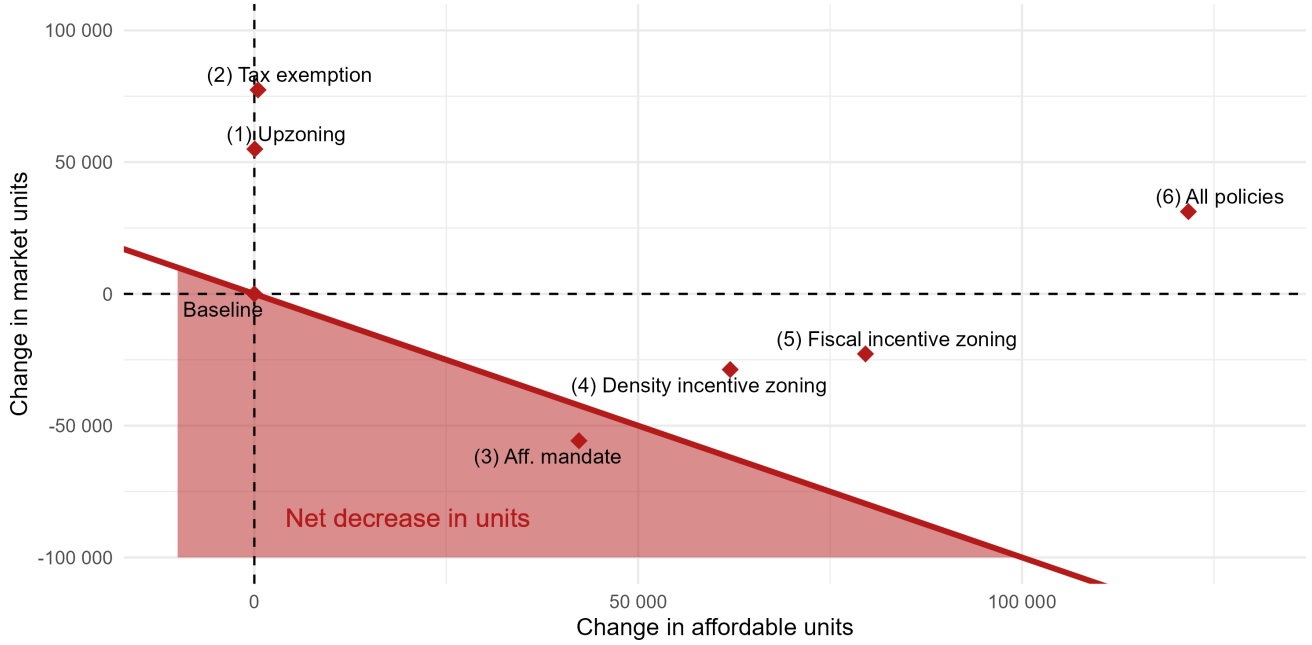
$$\Delta \text{UnitNet}(\mathcal{P}|h = 10) = \text{UnitNet}(\mathcal{P}|h = 10) - \text{UnitNet}(\mathcal{P}_0|h = 10)$$

We report monetary outcomes as percentage deviations to facilitate comparisons across policy bundles. Table D3 in the Appendix reports numerical outcomes for each policy.

Although the model allows any combination of policy instruments, we focus on six policy environments that span the main approaches used in practice. Policy (1) is upzoning, which raises allowable density by 50%. Policy (2) is a 20-year property tax exemption. Policy (3) is an affordability mandate that requires 30% of units in new construction to be affordable. Policy (4) is a Density Incentive Zoning, which combines the affordability mandate with the density bonus. Policy (5) is Fiscal Incentive Zoning, which combines the affordability mandate with the tax exemption. Policy (6) is the combined incentive policy that stacks all three instruments.

We organize the results in three blocks. We first study housing production and distinguish between market-rate and affordable units. We then study expected rents and decompose rent changes into four channels: the affordable share effect, the citywide supply effect, the local supply effect, and the replacement effect. We then turn to policy tradeoffs and show that policies that hold housing production fixed generally do not hold rents fixed.

Figure 6: Policy Counterfactuals — Housing Production



Notes: This scatter plot shows the expected change in affordable housing units (x-axis) and market-rate housing units (y-axis) relative to the baseline over a 10-year horizon. Policy (1) Upzoning corresponds to a 50% increase in allowable density; Policy (2) Tax exemption corresponds to a 20-year property tax exemption; and Policy (3) corresponds to a 30% affordability mandate. Policies (4), (5), and (6) combine these instruments.

6.1 Housing Production

6.1.1 Market-rate vs affordable units: aggregate effects

Figure 6 summarizes how policy affects aggregate housing production through the redevelopment margin by plotting changes in affordable units against changes in market-rate units. Each point corresponds to a policy regime. The negative 45-degree line provides a benchmark for interpreting how changes in affordable units map into changes in market-rate housing units. Downward movements along this line correspond to one-for-one reallocation from market-rate to affordable units. Consequently, points above the line indicate net expansions of the housing stock, while points below the line indicate net contractions.

Policy affects aggregate production primarily by shifting redevelopment thresholds $R_i^{e*}(\mathcal{P})$, which determine the set of parcels that redevelop. Policies that reduce $R_i^{e*}(\mathcal{P})$ activate redevelopment on marginal parcels and expand the housing stock. Policies that increase $R_i^{e*}(\mathcal{P})$ deactivate these parcels and contract supply. Unilateral upzoning ($\phi > 0$) and tax exemptions ($\tau > 0$) both increase market-rate housing, but they operate through distinct margins of the redevelopment decision. Tax exemptions lower the redevelopment threshold $R_i^{e*}(\mathcal{P})$ by increasing post-construction returns. This effect raises redevelopment probabilities $P_i^*(\mathcal{P})$ and shifts mass along the extensive

margin. Upzoning increases the value of redevelopment by allowing higher density $F_i^*(\mathcal{P})$. At the same time, it raises the option value of waiting. This effect increases $R_i^{e*}(\mathcal{P})$ for some parcels and delays redevelopment. As a result, both policies increase the supply of market-rate housing, but only tax exemptions systematically accelerate redevelopment across all parcels. This result confirms, in equilibrium, the parcel-level result developed in Section 3. By increasing the option value of waiting, upzoning can delay redevelopment on parcels near their redevelopment threshold even as it raises the value of eventual redevelopment.

A standalone affordability mandate ($\alpha > 0$) shifts production toward affordable units but lies below the negative 45-degree line. The mandate reduces the net return to redevelopment by affecting $\Gamma(F, z, \alpha, L)$, thereby lowering post-development rental income. This increases the redevelopment threshold and reduces the probability of redevelopment. The contraction in total housing supply arises from fewer redevelopment events rather than smaller projects. Affordable housing production increases only because it is bundled with redevelopment that still occurs on high-value parcels.

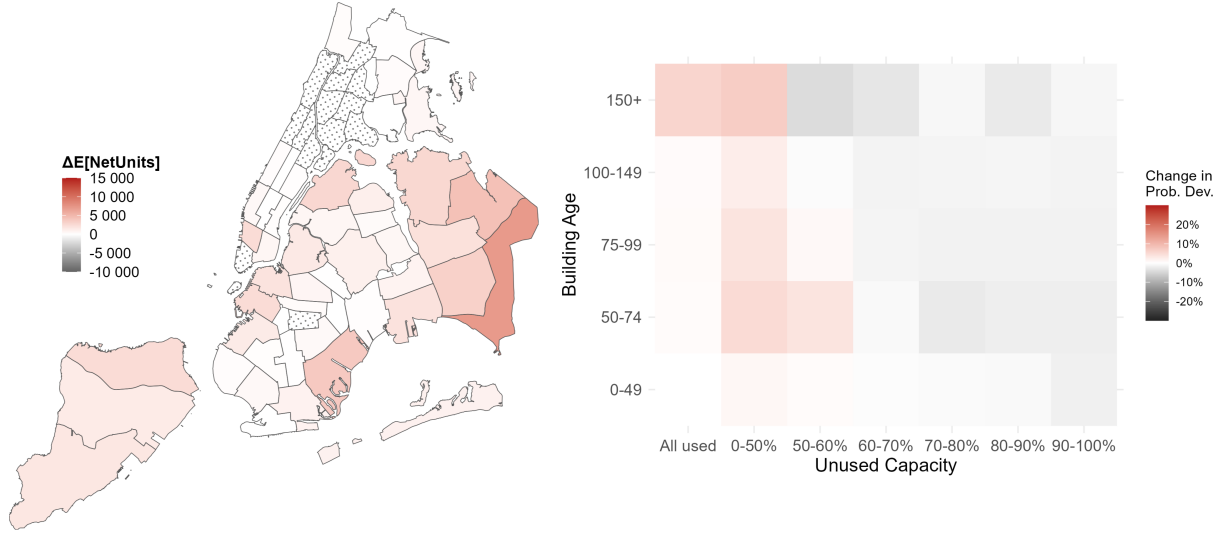
Policies that combine mandates with incentives occupy intermediate positions. Density Incentive Zoning (α, ϕ) and Fiscal Incentive Zoning (α, τ) both lie above the negative 45-degree line, indicating net expansions of the housing stock. In these regimes, incentives partially offset the increase in $R_i^{e*}(\mathcal{P})$ induced by the mandate. However, affordable housing gains arise primarily through reallocation. Increases in affordable units are accompanied by reductions in market-rate units within redeveloped projects, reflecting the constraint imposed by α . Only the Combined Incentive Policy (α, ϕ, τ) lies in the upper-right quadrant. This policy both lowers $R_i^{e*}(\mathcal{P})$ and increases $F_i^*(\mathcal{P})$ for most parcels. It activates redevelopment on a broad set of parcels while increasing density conditional on redevelopment. As a result, it expands both affordable and market-rate housing simultaneously.

6.1.2 Spatial heterogeneity is driven by parcel heterogeneity

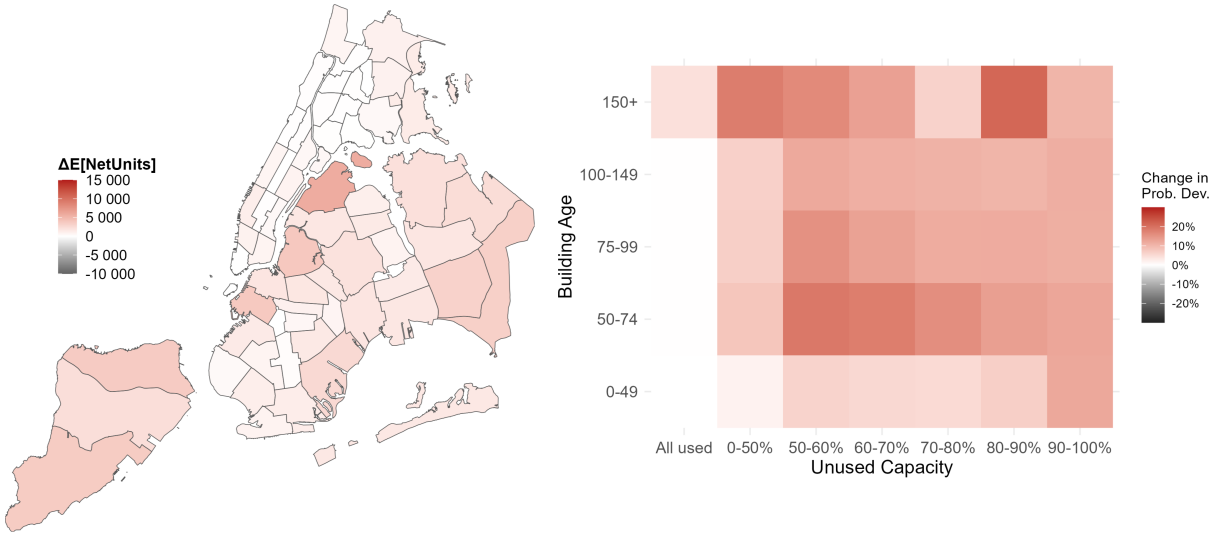
Figure 7 shows that aggregate housing production masks substantial spatial heterogeneity. This heterogeneity arises endogenously from parcel-level differences in redevelopment incentives. Even when policy is applied uniformly across the city, redevelopment responses vary because parcels differ in their redevelopment threshold, in their existing density, and in their level of depreciation.

The left-hand panels of Figure 7 map housing production changes across two benchmark policies: upzoning ($\phi > 0$) and tax exemptions ($\tau > 0$). Although both policies increase aggregate housing production, their spatial incidence differs sharply. Tax exemptions generate concentrated responses in areas with older buildings and substantial unused density, such as Western Brooklyn

Figure 7: Policy Counterfactuals — Housing Production Heterogeneity



A. Upzoning ($\phi = .50$)



B. Tax exemptions ($\tau = 20$)

Notes: The maps report the expected net change in the number of housing units under each counterfactual policy, aggregated at the community-district level. The right-hand panels report changes in redevelopment probabilities, grouping parcels by deciles of unused density ($\bar{F}_i - F_{B,i}$) and building age. Panel A reports results for an upzoning policy allowing a 50% increase in maximum density, and Panel B reports results for a 20-year exemption from property taxes on newly developed buildings.

and Queens. Upzoning reduces expected housing production in the Bronx and Harlem while increasing it mostly in Queens.

The right-hand panels of Figure 7 make this mechanism explicit by grouping parcels into deciles of unused density and building age. Tax exemptions raise redevelopment probabilities for parcels with older structures and substantial unused density. Upzoning, by contrast, reduces redevelopment probabilities for many parcels in the same deciles. This divergence reflects that

tax incentives generate additional housing production primarily on the extensive margin, while density bonuses operate on the intensive margin.

Thus, spatial heterogeneity arises because aggregate housing production results from individual redevelopment decisions. Policies do not shift all parcels uniformly. They move some parcels across their redevelopment threshold while leaving others unaffected. Tax incentives concentrate redevelopment in areas where parcels are already close to redevelopment. Density bonuses increase the value of redevelopment but often delay its timing. As a result, two policies that generate similar aggregate supply can produce very different spatial patterns of development.

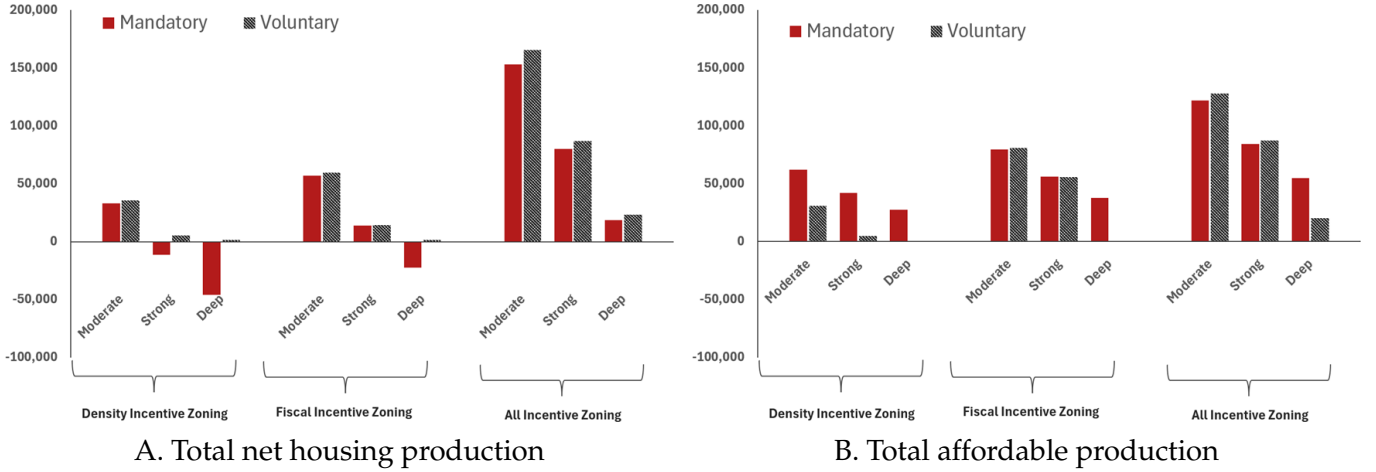
Targeted counterfactuals. To further highlight the importance of aligning policy instruments with parcel characteristics, we re-conduct our main unilateral policy counterfactuals on three targeted subsets: the 20% of parcels with the highest underlying demand (top quintile of R_i), the 20% most obsolete parcels (bottom quintile of $z_{B,i}$), and the 20% least utilized buildings (bottom quintile of $F_{B,i}/\bar{F}_i$). Appendix D.3 presents the full results and shows that instrument effectiveness depends critically on parcel characteristics. Upzoning generates the largest housing production when it targets high-value parcels, where additional floor area is most profitable. Tax exemptions are strongest when they target underbuilt parcels, where they lower the barrier to entry on the development margin.

6.1.3 Voluntary Incentive Policies and affordability levels

We next examine how voluntary adoption and the depth of affordability requirements affect housing production under the various incentive policies. We model voluntary regimes by allowing developers to adopt a policy bundle $\mathcal{P}$ only if it increases property value relative to not adopting it, that is, if $\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$. Parcels for which regulatory requirements dominate incentives remain under the baseline regulatory environment.²⁰ This adoption rule implies that policy affects outcomes not only by shifting redevelopment incentives, but also by determining which parcels are exposed to those incentives. We provide additional results on the types of parcels that adopt different policy bundles in Appendix D.2. We also vary the depth of affordability requirements through the rent discount ϵ_α , holding the affordable-share requirement fixed. Figure 8 summarizes the results. Panel A reports changes in total net housing production, and Panel B reports changes in net affordable housing production. Within each policy environment, outcomes are shown for moderate, strong, and deep affordability requirements.

²⁰The voluntary-adoption counterfactual introduces a binary participation decision that create discontinuities in the fixed-point iteration. To ensure numerical stability, we smooth adoption beliefs during the iteration using a logit approximation and recover the binary adoption rule once the algorithm converges. Appendix A.8 provides details.

Figure 8: Affordability Levels and *Voluntary* Policy Interventions



Notes: The figures report changes in expected housing production relative to the baseline calibration $\mathcal{P}_0$ over a 10-year horizon. Panel A reports the change in total housing units net of demolitions, while Panel B reports the change in affordable housing units. Bars are grouped by policy regime and by the depth of the affordability requirement. Affordability depth is parameterized by the rent discount ϵ_α , which scales the rental income of affordable units relative to market-rate units: $\epsilon_\alpha = 0.25$ (moderate), $\epsilon_\alpha = 0.50$ (strong), and $\epsilon_\alpha = 0.75$ (deep). Red bars correspond to mandatory policy regimes, in which all new construction must comply with the specified requirements. Stripped black bars correspond to voluntary policy regimes, in which developers choose whether to adopt the policy bundle or remain under the baseline regulatory environment.

Under mandatory regimes, increasing ϵ_α reduces the value of redevelopment by lowering effective rental income. This lower post-development income reduces redevelopment incentives for all parcels. As a result, both total and affordable housing production decline. In several cases, total housing production becomes negative relative to the baseline even as affordable production remains positive, indicating that the contraction is driven by a reduction in market-rate units rather than through a collapse of affordable construction.

Under voluntary adoption regimes, the same forces operate in addition to the participation condition. At moderate affordability levels, voluntary adoption yields housing production outcomes similar to mandatory regimes because most parcels that redevelop also satisfy the participation condition. As affordability requirements deepen, participation declines sharply as an increasing fraction of parcels no longer satisfies $\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$. The policy becomes ineffective because it fails to induce redevelopment on parcels near their threshold. As a result, both total and affordable housing production converge back toward the baseline.

The Combined Incentive Policy provides an exception. By increasing the value of redevelopment through both density bonus ϕ and tax exemptions τ , it offsets the decline in $\pi_i(\mathcal{P})$ induced by high ϵ_α . This keeps the participation condition satisfied for a broader set of parcels. As a result, expected housing production is still greater than the baseline even under deep affordability requirements ($\epsilon_\alpha = 0.75$). Relative to the other regimes, stacking incentives preserves redevelop-

ment across a broader set of parcels, allowing affordable production to remain positive despite stringent requirements.

6.2 Expected Rent

6.2.1 Average expected housing rent and its decomposition

Figure 9 reports the change in average citywide expected rent relative to the baseline $\Delta\text{Rent}(h; \mathcal{P}) = \text{Rent}(h; \mathcal{P})/\text{Rent}(h; \mathcal{P}_0) - 1$. Policies generate different rent responses even when they have similar effects on aggregate housing production. The difference arises because policy reshapes the housing stock through the redevelopment margin.

To formalize this mechanism, recall that expected rental income for parcel i at horizon h is given by:

$$\text{RentInc}_i(h, \mathcal{P}) = \mathbb{E}_{\mathbb{P}}\{R_{ih}\} \left[\left(1 - \hat{\mathbf{P}}_i^*(h; \mathcal{P})\right) \Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}, L_i) + \hat{\mathbf{P}}_i^*(h; \mathcal{P}) \Gamma(F_i^*, 1, \alpha, L_i) \right]$$

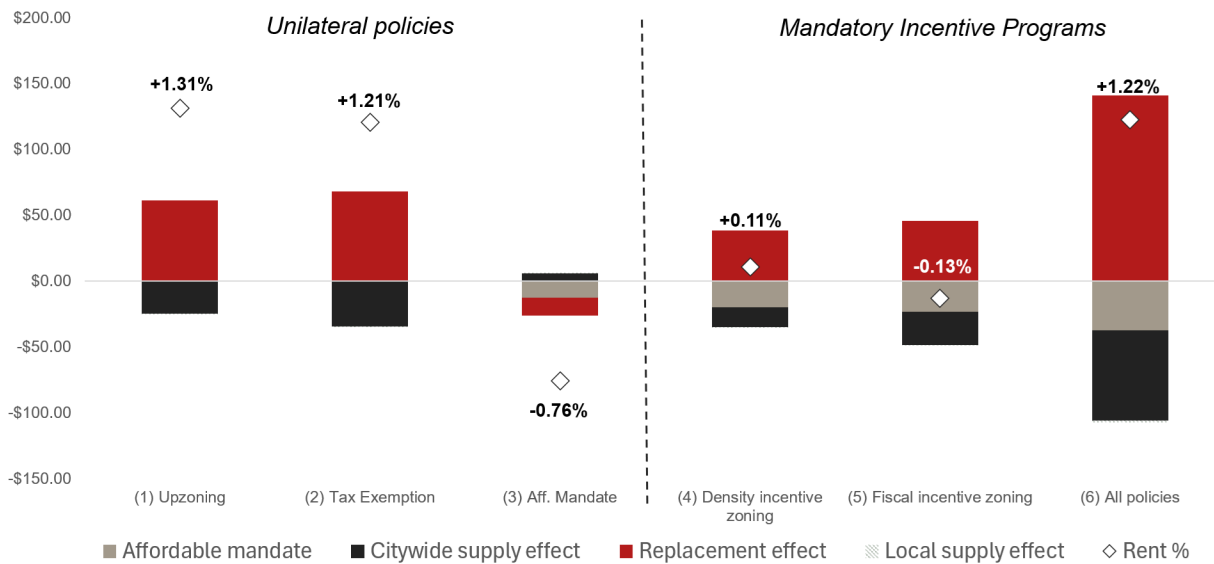
where $\hat{\mathbf{P}}_i^*(h; \mathcal{P})$ is the rent-weighted completion probability and $F_i^*(\mathcal{P})$ is the optimal redevelopment density. This expression shows that policy affects rents through three channels. First, changes in supply at the city-level and around parcel i impact expected potential rent $\mathbb{E}_{\mathbb{P}}\{R_{ih}\}$. Second, policy changes the probability that a parcel redevelops. Third, conditional on redevelopment, it replaces depreciated housing with new construction that generates higher rental income. We decompose changes in average expected rent into four components:

$$\Delta\text{Rent}(h; \mathcal{P}) = \Delta\text{Rent}^{\text{aff.}} + \Delta\text{Rent}^{\text{City.supply}} + \Delta\text{Rent}^{\text{Local.supply}} + \Delta\text{Rent}^{\text{Replacement}}.$$

The affordable-share effect reflects the inclusion of below-market units in new construction, which lowers average rents. The citywide supply effect captures downward pressure on rents from higher citywide housing supply governed by the inverse-demand elasticity $\zeta_C < 0$. The local supply effect captures local rent changes from nearby development and is governed by ζ_L . The replacement effect captures the rent increase that occurs when redevelopment replaces depreciated units with new units.

The expected rents we report are not adjusted for housing quality. In a hedonic framework, the price of housing, controlling for quality, typically falls as supply expands, which is embedded in our model through the endogenous changes of R_{ih} . Our object of interest is, however, different. It is the rent that prevails across the heterogeneous stock of units actually occupied, the rent a tenant pays, a developer underwrites, and a policymaker measures. This average can rise even as the quality-adjusted price falls, because redevelopment shifts the composition of the stock toward

Figure 9: Counterfactual Results — Expected Rents



Notes: The points report the average expected rent change under each counterfactual policy relative to the baseline at a 10-year horizon. The stacked bars decompose this change into four components. The Affordable Mandate component isolates the direct effect of the affordable share α . The Citywide Supply Effect captures the change in expected rents attributable to additional citywide housing supply. The Local Supply Effect is computed analogously and reflects the impact of local supply on neighborhood rents. The Replacement effect is defined as the residual component and captures rent changes associated with replacing the old with the new structure, including changes in redevelopment probabilities. Policies (1)–(3) correspond to unilateral interventions—upzoning, tax exemption, and affordability mandate—while policies (4)–(6) combine these instruments.

taller, non-depreciated units, constructed on higher-value parcels.

Figure 9 shows that the replacement effect is large enough to dominate the supply effect in many counterfactuals. For example, under upzoning, the replacement effect raises rents by approximately \$60 per unit per month, while the supply effect lowers rents by approximately \$25. The net effect is positive because redevelopment shifts the composition of the housing stock toward higher-rent units. Thus, under policies that strongly increase redevelopment—such as upzoning (Policy [1]), tax exemptions (Policy [2]), and the combined policy (Policy [6])—average expected rents increase by 1.31%, 1.21%, and 1.22%, respectively.²¹ In these cases, rents rise because redevelopment replaces a larger share of the housing stock with new, higher-rent units.

Introducing an affordability mandate dampens this increase by reallocating a share of new construction to below-market units, which, together with the supply effect, offsets the replacement effect. Under Density Incentive Zoning and Fiscal Incentive Zoning, average rents increase by 0.11% and decrease by 0.13%, respectively. These policies dampen the replacement effect by

²¹Figure D2 provides spatial context by mapping changes in average expected rent across community districts. Neighborhoods experiencing larger increases in redevelopment activity—such as parts of Brooklyn, Queens, and Staten Island—exhibit larger rent responses, while districts with limited redevelopment display more muted changes.

introducing below-market units, and also by weakening redevelopment incentives.

These results concern the composition and rent distribution of the housing stock rather than the housing outcomes of individual tenants. Whether a tenant of an old building subsequently faces a higher or lower rent depends on relocation, household sorting, and housing-search frictions that lie outside the supply-side framework studied here. Accordingly, the increase in average rent generated by the replacement effect should not be interpreted as a household-level welfare effect.

Unit normalization does not explain this result. Our model operates in per-square-foot terms, but we present results in housing units by dividing expected floor area by a fixed parcel-specific unit size. One might be concerned that the replacement effect is an artifact of this unit size normalization. We respond to this concern in two ways. First, we replicate the rent decomposition in square foot terms, which yields the same quantitative pattern, with the replacement effect dominating the supply effect across all policies (Figure D6). Second, we introduce a unit-size adjustment parameter β , defined as the ratio of the new unit size to the existing unit size, and reconduct our counterfactual analysis for different values of β . The results detailed in Appendix D.4 show that average expected rents always increase with additional supply induced by the policies. This holds even when new units are 20% smaller than the current stock.

6.2.2 Rent distribution

We next study how policy affects the rent distribution, not just its average. Table 4 reports the share of expected housing supply in each rent quintile under policy $\mathcal{P}$, where quintile cutoffs are defined using the baseline distribution of rent per unit under $\mathcal{P}_0$. This normalization lets us interpret distributional changes as shifts in the composition of the housing stock:

$$\Delta \text{shr}_q(\mathcal{P}) = \text{shr}_q(\mathcal{P}) - \text{shr}_q(\mathcal{P}_0),$$

where $\text{shr}_q(\mathcal{P})$ denotes the share of housing units in rent quintile q .

Policies without affordability mandates push the housing stock toward both tails of the distribution. Under upzoning (Policy [1]) and tax exemptions (Policy [2]), the share of units in the top quintile increases. This reflects the expansion of new market-rate construction, which enters the housing stock at higher rent levels due to the absence of depreciation. At the same time, the share of units in the bottom quintile also rises. That increase reflects the supply effect, which lowers rents for existing units everywhere.

Policies that include affordability mandates (Policies [4] - [6]) reshape the distribution in a

Table 4: Counterfactual Results – Change in Rent Distribution

	Q1	Q2	Q3	Q4	Q5
Range	\$0–\$1,857	\$1,857–\$2,268	\$2,268–\$2,697	\$2,697–\$3,568	\$3,568+
Baseline	20%	20%	20%	20%	20%
	Unilateral Policy				
(1) <i>Upzoning</i> ($\phi = 50\%$)	+0.51	+0.26	–0.68	–0.89	+0.81
(2) <i>Tax exemption</i> ($\tau = 20$)	+0.81	+0.12	–1.28	–1.17	+1.52
(3) <i>Affordability mandate</i> ($\alpha = 30\%$)	–0.10	0.00	+0.33	+0.35	–0.58
	Mandatory Incentive Programs				
(4) <i>Density Incentive Zoning</i>	+0.34	+0.22	–0.20	–0.38	+0.03
(5) <i>Fiscal Incentive Zoning</i>	+0.62	+0.18	–0.68	–0.40	+0.27
(6) <i>All Policies</i>	+1.45	+0.47	–1.67	–1.91	+1.65

Notes: This table reports the 10-year horizon change in the share of expected housing supply falling within each rent quintile relative to the baseline calibration P_0 , expressed in percentage points. Quintile thresholds are defined from the baseline rent distribution. A positive value indicates a relatively larger share of units in that rent segment compared to baseline. By construction, each row sums to zero.

similar way, but add a further distributional effect. Because of the requirement to have affordable units in new construction, the top quintile increases to a lesser extent. Yet the bottom quintile does not increase substantially despite the affordable mandate addition. To further examine this pattern, Figure D1 plots distributional changes in the expected number of housing units across \$500 rent bins where positive (negative) values indicate increases (decreases) in the mass of units in each rent interval relative to the baseline. The figure decomposes these changes into market-rate units and units subject to affordability mandates.

One key result is that newly created affordable units do not concentrate in the lower tail of the rent distribution. Under both Density Incentive Zoning and Fiscal Incentive Zoning, most additional affordable units are located above the mean rent. This pattern follows directly from the redevelopment decision rule. Redevelopment occurs only on parcels where it remains profitable, typically in high-demand locations. In these locations, even discounted rents remain well above the lower tail of the distribution. In our model, where redevelopment requires private profitability, the existing housing stock continues to supply most units in the lower end of the rent distribution. Newly built affordable units occupy intermediate portions of the distribution because they are tied to redevelopment on profitable parcels. As a result, increases in affordable housing supply do not necessarily improve affordability at the lower end of the rent distribution.

6.2.3 Robustness to demand elasticity assumptions

In the model, expected rents depend on both the housing stock and the demand responses that map supply into rent changes. Developers internalize these effects when making redevelopment decisions. Two parameters govern these responses: the citywide inverse demand elasticity $\zeta_C < 0$ and the local supply elasticity ζ_L .

Can the supply effect dominate? We first examine how alternative values of ζ_C affect rent outcomes. Figure 10 reports the change in average expected rent $\Delta\text{Rent}(h; \mathcal{P})$ under the Upzoning and Tax Exemption counterfactuals as functions of ζ_C . Under the baseline calibration, both policies raise average expected rents by roughly 1.25%, even though each expands housing supply. This reflects that the housing replacement effect dominates the aggregate supply effect. Absent any supply effect, the rent increase would approach 3%, since developers would face no offsetting disincentive to supply more expensive housing.

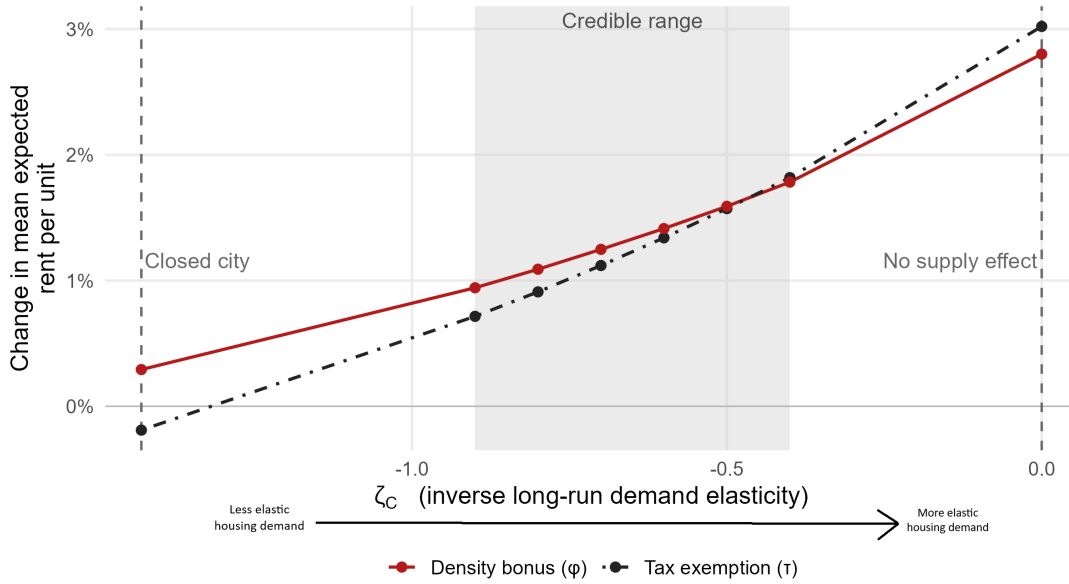
As the long-run housing elasticity falls ($|\zeta_C|$ rises), the downward pressure from aggregate supply strengthens. Yet across the credible range, $\zeta_C \in [-0.90, -0.40]$, the average rent response to both policies stays positive, between 1% and 2%. Only substantially larger values of $|\zeta_C|$ produce a supply effects that offset the replacement effect and push average rents down. This occurs, however, only as $|\zeta_C|$ approaches 1.43, a parametrization consistent only with a closed city framework.

At these values, redevelopment incentives also weaken. Lower expected future rents raise the redevelopment threshold and reduce redevelopment probabilities and housing production. The parameter values that could generate large rent declines therefore also generate weak housing production responses. This constraint comes from endogenous rent feedback and developers' rationality: policies that put strong downward pressure on rents also reduce the expected return to redevelopment.

Local spillovers and spatial clustering. We next turn to local spillover effects from new construction and show how the value of ζ_L alters the clustering of construction and, in turn, the clustering of rents. We consider $\zeta_L \in [-0.10, 0.10]$, a range spanning the region where local competition from nearby construction dominates at one end and positive endogenous amenities dominate at the other. To quantify spatial clustering, we compute Moran's I using inverse-distance weights within a 500-foot radius. Under the baseline, Moran's I equals 0.78 for rents and 0.18 for construction probabilities. Rents are therefore strongly clustered across housing units, while construction is much less so.

Panel A of Figure D9 reports how the spatial correlation of construction changes under up-

Figure 10: Citywide Demand Elasticity and Average Expected Rents

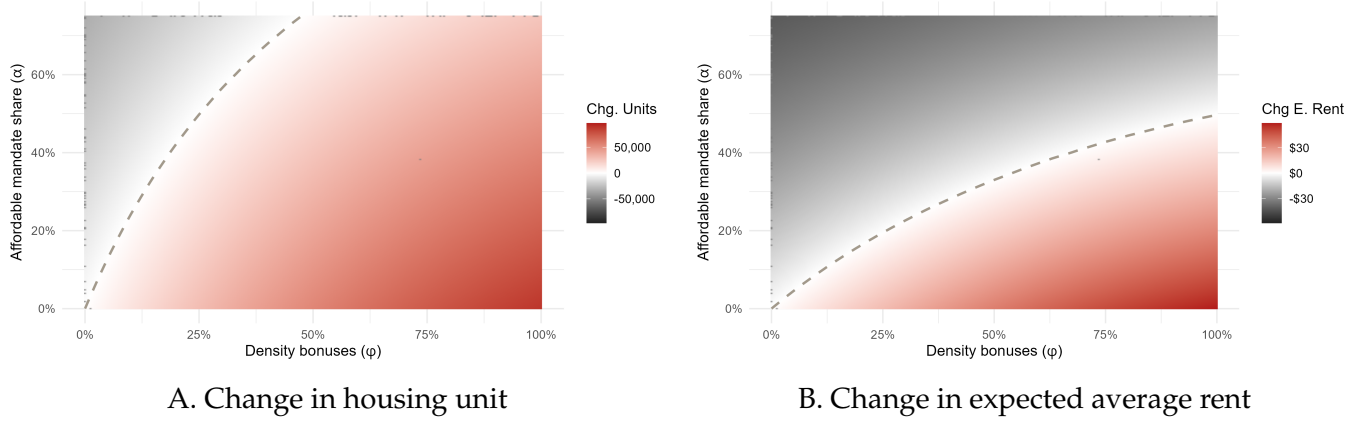


Notes: This figure reports the change in the mean expected rent per unit at the ten-year horizon under two policies, upzoning ($\phi = 50\%$) and a post-redevelopment tax exemption ($\tau = 20$), as a function of the inverse long-run price elasticity of housing demand ζ_C . The shaded band, $\zeta_C \in [-0.90, -0.40]$, spans the range implied by the empirical estimates. The right dashed line at $\zeta_C = 0$ is the no-supply-effect economy, in which the citywide and local channels are both inactive ($\zeta_C = \zeta_L = 0$). The left dashed line at $\zeta_C = -1.43$ is the closed-city case.

zoning and tax exemptions. Both policies increase construction clustering, and the effect strengthens as local spillovers increase. At $\zeta_L = 0.10$, which corresponds to positive endogenous amenities, the correlation rises by more than 10 percentage points under both policies, an increase of roughly 60 to 70 percent relative to the baseline of 0.18. Stronger local spillovers make redevelopment near recently developed parcels more attractive, so construction concentrates in space.

The spatial correlation of rents follows the same pattern with respect to ζ_L . The two policies, however, move rent clustering in opposite directions relative to the no-policy baseline, as shown in Panel B of Figure D9. Upzoning raises the spatial correlation of rents, while tax exemptions lower it. This divergence reflects the different margins on which the two instruments operate. Upzoning concentrates redevelopment in high-value areas, where new construction clusters near existing high-rent units. Tax exemptions instead target older, lower-density parcels, which are more common in lower-rent neighborhoods, so new supply appears away from the existing high-rent clusters. At $\zeta_L = 0.10$, upzoning raises the spatial correlation of rents by about two percentage points, roughly a 3 percent increase relative to the baseline level of 0.78.

Figure 11: Continuous Policy Outcomes - Density Incentive Zoning



Notes: These heatmaps report changes in model outcomes under a Density Incentive Zoning program across continuous values of the density bonus ϕ (horizontal axis) and the affordable housing requirement α (vertical axis). All changes are measured relative to the status-quo policy $\mathcal{P}_0$ at the 10-year horizon. Panel A shows the net change in the expected number of housing units $\text{UnitNet}(\mathcal{P})$. Panel B reports the change in average expected rents per housing unit. Dotted contour lines indicate combinations of (ϕ, α) that leave the corresponding outcome equal to its baseline level under the status-quo policy $\mathcal{P}_0$.

6.3 Rent and Housing Production Trade-offs

The preceding analysis evaluates a discrete set of policy bundles. We now describe the trade-offs that arise when policy instruments vary continuously. Let policy be defined by $\mathcal{P} = [\alpha, \phi, \tau]$. The model maps each policy into a pair of outcomes for housing production and rents: $\{\Delta\text{UnitNet}(\mathcal{P}); \Delta\text{Rent}(\mathcal{P})\}$. Figure 11 plots these outcomes for Density Incentive Zoning as a function of the density bonus ϕ (horizontal axis) and the affordability requirement α (vertical axis). Figure D10 reports analogous results for policies that combine tax exemptions τ with affordability mandates.

Figure 11 shows that there is a trade-off between housing production and rent outcomes. Policies that strongly increase redevelopment also increase the replacement of existing housing stock, which raises average rents through the replacement effect. Policies that limit redevelopment weaken the replacement effect but also reduce total housing production. This trade-off implies that policy cannot simultaneously maximize housing production and minimize average rents. Increasing supply through redevelopment raises the share of new, higher-rent units in the housing stock. Reducing rents instead requires limiting redevelopment or imposing affordability requirements that weaken the replacement effect. These objectives conflict because they operate through the same underlying decision: whether and when parcels redevelop. Figure 11 also illustrates that policy effects are not linear. Increasing ϕ and α proportionally does not produce proportional changes in outcomes. Nonlinearity arises because redevelopment responds to threshold crossings rather than marginal adjustments. Once a parcel crosses the redevelopment threshold,

further increases in incentives have limited effects on its decision, while small changes near the threshold can have large effects.

Consequently, policy design involves a trade-off between activating redevelopment and controlling the composition of the housing stock. Policies that stimulate redevelopment increase supply but also speed up the replacement of low-rent units with higher-rent new construction. Policies that constrain redevelopment preserve lower-rent units but limit supply expansion. This trade-off is a direct consequence of the redevelopment margin: the same redevelopment decisions determine both housing production and rent outcomes.

7 Policy Incidence and the Cost of Mandated Affordable Units

Having established how the redevelopment margin affects housing production, and rents, we now examine who ultimately bears the costs of the housing policies. Rather than conducting a full welfare analysis, we focus on the economic incidence of alternative housing policies by quantifying how policy-induced changes in redevelopment decisions translate into changes in private parcel values and the city’s fiscal position. Because different policy instruments alter redevelopment incentives in different ways, they generate distinct distributions of costs across landowners and the public sector.

7.1 Public and Private Costs of Policy Interventions

We evaluate how each policy shifts the distribution of costs between the public sector and private landowners. The empirical objects are the present value of property tax revenues and parcel values, which together capture how redevelopment responses are reflected in fiscal outcomes and in land value capitalization.

$$\Delta PV^{\text{Tax}}(\mathcal{P}) = \frac{\sum_i PV_i^{\text{Tax}}(\mathcal{P})}{\sum_i PV_i^{\text{Tax}}(\mathcal{P}_0)} - 1 \quad ; \quad \Delta PV(\mathcal{P}) = \frac{\sum_i \pi_i(\mathcal{P})}{\sum_i \pi_i(\mathcal{P}_0)} - 1. \quad (16)$$

Table 5 reports both dimensions for each counterfactual policy. All policies that increase supply reduce tax revenues and also aggregate residential asset values. This decrease reflects the downward pressure of additional supply on existing properties, consistent with land-value dilution concerns following land-use deregulation (Duranton and Puga, 2023; Ahlfeldt et al., 2026). Tax exemptions (Policy [2]) generate the largest fiscal cost among unilateral interventions (−4.55%), reflecting foregone tax revenues during the exemption window. The affordability mandate (Policy [3]) has limited fiscal effects (−0.07%) but the reduced redevelopment activity limits supply expansion. The Combined Incentive Zoning (Policy [6]) yields the largest decline in both

Table 5: Counterfactual Results – Costs of Interventions

		Private Gain/Loss				Public
		ΔPV	Shr. PV-losers	Avg. Gain	Avg. Loss	ΔPV^{Tax}
		(%)	(%)	(000's \$ per parcel)		(%)
Unilateral Policy						
(1)	Upzoning ($\phi = 50\%$)	-1.74	98.12	62.86	-37.27	-1.04
(2)	Tax exemption ($\tau = 20$)	-1.84	92.78	158.16	-52.63	-4.55
(3)	Affordability mandate ($\alpha = 30\%$)	0.34	5.46	9.41	-36.76	-0.07
Mandatory Incentive Programs						
(4)	Density Incentive Zoning	-1.17	98.78	40.11	-24.57	-1.04
(5)	Fiscal Incentive Zoning	-1.41	93.74	117.72	-38.46	-3.98
(6)	Combined Incentive Zoning	-4.23	93.89	128.46	-100.09	-6.69

Notes: This table reports the private and public cost outcomes under each counterfactual policy relative to the baseline calibration P_0 with a horizon of 10 years. Each row corresponds to a counterfactual regulatory environment $\mathcal{P}$. ΔPV reports the percentage change in the aggregate present value of residential parcels. The share of PV-losers is the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(P_0)$). Average gain and average loss report the average change in present value (in thousands of dollars per parcel) among PV-winners and PV-losers, respectively. ΔPV^{Tax} reports the percentage change in the present value of property tax revenues collected by the city.

parcel values (-4.23%) and tax revenues (-6.69%), alongside the widest dispersion of outcomes, with large gains among a small set of redeveloping parcels (6.1% of parcels with positive land value increase) and substantial losses among the majority. This result reflects the redistribution induced by changes in redevelopment incentives.

Heterogeneous Impact on Landowners. Aggregate statistics in Table 5 hide substantial heterogeneity in how policies affect individual landowners. Across policies, except the pure affordable mandate, more than 90% of parcels experience declines in present value, while a small minority capture gains. Two forces account for this pattern. The supply effect operates broadly, reducing expected rental income everywhere and generating diffuse losses. In contrast, the redevelopment option generates concentrated gains for parcels near the redevelopment threshold, where the policy shifts the optimal timing decision.

Figure D11 plots the distribution of the change in present value across parcels, illustrating this asymmetry for upzoning (Panel A) and the tax exemption (Panel B). Both distributions are highly skewed. Losses cluster near zero, while gains form a long right tail, reflecting large increases in option value for parcels close to their redevelopment thresholds. Under upzoning, 98.1% of parcels lose present value with an average loss of \$341 thousand per acre among losers and an average gain of \$728.07 thousand per acre among winners. The tax exemption produces a larger set of winners, 7.2% , whose gains are also more pronounced, averaging \$1,470 thousand

per acre. Gains are therefore concentrated among a limited set of parcels against which a mass of parcels lose. This uneven distribution aligns with political economy constraints emphasized in [Fischel \(2002\)](#) and [Diamond and McQuade \(2019\)](#).

7.2 Cost of Mandated Affordable Units

The previous incidence results might not be directly comparable across policies, as interventions differ in their effects on total housing supply. To isolate the cost of mandated affordable units, we focus on policy combinations that hold total expected net housing supply fixed, $\Delta\text{NetUnit}(h; \mathcal{P}) = 0$. Under this restriction, policies reallocate units from market-rate to affordable without changing aggregate supply. This isolates the incidence of one additional affordable unit from broader effects. Abstracting from supply growth allows us to set aside changes in agglomeration externalities from densification ([Duranton and Puga, 2020](#)).

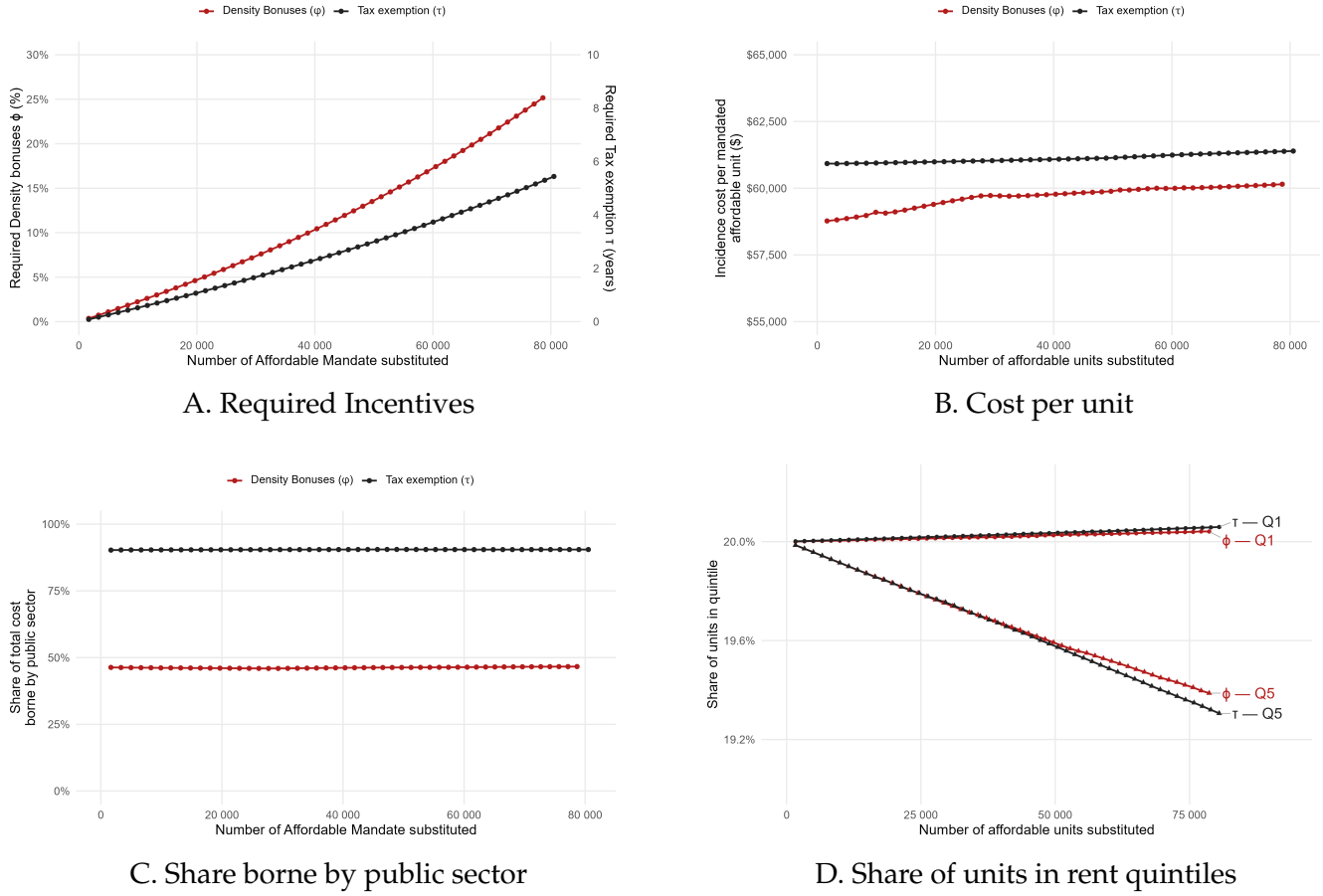
We construct the cost measure as follows. For each affordability mandate $\alpha \in \{0.01, \dots, 0.50\}$, we search over the incentive ϕ (holding $\tau = 0$) or τ (holding $\phi = 0$) that restores total housing supply to its baseline level. The existence of such a compensating incentive follows from the monotonicity of redevelopment incentives established in [Appendix A.9](#). For a fixed mandate α , increasing either ϕ or τ weakly raises parcel values and expands the set of parcels that redevelop. As a result, aggregate housing supply is weakly increasing in incentives and continuous in aggregate. [Panel A of Figure 12](#) shows the level of incentives required to maintain supply as the affordability mandate increases. For instance, replacing roughly 40,000 market-rate units with affordable ones (equivalent to an α of 25%) requires a density bonus of 10% or a tax exemption of 2.40 years. Both required density bonuses and tax exemptions increase with the mandate at an increasing rate. The convexity of both schedules indicates that maintaining supply neutrality requires progressively larger incentives as affordability requirements increase.

With each policy bundle, we compute $\Delta\text{Aff.Net}(\mathcal{P})$, the change in expected affordable units, together with the private cost $\Delta PV(\mathcal{P})$ and the fiscal cost $\Delta PV^{\text{Tax}}(\mathcal{P})$. The cost per mandated affordable unit is then summarized as

$$\omega(\mathcal{P} | \Delta\text{NetUnit}(\mathcal{P}) = 0) = \frac{\Delta PV(\mathcal{P}) + \Delta PV^{\text{Tax}}(\mathcal{P})}{\Delta\text{Aff.Net}(\mathcal{P})}, \quad \Delta\text{Aff.Net}(\mathcal{P}) > 0. \quad (17)$$

The wedge ω measures the total cost of generating one affordable unit while holding aggregate supply fixed. It therefore isolates the cost of the mandate from broader rent or density effects, and provides a metric for comparing policies that differ in design (ϕ versus τ) but yield similar housing production. [Panel B](#) reports the cost per mandated affordable unit, $\omega(\mathcal{P})$. The per-unit cost remains relatively stable, yet slightly increases with the number of units obtained, across

Figure 12: Cost of Mandated Affordable Housing



Notes: These figures show outcomes against the number of affordable units substituted for market-rate units, holding total net housing supply fixed ($\Delta \text{NetUnit}(h; \mathcal{P}) = 0$). Panel A shows the level of the incentives required to achieve supply neutrality under various affordability mandates α . Panel B reports the welfare cost per mandated affordable unit, $\omega(\mathcal{P})$. Panel C shows the share of this cost borne by the public sector. Panel D reports the share of units in the bottom and top quintiles of the baseline rent distribution.

mandate levels and policy instruments. It averages \$59,900 under density-based policies and \$61,200 under fiscal policies.

We further decompose ω into private and public components. The private component captures the reduction in parcel values borne by landowners, and the public component captures the fiscal cost of the incentive:

$$\omega^{\text{private}}(\mathcal{P}) = \frac{\Delta PV(\mathcal{P})}{\Delta \text{Aff.Net}(\mathcal{P})}, \quad \omega^{\text{public}}(\mathcal{P}) = \frac{\Delta PV^{\text{Tax}}(\mathcal{P})}{\Delta \text{Aff.Net}(\mathcal{P})}.$$

Panel C shows how this cost is distributed between the public and private sectors. Under density-based policies, the burden is roughly evenly split. Landowners lose about \$30,000 per unit

through negative capitalization, while the city experiences a comparable reduction in the present value of tax revenues. In contrast, under fiscal policies, most of the cost is borne by the public sector, where up to 90% of the cost arises from reduced tax revenues. This asymmetry reflects differences in how policy instruments allocate costs across agents. Fiscal instruments benefit landowners but are fiscally costly, while regulatory instruments preserve public revenues at the expense of private asset values.

Finally, Panel D shows the share of housing units in the bottom and top quintiles of the baseline rent distribution. Despite the increase in mandated affordable units, the share of units in the bottom quintile does not increase substantially. At most, this share rises to 20.06%, which highlights that increases in affordable housing supply through this type of mandate do not translate into large changes at the lower end of the rent distribution. In contrast, the share of units in the highest quintile decreases systematically with the strength of the affordability mandate. At $\alpha = 50\%$, nearly 24,000 units that would have fallen in the top quintile of the baseline were “transferred” to lower segments. This result reflects that new affordable units are obtained through redevelopment on higher-value parcels, and therefore do not substantially increase the mass of units in the lowest segment of the rent distribution but deter the development of higher-end units.

The incidence results of this section reinforce the paper’s central message. Housing policies operate through the redevelopment margin, but they also reveal that policy instruments differ systematically. Density bonuses expand supply but alter the composition of the housing stock, fiscal incentives accelerate redevelopment at a fiscal cost, and affordability mandates deliver targeted affordable housing while generating substantial private and fiscal costs. Choosing among these policies is therefore a choice about which costs to bear, not simply about how much housing to produce.

8 When Does the Replacement Effect Dominate?

The redevelopment margin we document in this study is a fundamental aspect of housing production in mature cities, because redevelopment always consists of replacing a depreciated low-rise structure with one taller that enters without depreciation. Its effect on rents, or the housing replacement effect, relative to the supply effect, however, varies with the built environment, demand conditions, and regulatory constraints. We discuss these boundary conditions here.

First, the housing replacement effect—that is, the upward pressure on average rents from replacing older, lower-rent housing with new construction—becomes weaker when development occurs on underutilized or vacant parcels, where there is little existing stock to replace. This distinction maps onto *mature* versus *expanding* housing markets. Table D2 exemplifies this dis-

tion, showing that targeting policy at the bottom quintile of unused capacity is the policy in which rents increase the least.

Second, it is also weaker when demand is less elastic. Yet under reasonable values of ζ_C , the supply effect never outweighs the replacement effect, a result discussed in Section 6.2.3. The same conclusion holds when the density constraint is removed entirely. The last row of Table 3 shows that construction rises by a factor of 1.8 once $\bar{F}_i = \infty$, yet average rents still rise. Developers redevelop when it is privately optimal, not simply because they are permitted to build taller, and the redevelopment threshold continues to govern timing even absent a binding density cap. Because redevelopment and rent growth are jointly determined, no parameter region lets developers build without regard to the rents their construction will support. Consequently, the forces that would deliver a dominant supply effect are the same ones that keep developers from overbuilding when demand is inelastic or expected future supply is high.

Third, the replacement incentives are not uniform. The effect of a policy on a given parcel depends critically on that parcel's built structure. We document this feature, for instance, in Section 6.1.2 in the context of housing production heterogeneity, and in Section 7.1 in relation to policy incidence. Parcels far from R_i^{e*} do not directly respond to a policy change, but they are affected by the redevelopment of others. Parcels near the threshold, in contrast, can be pushed into redevelopment, generating the concentrated gains and diffuse losses shown there.

Finally, we want to reiterate that the effect as measured per housing unit reflects a reporting convention rather than the underlying mechanism. Figure D6 shows the replacement effect also dominates when changing the reporting convention to square foot. Figure D8 shows expected rents rise under both upzoning and tax exemptions regardless of our square-foot-to-units normalization, including units that would be 20 percent smaller than the existing stock.

9 Conclusion

In mature urban markets, housing supply is better understood through the replacement of the existing housing stock rather than through the evolution of an aggregate housing supply. We develop a forward-looking spatial model of housing supply in which this process emerges from parcel-level redevelopment decisions: the redevelopment margin. By embedding these decisions in a spatial equilibrium with endogenous rent dynamics, the framework explains how housing policies affect redevelopment timing, construction intensity, housing composition, rent dynamics, and policy incidence.

Viewing housing supply through the lens of the redevelopment margin changes how housing

policies should be evaluated. Policies that appear similar in terms of aggregate housing production can generate markedly different outcomes because they alter redevelopment incentives in different ways. Our results show that policies that increase construction need not reduce average rents once replacement of lower-rent housing is taken into account, that affordability mandates expand affordable housing primarily by crowding out market-rate construction rather than through net housing growth, and that the costs of providing affordable housing depend critically on the policy instrument through which those costs are financed and distributed.

More broadly, the model we develop in this study offers a unified structural tool for studying urban real estate markets. When a policy or market shock changes the incentive to redevelop, its effects propagate through the redevelopment margin, from the decision on each parcel to citywide supply and rents. Evaluating them therefore requires understanding not only how much housing is produced, but also when redevelopment occurs, which parcels redevelop, how intensively parcels are rebuilt, what replaces the existing housing stock, and who ultimately bears the costs of those changes. These are the elements our model is built to quantify.

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Online Appendix

A Structural Model Details and Proofs

In this Appendix, we develop the analytical foundations of the model. We first formalize the real-options redevelopment problem introduced in Section 2. We establish the pricing framework, derive parcel values, characterize the redevelopment threshold, determine the resulting redevelopment probability, and solve for optimal redevelopment density. We then show how these parcel-level redevelopment decisions aggregate into a spatial equilibrium, closing the model through a fixed point that links developers' expectations, redevelopment behavior, and rent growth. We define this numerical fixed point and then derive the implied rental incomes that result from this equilibrium. Together, these results characterize when redevelopment occurs, the probability of redevelopment, how intensively parcels are rebuilt, and how individual redevelopment decisions aggregate into a citywide equilibrium.

A.1 Foundations for Risk-Neutral Pricing of Rents.

To obtain analytical valuations of development options, we represent rent R_t as a geometric Brownian motion under the risk-neutral measure $\mathbb{Q}$. Rents are not traded claims, but their risk can be priced consistently through a common kernel linked to traded real-estate instruments. Residential property values are therefore the present value of net rents:

$$V_t = \mathbf{E}_{\mathbb{Q}} \left[\int_0^{\infty} e^{-ru} R_{t+u} du \right] = \frac{R_t}{r - \delta}, \quad \delta < r.$$

This parallels rent-capitalization approaches used to relate the rental and asset markets through a capitalization rate – or rent-to-price ratio (DiPasquale and Wheaton, 1992; Büchler et al., 2021). Thus, with V_t inheriting rent dynamics and applying Itô's Lemma gives

$$dV_t = \delta V_t dt + \sigma V_t dW_t.$$

Four assumptions underlie this representation: (1) asset value derives solely from net rental income; (2) rents equal net operating income, so there is no wedge with value; (3) rents are discounted at a constant r ; and (4) rent risk is spanned by traded real-estate instruments such as REITs or CMBS. The measure $\mathbb{Q}$ is thus viewed as a pricing device calibrated to observed valuations. Although real estate markets are incomplete, *good-deal bounds* narrow admissible measures, permitting valuation under $\mathbb{Q}$ without perfect replication (Cochrane and Saa-Requejo, 2000). It will follow from the empirical work that we calibrate δ from observed capitalization rates c_{obs} via $\delta = r - c_{\text{obs}}$, treating δ as a reduced-form risk premium.

A.2 Proof of Proposition 1: Parcel Value

Let $T_{R^e} \equiv \inf\{t \geq 0 : R_t = R^e\}$ with $R^e > R$, and $H_{R^e} \equiv T_{R^e} + d$, with $d > 0$.

Under $\mathbb{Q}$, $dR_t = \delta R_t dt + \sigma R_t dW_t^{\mathbb{Q}}$ with $r > \delta$, so

$$R_t = R \exp\left(\left(\delta - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{Q}}\right) = R e^{\sigma Z_t}, \quad Z_t \equiv bt + W_t^{\mathbb{Q}}, \quad b \equiv \frac{\delta - \frac{1}{2}\sigma^2}{\sigma}.$$

Define

$$a \equiv \frac{1}{\sigma} \ln\left(\frac{R^e}{R}\right) > 0, \quad T_{R^e} = \inf\{t \geq 0 : Z_t = a\} \equiv T_a.$$

Parcel value is

$$\begin{aligned} \pi(R, F_B, z_B, \alpha_B, F; \mathcal{P}, \Theta) = & \underbrace{\mathbf{E}_{\mathbb{Q}}\left\{\int_0^{T_{R^e}} e^{-ru}(1-\psi)\Gamma(F_B, z_B, \alpha_B)R_u du\right\}}_{\text{Value of existing property}} \\ & - \underbrace{\mathbf{E}_{\mathbb{Q}}\left\{e^{-rT_{R^e}}\gamma(F_B)\right\}}_{\text{Demolition costs}} - \underbrace{\mathbf{E}_{\mathbb{Q}}\left\{\int_{T_{R^e}}^{H_{R^e}} e^{-ru}\kappa(F, \alpha) du\right\}}_{\text{Construction costs}} \\ & + \underbrace{\mathbf{E}_{\mathbb{Q}}\left\{\int_{H_{R^e}}^{\infty} e^{-ru}\Gamma(F, 1, \alpha)R_u du\right\}}_{\text{Value of new property gross of tax}} - \underbrace{\mathbf{E}_{\mathbb{Q}}\left\{\int_{H_{R^e}+\tau}^{\infty} e^{-ru}\psi\Gamma(F, 1, \alpha)R_u du\right\}}_{\text{Property taxes after } \tau \text{ years of exemption}}. \quad (\text{A.1}) \end{aligned}$$

Drift removal. Define an auxiliary measure $\tilde{\mathbb{P}}$ by

$$\left.\frac{d\mathbb{Q}}{d\tilde{\mathbb{P}}}\right|_{\mathcal{F}_t} = \exp\left(bZ_t - \frac{1}{2}b^2t\right), \quad t \geq 0, \quad (\text{A.2})$$

so Z_t is standard Brownian motion under $\tilde{\mathbb{P}}$. Let

$$\lambda \equiv \sqrt{2r + b^2}, \quad \xi \equiv \frac{b - \lambda}{\sigma}.$$

1. Existing cash flows.

$$\mathbf{E}_{\mathbb{Q}}\left[\int_0^{T_{R^e}} e^{-ru}(1-\psi)\Gamma(F_B, z_B, \alpha_B)R_u du\right] = (1-\psi)\Gamma(F_B, z_B, \alpha_B)R \mathbf{E}_{\mathbb{Q}}\left[\int_0^{T_a} e^{-ru}e^{\sigma Z_u} du\right].$$

Using (A.2),

$$\mathbf{E}_{\mathbb{Q}}\left[\int_0^{T_a} e^{-ru}e^{\sigma Z_u} du\right] = \mathbf{E}_{\tilde{\mathbb{P}}}\left[\int_0^{T_a} e^{-\frac{\lambda^2}{2}u}e^{(\sigma+b)Z_u} du\right] = \mathbf{E}_{\tilde{\mathbb{P}}}\left[\int_0^{\infty} \cdots du\right] - \mathbf{E}_{\tilde{\mathbb{P}}}\left[\int_{T_a}^{\infty} \cdots du\right].$$

The resolvent identity yields

$$\mathbf{E}_{\tilde{\mathbb{P}}}\left[\int_0^{\infty} e^{-\frac{\lambda^2}{2}u}e^{(\sigma+b)Z_u} du\right] = \frac{1}{\lambda} \int_{-\infty}^{\infty} e^{-|y|\lambda}e^{(\sigma+b)y} dy = \frac{1}{r - \delta}. \quad (\text{A.3})$$

where the integral converges under the maintained assumption $\delta < r$, equivalently $\lambda > \sigma + b$. By the strong Markov property at T_a and $Z_{T_a} = a$,

$$\mathbf{E}_{\tilde{\mathbb{P}}}\left[\int_{T_a}^{\infty} e^{-\frac{\lambda^2}{2}u}e^{(\sigma+b)Z_u} du\right] = e^{(\sigma+b)a}\mathbf{E}_{\tilde{\mathbb{P}}}\left[e^{-\frac{\lambda^2}{2}T_a}\right] \cdot \frac{1}{r - \delta},$$

and the Laplace transform of T_a gives $\mathbf{E}_{\tilde{\mathbb{P}}}[e^{-(\lambda^2/2)T_a}] = e^{-\lambda a}$. Hence

$$\mathbf{E}_{\mathbb{Q}} \left[\int_0^{T_a} e^{-ru} e^{\sigma Z_u} du \right] = \frac{1}{r - \delta} [1 - e^{(\sigma + b - \lambda)a}] = \frac{1}{r - \delta} \left[1 - \left(\frac{R^e}{R} \right)^{1+\xi} \right].$$

Therefore,

$$\mathbf{E}_{\mathbb{Q}} \left[\int_0^{T_{R^e}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right] = \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta} \left[R - R^e \left(\frac{R^e}{R} \right)^{\xi} \right]. \quad (\text{A.4})$$

2. Demolition.

$$\mathbf{E}_{\mathbb{Q}}[e^{-rT_{R^e}}] = \mathbf{E}_{\tilde{\mathbb{P}}}[e^{-\frac{\lambda^2}{2}T_a}] = e^{-\lambda a} = \left(\frac{R^e}{R} \right)^{\xi},$$

so

$$\mathbf{E}_{\mathbb{Q}}[e^{-rT_{R^e}} \gamma(F_B)] = \gamma(F_B) \left(\frac{R^e}{R} \right)^{\xi}. \quad (\text{A.5})$$

3. Construction costs on $[T_{R^e}, T_{R^e} + d]$.

$$\mathbf{E}_{\mathbb{Q}} \left[\int_{T_{R^e}}^{T_{R^e} + d} e^{-ru} \kappa(F, \alpha) du \right] = \kappa(F, \alpha) \mathbf{E}_{\mathbb{Q}}[e^{-rT_{R^e}}] \int_0^d e^{-rs} ds = \frac{\kappa(F, \alpha)}{r} (1 - e^{-rd}) \left(\frac{R^e}{R} \right)^{\xi}.$$

4. New value after completion. Conditioning on $\mathcal{F}_{T_{R^e}}$ and using $R_{T_{R^e}} = R^e$,

$$\mathbf{E}_{\mathbb{Q}} \left[\int_{T_{R^e} + d}^{\infty} e^{-ru} R_u du \right] = \mathbf{E}_{\mathbb{Q}}[e^{-rT_{R^e}}] \int_d^{\infty} e^{-rs} \mathbf{E}_{\mathbb{Q}}[R_{T_{R^e} + s} | R_{T_{R^e}} = R^e] ds = \left(\frac{R^e}{R} \right)^{\xi} \frac{R^e}{r - \delta} e^{-(r - \delta)d},$$

hence

$$\mathbf{E}_{\mathbb{Q}} \left[\int_{H_{R^e}}^{\infty} e^{-ru} \Gamma(F, 1, \alpha) R_u du \right] = \Gamma(F, 1, \alpha) \frac{R^e}{r - \delta} e^{-(r - \delta)d} \left(\frac{R^e}{R} \right)^{\xi}. \quad (\text{A.6})$$

5. Taxes after τ . By the same argument with lower limit $d + \tau$,

$$\mathbf{E}_{\mathbb{Q}} \left[\int_{H_{R^e} + \tau}^{\infty} e^{-ru} \psi \Gamma(F, 1, \alpha) R_u du \right] = \psi \Gamma(F, 1, \alpha) \frac{R^e}{r - \delta} e^{-(r - \delta)(d + \tau)} \left(\frac{R^e}{R} \right)^{\xi}. \quad (\text{A.7})$$

Closed form. Combining (A.4), (A.5), Step 3, (A.6), and (A.7) gives

$$\begin{aligned} \pi(R, R^e, F, F_B, z_B, \alpha_B; \mathcal{P}, \Theta) &= \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta} \left[R - R^e \left(\frac{R^e}{R} \right)^{\xi} \right] - \gamma(F_B) \left(\frac{R^e}{R} \right)^{\xi} \\ &\quad - \frac{\kappa(F, \alpha)}{r} (1 - e^{-rd}) \left(\frac{R^e}{R} \right)^{\xi} + \Gamma(F, 1, \alpha) \frac{R^e}{r - \delta} e^{-(r - \delta)d} \left(\frac{R^e}{R} \right)^{\xi} \end{aligned}$$

$$- \psi \Gamma(F, 1, \alpha) \frac{R^e}{r - \delta} e^{-(r-\delta)(d+\tau)} \left(\frac{R^e}{R} \right)^\xi. \quad (\text{A.8})$$

A.3 Proof of Proposition 2: Optimal Expansion Barrier

Fix $F \leq \bar{F}$ and regime $\mathcal{P}$. Under deterministic time-to-build d , parcel value is

$$\begin{aligned} \pi = & \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta} \left[R - R^e \left(\frac{R^e}{R} \right)^\xi \right] - \gamma(F_B) \left(\frac{R^e}{R} \right)^\xi \\ & - \frac{\kappa(F, \alpha)}{r} (1 - e^{-rd}) \left(\frac{R^e}{R} \right)^\xi + \Gamma(F, 1, \alpha) \frac{R^e}{r - \delta} e^{-(r-\delta)d} \left(\frac{R^e}{R} \right)^\xi \\ & - \psi \Gamma(F, 1, \alpha) \frac{R^e}{r - \delta} e^{-(r-\delta)(d+\tau)} \left(\frac{R^e}{R} \right)^\xi, \end{aligned} \quad (\text{A.9})$$

with $\xi = \frac{b-\lambda}{\sigma}$, $b = \frac{\delta - \frac{1}{2}\sigma^2}{\sigma}$, $\lambda = \sqrt{2r + b^2}$.

Define

$$\begin{aligned} K(F) &= \gamma(F_B) + \frac{\kappa(F, \alpha)}{r} (1 - e^{-rd}), \\ \vartheta &= \Gamma(F, 1, \alpha) \frac{e^{-(r-\delta)d} - \psi e^{-(r-\delta)(d+\tau)}}{r - \delta} - \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta}. \end{aligned}$$

Then

$$\pi = \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta} R + \left[\vartheta R^e - K(F) \right] \left(\frac{R^e}{R} \right)^\xi.$$

Maximizing in R^e is equivalent to maximizing

$$g(R^e) = \left[\vartheta R^e - K(F) \right] \left(\frac{R^e}{R} \right)^\xi.$$

FOC.

$$g'(R^e) = \vartheta \left(\frac{R^e}{R} \right)^\xi + \left[\vartheta R^e - K(F) \right] \frac{\xi}{R} \left(\frac{R^e}{R} \right)^{\xi-1}.$$

Setting $g'(R^e) = 0$ yields

$$\vartheta(1 + \xi)R^e = \xi K(F),$$

so, for $\vartheta > 0$,

$$R^{e*}(F) = \frac{\xi}{1 + \xi} \frac{K(F)}{\vartheta}.$$

SOC. Since

$$\begin{aligned} g(R^e) &= R^{-\xi} \left(\vartheta (R^e)^{1+\xi} - K(F) (R^e)^\xi \right), \\ g''(R^e) &= R^{-\xi} \left[\vartheta(1 + \xi)\xi (R^e)^{\xi-1} - K(F)\xi(\xi - 1)(R^e)^{\xi-2} \right]. \end{aligned}$$

Using the FOC,

$$g''(R^{e*}) = R^{-\xi} \vartheta(1 + \xi) (R^{e*})^{\xi-1}.$$

Under $r > \delta$, $\xi < -1$, hence $(1 + \xi) < 0$ and $g''(R^{e*}) < 0$. Thus R^{e*} is a strict local maximizer.

A.4 Proof of Proposition 3: Probability of Development

Under the physical measure $\mathbb{P}$, rent follows

$$dR_t = gR_t dt + \sigma R_t dW_t^{\mathbb{P}}, \quad R_t = R \exp\left((g - \frac{1}{2}\sigma^2)t + \sigma W_t^{\mathbb{P}}\right).$$

Define

$$X_t = \frac{1}{\sigma} \ln\left(\frac{R_t}{R}\right) = mt + W_t^{\mathbb{P}}, \quad m = \frac{g - \frac{1}{2}\sigma^2}{\sigma}.$$

Consider first a parcel whose current rent lies below its redevelopment threshold, $R < R^{e*}$, and let

$$a^* = \frac{1}{\sigma} \ln\left(\frac{R^{e*}}{R}\right) > 0.$$

The initiation time is

$$T_{R^{e*}} = \inf\{t \geq 0 : R_t = R^{e*}\} = \inf\{t \geq 0 : X_t = a^*\} \equiv T_{a^*}.$$

For Brownian motion with drift, the first-passage distribution satisfies

$$\mathbf{P}_{\mathbb{P}}(T_{a^*} \leq t) = \mathcal{N}\left(\frac{mt - a^*}{\sqrt{t}}\right) + \exp(2ma^*)\mathcal{N}\left(-\frac{mt + a^*}{\sqrt{t}}\right), \quad t > 0.$$

Applying this result to the stopping time $T_{R^{e*}}$ gives the redevelopment probability $\mathbf{P}_{\mathbb{P}}(T_{R^{e*}} \leq t)$. With deterministic construction length d , completion occurs at $H = T + d$. Therefore, for horizon h ,

$$\mathbf{P}_{\mathbb{P}}(H \leq h) = \begin{cases} \mathbf{P}_{\mathbb{P}}(T \leq h - d), & h > d, \\ 0, & h \leq d. \end{cases}$$

Substituting $t = h - d$ yields

$$\mathbf{P}_{\mathbb{P}}(H \leq h) = \mathcal{N}\left(\frac{m(h - d) - a^*}{\sqrt{h - d}}\right) + \exp(2ma^*)\mathcal{N}\left(-\frac{m(h - d) + a^*}{\sqrt{h - d}}\right),$$

which establishes the result for $R < R^{e*}$.

Finally, if $R \geq R^{e*}$ then $X_0 = \frac{1}{\sigma} \ln(R/R^{e*}) \geq 0 = a^*$, so the threshold is already attained at $t = 0$ and $T_{R^{e*}} = 0$. Redevelopment is initiated immediately, and completion occurs deterministically at $t = d$. Hence $\mathbf{P}_{\mathbb{P}}(H_{R^{e*}} \leq h) = 1$ for $h > d$ and 0 for $h \leq d$.

A.5 Optimal Density Conditional on a Fixed-expansion Threshold

Fix R^e and deterministic time-to-build d (so $H_{R^e} = T_{R^e} + d$). From (A.8), the terms that depend on F can be written as

$$\pi(F; R^e) = \text{const} - D(d) \kappa(F, \alpha) + M(d, \tau) \Gamma(F, 1, \alpha),$$

where const is independent of F and

$$D(d) \equiv \frac{1 - e^{-rd}}{r} \left(\frac{R^e}{R} \right)^\xi, \quad M(d, \tau) \equiv \frac{e^{-(r-\delta)d}(1 - \psi e^{-(r-\delta)\tau})}{r - \delta} R^e \left(\frac{R^e}{R} \right)^\xi,$$

with $\xi = (b - \lambda)/\sigma$ as in the main text. Using the calibration functional forms from subsection 4.1

$$\kappa(F, \alpha) = \frac{k_c L}{d} (1 - \alpha \eta_\alpha) F^{1+\eta_F}, \quad \Gamma(F, 1, \alpha) = L (1 - \alpha \epsilon_\alpha) F^{1+\epsilon_F},$$

the objective becomes $\pi(F; R^e) = \text{const} - K_1 F^{1+\eta_F} + K_2 F^{1+\epsilon_F}$ with

$$K_1 = D(d) \frac{k_c L}{d} (1 - \alpha \eta_\alpha), \quad K_2 = M(d, \tau) L (1 - \alpha \epsilon_\alpha).$$

Assuming an interior optimum with $\eta_F \neq \epsilon_F$, the first-order condition is

$$-(1 + \eta_F) K_1 F^{\eta_F} + (1 + \epsilon_F) K_2 F^{\epsilon_F} = 0,$$

which yields

$$\begin{aligned} F^*(R^e) &= \left[\frac{(1 + \epsilon_F) K_2}{(1 + \eta_F) K_1} \right]^{\frac{1}{\eta_F - \epsilon_F}} \\ &= \left[\frac{rd}{k_c(1 - e^{-rd})} \cdot \frac{e^{-(r-\delta)d}(1 - \psi e^{-(r-\delta)\tau})}{r - \delta} \cdot R^e \cdot \frac{(1 + \epsilon_F)(1 - \alpha \epsilon_\alpha)}{(1 + \eta_F)(1 - \alpha \eta_\alpha)} \right]^{\frac{1}{\eta_F - \epsilon_F}} \end{aligned}$$

As in the main text, once R^{e*} is substituted back as a function of F , the joint choice of (R^e, F) generally does not admit a closed-form solution. The optimal density must therefore be obtained numerically from the reduced-form objective $\pi(R, R^{e*}(F), F, \cdot)$.

A.6 Iterative Algorithm for the Spatial Fixed Point

The equilibrium drift vector is defined as the fixed point of a mapping from expected housing supply to parcel-level redevelopment decisions and back to aggregate supply. Policy-induced construction alters expected supply, which modifies the rent dynamics entering both the valuation problem and the redevelopment probability. These changes affect optimal redevelopment timing and density and therefore expected supply. Equilibrium requires consistency between supply adjustments and developers' responses. This appendix describes the fixed-point algorithm under policy $\mathcal{P}$ at horizon h .

Measures. Under the pricing measure $\mathbb{Q}$, rent evolves as $dR_t = \delta_i R_t dt + \sigma_i R_t dW_t^{\mathbb{Q}}$, where the risk-neutral drift δ_i governs valuation and determines optimal redevelopment thresholds and densities.

Under the physical measure $\mathbb{P}$, rent evolves as $dR_t = g_i R_t dt + \sigma_i R_t dW_t^{\mathbb{P}}$, where the physical growth rate g_i governs expected rent growth and redevelopment probabilities.

Initialization. Let δ'_i denote the baseline pricing drift implied by the cap rate and let g'_i denote the baseline physical rent growth rate. Initialize

$$\delta_i^{(0)} = \delta'_i, \quad g_i^{(0)} = g'_i, \quad \Delta S^{(0)} = 0, \quad \Delta N_i^{(0)} = 0.$$

Iteration $k \geq 1$. Given the previous supply adjustments, compute the common supply-induced drift adjustment

$$\Delta \delta_i^{(k)} = \frac{1}{h} \zeta_C \Delta S^{(k-1)} + \frac{1}{h} \zeta_L \Delta N_i^{(k-1)}.$$

Update both the pricing drift and the physical growth rate:

$$\delta_i^{(k)} = \delta'_i + \Delta \delta_i^{(k)}, \quad g_i^{(k)} = g'_i + \Delta \delta_i^{(k)}.$$

Using $\delta_i^{(k)}$, solve the parcel valuation problem under $\mathbb{Q}$ to obtain the optimal redevelopment trigger and density, $R_i^{e*,(k)}$ and $F_i^{*,(k)}$. Using the physical growth rate $g_i^{(k)}$, compute the probability of redevelopment

$$\mathbf{P}_i^{(k)}(H \leq h) = \mathbf{P}_{\mathbb{P}}\left(T_i^{(k)} \leq h - d\right).$$

Expected supply on parcel i is therefore

$$s_i^{(k)} = \mathbf{P}_i^{(k)} F_i^{*,(k)} L_i + (1 - \mathbf{P}_i^{(k)}) F_{B,i} L_i.$$

Aggregate and neighborhood supply changes are updated as

$$\Delta S^{(k)} = \ln\left(\sum_i s_i^{(k)}\right) - \ln\left(\sum_i s_i^{(0)}\right), \quad \Delta N_i^{(k)} = \ln\left(\sum_{j \in \mathcal{L}_i} s_j^{(k)}\right) - \ln\left(\sum_{j \in \mathcal{L}_i} s_j^{(0)}\right).$$

where $\mathcal{L}_i = \{j : \|x_j - x_i\| \leq 500 \text{ ft}\}$ denotes the set of parcels constituting the neighborhood of parcel i , where $\|x_j - x_i\|$ is the euclidean distance between centroids.

Convergence. The algorithm iterates until $|\Delta S^{(k)} - \Delta S^{(k-1)}| < \text{tol}$. At convergence the equilibrium parameters satisfy

$$\delta_i(h; \mathcal{P}) = \delta'_i + \Delta \delta_i(h; \mathcal{P}), \quad g_i(h; \mathcal{P}) = g'_i + \Delta \delta_i(h; \mathcal{P}),$$

which aligns supply adjustments with developers' optimal redevelopment responses.

A.7 Proof of Corollary 1: Expected Rental Incomes

Fix parcel i and suppress the index. Rental income at horizon h is the income of whichever structure is in place: the existing structure if redevelopment has not completed by h , and the redeveloped structure otherwise. Writing this state by state and taking the physical expectation,

$$\text{RentInc}_i(h, \mathcal{P}) = \mathbf{E}_{\mathbb{P}} \left\{ \underbrace{\mathbf{1}_{\{H_{Re*} > h\}} \Gamma(F_B, z_B(h), \alpha_B) R_{ih}}_{\text{Income if existing structure remains}} + \underbrace{\mathbf{1}_{\{H_{Re*} \leq h\}} \Gamma(F^*, 1, \alpha) R_{ih}}_{\text{Income if redevelopment completed}} \right\}. \quad (\text{A.10})$$

Because the depreciation index $z_B(h)$ is deterministic, the rent-scaling functions are constants and factor out of each term, leaving

$$\text{RentInc}_i(h, \mathcal{P}) = \Gamma(F_B, z_B(h), \alpha_B) \mathbf{E}_{\mathbb{P}}[\mathbf{1}_{\{H_{Re^*} > h\}} R_{ih}] + \Gamma(F^*, 1, \alpha) \mathbf{E}_{\mathbb{P}}[\mathbf{1}_{\{H_{Re^*} \leq h\}} R_{ih}].$$

These expectations couple the redevelopment event to the terminal rent and do *not* factor into a probability times $\mathbf{E}_{\mathbb{P}}\{R_{ih}\}$, because redevelopment occurs precisely on the paths along which rent has risen to the threshold. We evaluate the redevelopment term exactly; the no-development term then follows by subtraction, since the two sum to $\mathbf{E}_{\mathbb{P}}\{R_{ih}\} = R_i e^{g_i(h; \mathcal{P})h}$.

Under the physical measure $\mathbb{P}$, rent evolves with the policy-adjusted drift $g \equiv g_i(h; \mathcal{P})$, held constant over $[0, h]$ within the counterfactual, as $dR_t = g R_t dt + \sigma R_t dW_t^{\mathbb{P}}$, so that

$$R_t = R \exp\left((g - \frac{1}{2}\sigma^2)t + \sigma W_t^{\mathbb{P}}\right) = R e^{\sigma X_t}, \quad X_t = mt + W_t^{\mathbb{P}}, \quad m = \frac{g - \frac{1}{2}\sigma^2}{\sigma}.$$

Let $\tau = h - d$ and $a^* = \frac{1}{\sigma} \ln(R^{e^*}/R)$, and consider a parcel with $R < R^{e^*}$, so $a^* > 0$. Redevelopment is completed by h if and only if initiation occurs by τ , that is, if and only if $M_\tau \equiv \max_{s \leq \tau} X_s \geq a^*$; equivalently $\{H_{Re^*} \leq h\} = \{M_\tau \geq a^*\}$. Using $R_h = R e^{\sigma X_h}$ and splitting $X_h = X_\tau + (X_h - X_\tau)$, where the increment over $[\tau, h]$ is independent of $\mathcal{F}_\tau$ with $\mathbf{E}_{\mathbb{P}}[e^{\sigma(X_h - X_\tau)}] = e^{\sigma m d + \sigma^2 d/2} = e^{gd}$,

$$\mathbf{E}_{\mathbb{P}}[R_h \mathbf{1}\{M_\tau \geq a^*\}] = R e^{gd} \mathbf{E}_{\mathbb{P}}[e^{\sigma X_\tau} \mathbf{1}\{M_\tau \geq a^*\}].$$

Define the tilted measure $\mathbb{P}^\sigma$ by $\frac{d\mathbb{P}^\sigma}{d\mathbb{P}} \Big|_{\mathcal{F}_\tau} = \exp(\sigma W_\tau^{\mathbb{P}} - \frac{1}{2}\sigma^2 \tau)$. Under $\mathbb{P}^\sigma$, the process $W_t^{\mathbb{P}} - \sigma t$ is a Brownian motion, so X has drift $m + \sigma$. Hence

$$\mathbf{E}_{\mathbb{P}}[e^{\sigma X_\tau} \mathbf{1}\{M_\tau \geq a^*\}] = e^{\sigma m \tau} \mathbf{E}_{\mathbb{P}}[e^{\sigma W_\tau^{\mathbb{P}}} \mathbf{1}\{M_\tau \geq a^*\}] = e^{(\sigma m + \sigma^2/2)\tau} \mathbb{P}^\sigma(M_\tau \geq a^*) = e^{g\tau} \mathbb{P}^\sigma(T_{a^*} \leq \tau),$$

where $\sigma m + \sigma^2/2 = g$ and $T_{a^*} = \inf\{t \geq 0 : X_t = a^*\}$. Collecting terms, $e^{gd} e^{g\tau} = e^{gh}$, so

$$\mathbf{E}_{\mathbb{P}}[R_h \mathbf{1}\{M_\tau \geq a^*\}] = R e^{gh} \mathbb{P}^\sigma(T_{a^*} \leq h - d) = \mathbf{E}_{\mathbb{P}}\{R_h\} \hat{\mathbf{P}}^*, \quad \hat{\mathbf{P}}^* \equiv \mathbb{P}^\sigma(T_{a^*} \leq h - d).$$

Applying the first-passage law for a Brownian motion with drift $m + \sigma$ to level $a^* > 0$ yields the closed form of Proposition 3 (Equation (7)) with m replaced by $m + \sigma$. The no-development term is $\mathbf{E}_{\mathbb{P}}\{R_{ih}\}(1 - \hat{\mathbf{P}}^*)$ by subtraction, and substituting both terms into (A.10) establishes (10). If instead $R \geq R^{e^*}$, redevelopment is immediate and $\hat{\mathbf{P}}^* = 1$ for $h > d$.

A.8 Fixed-Point Computation under Voluntary Policy Adoption

In the voluntary-adoption counterfactual, developers choose whether to adopt the policy bundle $\mathcal{P}$ or remain under the baseline regime $\mathcal{P}_0$. Parcel i adopts the policy if and only if it increases project value:

$$\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0).$$

Because equilibrium is computed as a fixed point, adoption decisions depend on expectations about other parcels' participation. The resulting binary rule introduces a discontinuity in the mapping from supply adjustments to aggregate supply, which can generate oscillations in the

iteration.

To stabilize the algorithm, we smooth adoption beliefs during the fixed-point iteration using a logit approximation. The perceived probability that parcel i adopts the policy is

$$q_i(\mathcal{P}) = \frac{1}{1 + \exp\left(-\frac{\pi_i(\mathcal{P}) - \pi_i(\mathcal{P}_0)}{\nu}\right)}. \quad (\text{A.11})$$

The probabilities $q_i^{(k)}(\mathcal{P})$ enter the supply update step of the fixed-point iteration described in Appendix A.6. Expected supply on parcel i during iteration k is

$$s_i^{(k)} = q_i^{(k)}(\mathcal{P}) s_i^{(k)}(\mathcal{P}) + (1 - q_i^{(k)}(\mathcal{P}_0)) s_i^{(k)}(\mathcal{P}_0).$$

This smoothing replaces the discontinuous adoption rule during the iteration and yields a continuous mapping from drift adjustments to aggregate supply. Once the fixed point has converged, the smoothing device is removed and the final adoption decision reverts to the deterministic rule

$$\text{parcel } i \text{ adopts } \mathcal{P} \iff \pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0).$$

The parameter ν governs the degree of smoothing. As $\nu \rightarrow 0$, equation (A.11) converges to the deterministic adoption rule.

A.9 Feasibility and Compensating Incentives

This appendix characterizes the properties of parcel value that underlie the feasibility and participation results discussed in Section 7.

Proposition 1 (Continuity, Monotonicity, and Feasibility Thresholds) *Fix a mandate intensity $\alpha > 0$. Let parcel value be defined as in Proposition 1, where for each (α, ϕ, τ) the developer chooses optimal redevelopment timing and density:*

$$\pi_i(\alpha, \phi, \tau) = \max_{0 \leq F \leq \bar{F}(\phi)} \pi(R, R^{e*}(F; \alpha, \phi, \tau), F, F_B, z_B, \alpha_B).$$

Then: (i) $\pi_i(\alpha, \phi, \tau)$ is continuous in (ϕ, τ) on the admissible policy domain; (ii) $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in the fiscal incentive τ ; (iii) $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in the density bonus ϕ through expansion of the feasible set $F \leq \bar{F}(\phi)$; (iv) Define $\varsigma_i(\phi, \tau) = \pi_i(\alpha, \phi, \tau) - \pi_i(\mathcal{P}_0)$. Then the compensating set $\mathcal{C}_i(\alpha) = \{(\phi, \tau) : \varsigma_i(\phi, \tau) \geq 0\}$ is monotone in incentives: if $(\phi, \tau) \in \mathcal{C}_i(\alpha)$ and $(\phi', \tau') \geq (\phi, \tau)$ componentwise, then $(\phi', \tau') \in \mathcal{C}_i(\alpha)$; (v) Feasibility is characterized by parcel-specific thresholds: the boundary $\{(\phi, \tau) : \varsigma_i(\phi, \tau) = 0\}$ separates infeasible and feasible policy combinations.

The proof proceeds in steps:

Step 1: Continuity. For any fixed construction density F , parcel value $\pi(\cdot)$ is given in Proposition 1. This expression is continuous in (ϕ, τ) on the admissible domain. The optimized value $\pi_i(\alpha, \phi, \tau)$ is obtained by maximizing this continuous function over the compact set $F \in [0, \bar{F}(\phi)]$. Standard arguments imply that the resulting value function is continuous in (ϕ, τ) .

Step 2: Monotonicity in τ . From (4), a longer tax exemption reduces the present value of

property taxes paid after redevelopment. Holding F fixed, this increases project value. Since the developer optimizes over F , the optimized value $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in τ .

Step 3: Monotonicity in ϕ . A higher density bonus increases the upper bound $\bar{F}(\phi)$ on feasible construction density. This expands the feasible set of the developer's optimization problem. Since the objective is maximized over this set, enlarging the feasible set cannot reduce the optimized value. Hence $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in ϕ . Note that although a higher density bonus raises optimized parcel value, it may also increase the optimal expansion threshold $R^{e*}(F)$ and delay redevelopment. This affects development probabilities but does not overturn the monotonicity of the value function.

Step 4: Compensating set. Define $\varsigma_i(\phi, \tau) = \pi_i(\alpha, \phi, \tau) - \pi_i(\mathcal{P}_0)$. By Steps 1–3, ς_i is continuous and weakly increasing in each argument. The compensating set $\{(\phi, \tau) : \varsigma_i(\phi, \tau) \geq 0\}$ is therefore a superlevel set of a continuous, monotone function. It follows that if $(\phi, \tau) \in \mathcal{C}_i(\alpha)$, any $(\phi', \tau') \geq (\phi, \tau)$ also belongs to $\mathcal{C}_i(\alpha)$.

Step 5: Threshold structure. Continuity of ς_i implies that the boundary $\{(\phi, \tau) : \varsigma_i(\phi, \tau) = 0\}$ separates the feasible and infeasible regions. This boundary defines parcel-specific thresholds for incentives required to restore feasibility.

B Additional Parcel-Level Comparative Statics

In this Appendix, we extend the parcel-level analysis developed in Section 3 by providing additional comparative statics on policy instruments, examining how redevelopment incentives vary across heterogeneous parcels, showing the trade-offs between policy instruments, and illustrating how spatial spillovers transmit redevelopment shocks across the city.

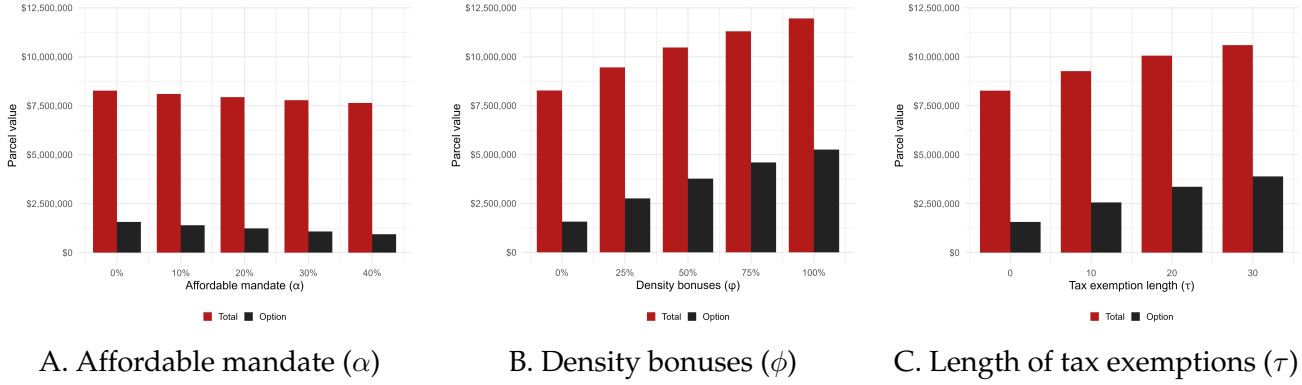
Unless specified otherwise, the comparative statics for the single-parcel are obtained for a parcel built at 66% of its capacity, with $F_B = 2$ and $\bar{F} = 3$, an age of 150 years, an expected rental growth rate of 4% (total return of 8% and a caprate of 4%), an annual potential rent per sqft of 50, and a lot size of 2,500 sqft. All other elasticities correspond to the values shown in Table 1.

B.1 Policy Instruments Effects

In Table B1, we report changes in parcel value and in the probability of redevelopment associated with the enactment of a single policy. The table shows how these results vary with key elasticity assumptions. Figure B1 highlights the effects of each policy instrument on the value of the redevelopment option, and Figure B2 shows the outcomes when policies are combined.

Affordable mandate (α). Mandating affordable units on a given parcel unambiguously reduces both parcel value and the probability of redevelopment, because of the decline in the redevelopment option value (Panel A of Figure B1). For instance, Table B1 shows that a 20% mandate lowers parcel value by 0.92% and development probability by 2.31 percentage points. The impact is stronger in high-demand areas where the same 20% mandate leads to a 1.52% drop in value and a 4.97 percentage-point reduction in development probability. Effects are also stronger for parcels closer to the redevelopment threshold (i.e., with more unused density and older buildings). If the rental discount on affordable units doubles (ϵ_α), a 20% mandate reduces property values by

Figure B1: Policy Instruments Influence the redevelopment Option Value



Notes: These figures show the effects the policy instruments on the present value of the current building and the present values of the redevelopment option. This option value is calculated as the difference between the total parcel value and its counterpart, where no development is allowed. These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, a potential rent of 100, and a lot size of 2,500 sqft. All other parameters and elasticities are shown in Table 1.

1.98% and development probability by 5.24 percentage points, because deeper rent discounts further reduce the incentive to develop. In contrast, if developers can reduce construction costs for affordable units (higher η_α), the 20% mandate only modestly lowers property values, suggesting that cost savings can partially offset the mandate's negative effects.

Density bonuses (ϕ). Although density bonuses unambiguously increase parcel value through a higher redevelopment option value (Figure B1), they may also delay redevelopment as developers may require stronger market potential to justify the higher cost of construction. As an example, the second panel of Table B1 shows that a 50% increase in allowable density ($\phi = 0.50$) raises parcel value by 6.08% and development probability by 3.68 percentage points. In high-demand areas, property values and development probability increase more strongly with ϕ . For parcels closer to the redevelopment threshold, the parcel value still increases, but the probability of redevelopment declines sharply as developers would rationally wait to construct a taller building rather than build a lower one earlier. If the marginal benefit of height is higher (ϵ_F), a ϕ of 50% raises property values by 7.57% and development probability by 5.38 percentage points. In contrast, if we set a higher height elasticity of construction cost, the parcel value increases modestly with the density restriction relaxation, and redevelopment probabilities remain essentially unchanged. These results highlight that the impact of density policy depends critically on construction technology.

Tax exemptions (τ). Offering tax exemptions consistently increases parcel values and accelerates redevelopment. The increase in value arises entirely from the redevelopment option, since the exemption applies only to new construction (Figure B1). As exemplified by the third panel of Table B1, a 20-year exemption increases parcel value by 5.15% and development probability by 15.57 percentage points. These effects are amplified for parcels closer to redevelopment, where probabilities rise by 21.82 percentage points, and in high-demand areas, where values increase by 8.48% and redevelopment probabilities by 26.42 percentage points. As expected, the impact of exemptions grows with higher property tax rates.

Table B1: Policy's Effects on Single Parcel Outcomes and Sensitivity Analyses

	Change in Property Value (%)	Change in Prob. Dev. (p.p.)
<u>Mandating 20% of Affordable Housing ($\alpha = 0.20$)</u>		
<i>Baseline</i>	-0.92	-2.31
<i>Higher potential rent ($R = 1.25R$)</i>	-1.52	-4.97
<i>Parcel closer to redevelopment ($F_0 = F_0/2$ and $age = 2age$)</i>	-3.15	-3.19
<i>Higher affordable rental discount ($\epsilon_\alpha = 2\epsilon_\alpha$)</i>	-1.98	-5.24
<i>Higher affordable construction discount ($\eta_\alpha = 2\eta_\alpha$)</i>	-0.58	-0.78
<u>Increase Maximum Allowed Density by 50% ($\phi = 0.50$)</u>		
<i>Baseline</i>	6.08	3.68
<i>Higher potential rent ($R = 1.25R$)</i>	10.02	7.16
<i>Parcel closer to redevelopment ($F_0 = F_0/2$ and $age = 2age$)</i>	8.33	-12.94
<i>Higher return to height ($\epsilon_F = 1.5\epsilon_F$)</i>	7.57	5.38
<i>Higher cost from height ($\eta_F = 1.5\eta_F$)</i>	1.64	-0.39
<u>20-years Property Tax Exemption ($\tau = 20$)</u>		
<i>Baseline</i>	5.15	15.57
<i>Higher potential rent ($R = 1.25R$)</i>	8.48	26.42
<i>Parcel closer to redevelopment ($F_0 = F_0/2$ and $age = 2age$)</i>	20.43	21.82
<i>Higher property tax rate ($\psi = 1.5\psi$)</i>	7.93	20.39

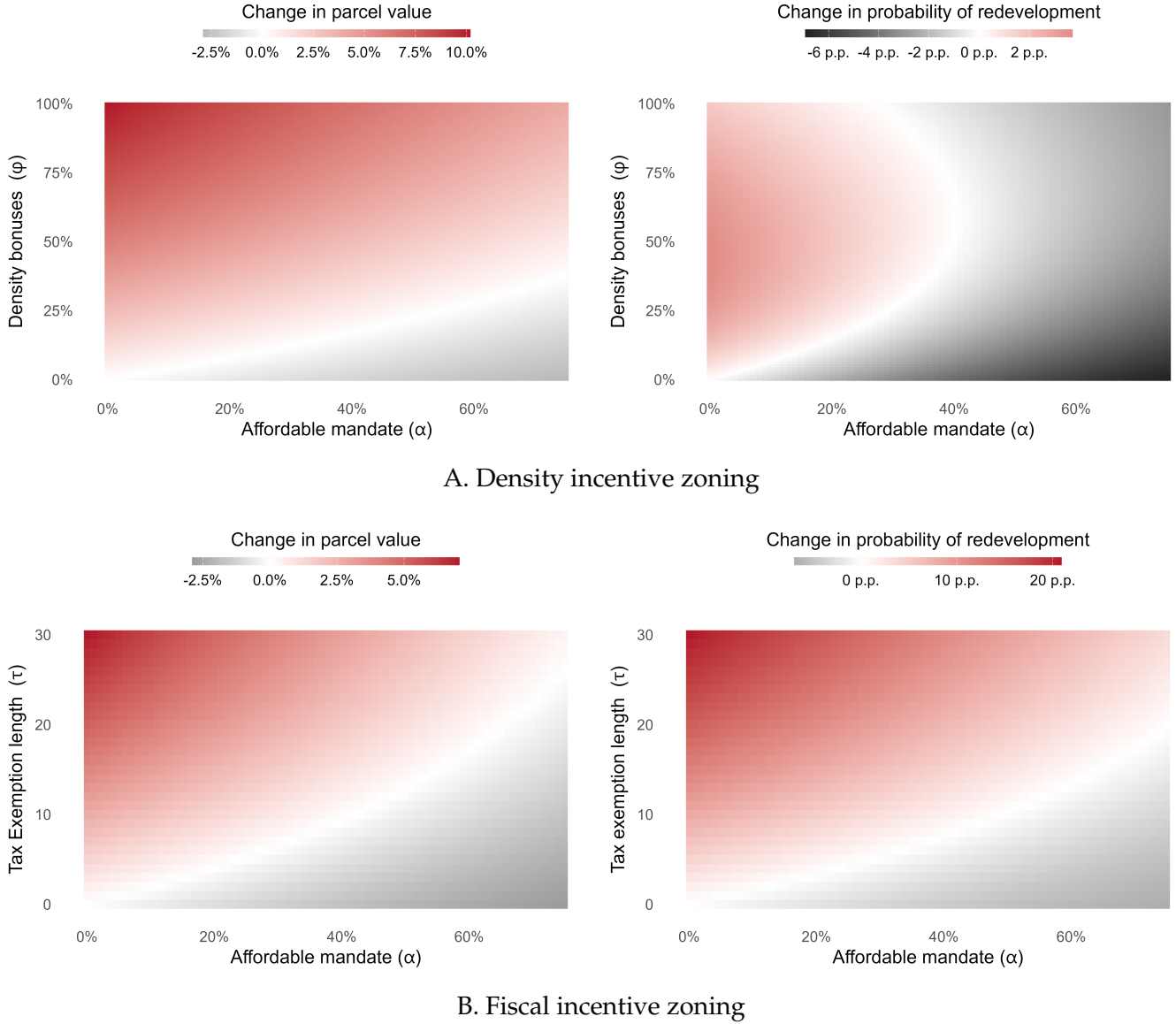
Notes: This table reports the change in property value and the change in the probability of development within 10 years with respect to the policy instruments. These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and an annual potential rent of \$50 per square foot. All other parameters and elasticities are shown in Table 1.

Trade-offs. We then study how affordability mandates interact with density bonuses and tax incentives. The left panels of Figure B2 show that, for any affordability mandate, one can choose a density bonus or a tax exemption that keeps property values unchanged. At those combinations, developers are indifferent between the status-quo and the combination of instruments. The right panels also show that there are combinations that keep redevelopment incentives unchanged. The two indifference curves, however, do not coincide. Financial indifference does not imply housing production indifference. Affordability mandates cannot generally be offset by a single incentive policy that simultaneously restores both developer value and redevelopment behavior. This distinction explains why policy instruments that appear similar from a developer's perspective can produce different redevelopment decisions. Although zoning in New York is primarily as-of-right (i.e., development allowed without discretionary review), these parcel-level results illustrate how the model extends naturally to discretionary zoning systems in other cities.

B.2 Parcel Heterogeneity and the Probability of Development

In this subsection, we show how the model predicts parcel value and redevelopment probability as functions of the parcel's characteristics. These results highlight that heterogeneity in

Figure B2: Trade-offs Between Mandate and Incentive in Simple Parcel Analysis

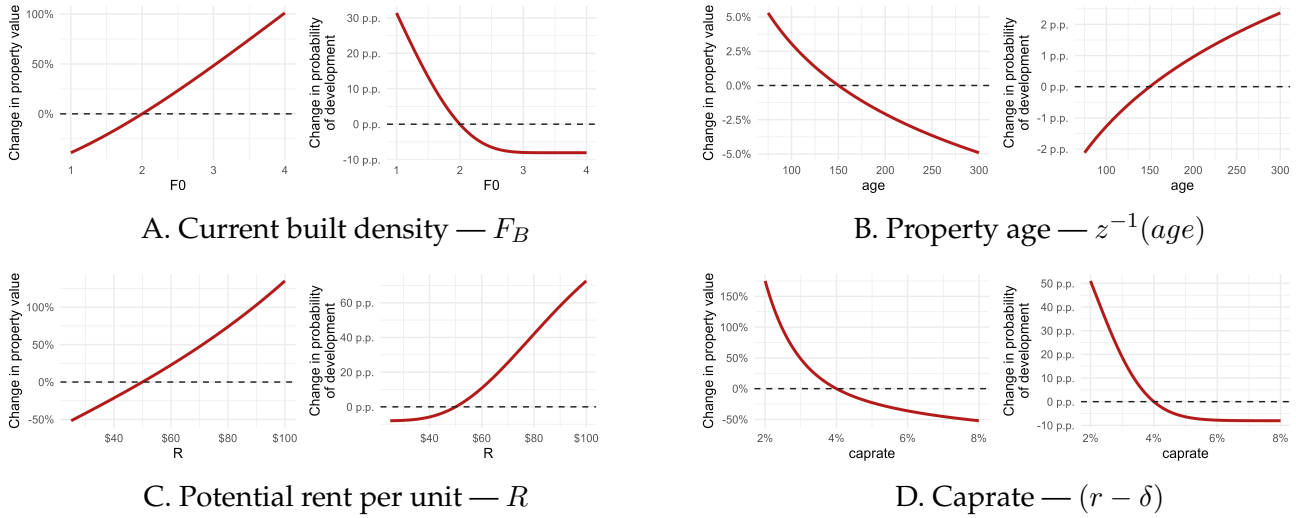


Notes: These figures show the effects of combining an affordable mandate with either density bonuses (Panel A) or with a property tax exemption (Panel B) on the change in the property value (left panels) and the change in the probability of development completion within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and an annual potential rent of 50. All other parameters and elasticities are shown in Table 1.

age, density, and local demand fundamentals is central to explaining spatial variation in redevelopment activity and, therefore, essential for understanding both the distributional and aggregate effects of policy interventions. Figure B4 further illustrates marginal effects with respect to the remaining parameters of the parcel-level model.

Panel A of Figure B3 shows that parcels with low initial built density (F_B) have lower property values than more intensively developed sites but exhibit the highest redevelopment prob-

Figure B3: Effects of key Parcel Heterogeneity on Value and Probability of Development



Notes: These figures show the effects of changing the one input parameter on the change in the property value (left panels) and the change in the probability of development starting within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and a potential rent of 50. All other parameters and elasticities are shown in Table 1.

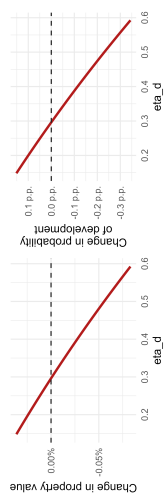
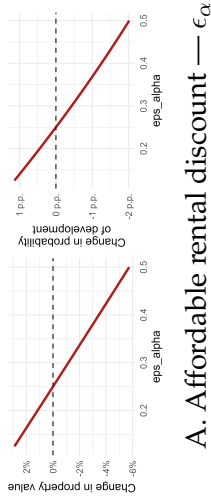
abilities, reflecting the larger embedded option value of underbuilt parcels. As current density approaches its maximum, the potential for value creation through rebuilding declines sharply, even though the standing property retains a high value. Panel B shows that redevelopment incentives rise with building age. Older structures face greater physical and economic obsolescence, which increases the value of the redevelopment option, while property values decline with age as rental income from existing buildings deteriorates. Together, consistent with the evidence in Section 1, these panels show that redevelopment becomes more likely when the expected gains from rebuilding outweigh the opportunity cost of preserving the existing structure—an essential mechanism for understanding redevelopment activity across the city.

Spatial heterogeneity further amplifies these dynamics. In our framework, spatial variation operates through potential rent (R) and caprate (c), which together summarize local demand fundamentals. Panels C and D show that higher rent levels (R) or lower caprate (i.e., higher expected rental growth) both increase parcel value and sharply raise redevelopment probabilities, with effects that intensify at higher R and δ . Small differences in neighborhood demand can therefore translate into disproportionately large differences in redevelopment activity, illustrating how micro-level heterogeneity aggregates into uneven spatial patterns of urban growth.

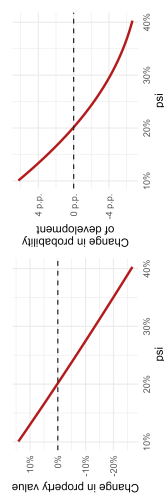
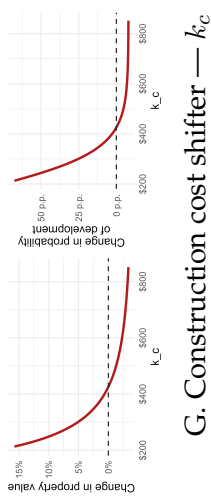
B.3 Spatial Responses to an Exogenous Supply Shock

We show how an exogenous increase in supply on a subset of parcels propagates across space through the local spillover and citywide supply channels. In a stylized city, we force 1% of parcels to redevelop. This shock raises housing supply by 0.91% of the existing stock. We then study how this shock changes construction probabilities on all other parcels. Developers endogenously update their redevelopment decisions in response to both aggregate and local supply effects. We run this experiment in a stylized city where potential rents R_i rise from South to North, while the

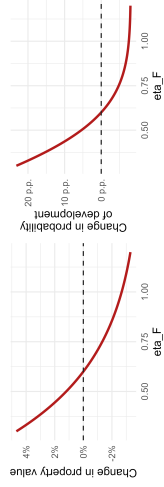
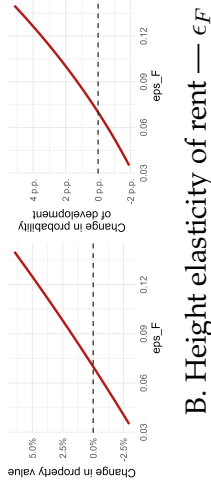
Figure B4: Effects of Parameter Changes on Parcel Value and Probability of Development



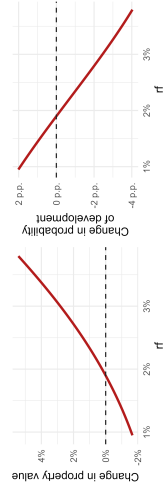
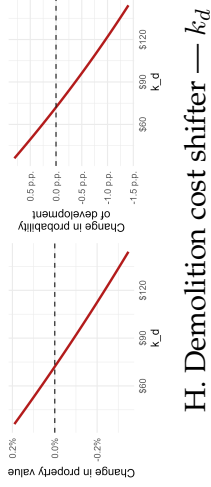
D. Height elasticity of demolition — η_D



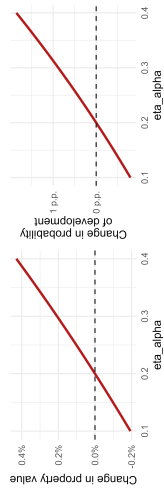
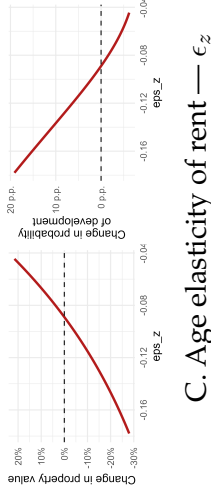
J. Property tax rate — ψ



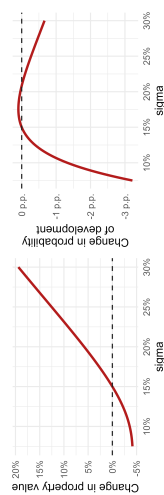
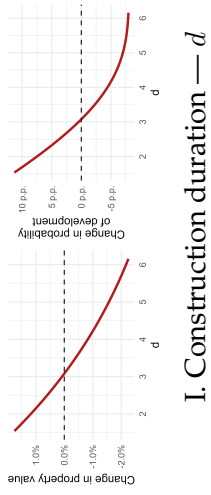
E. Height elasticity of construction — η_F



K. Risk-free rate — r



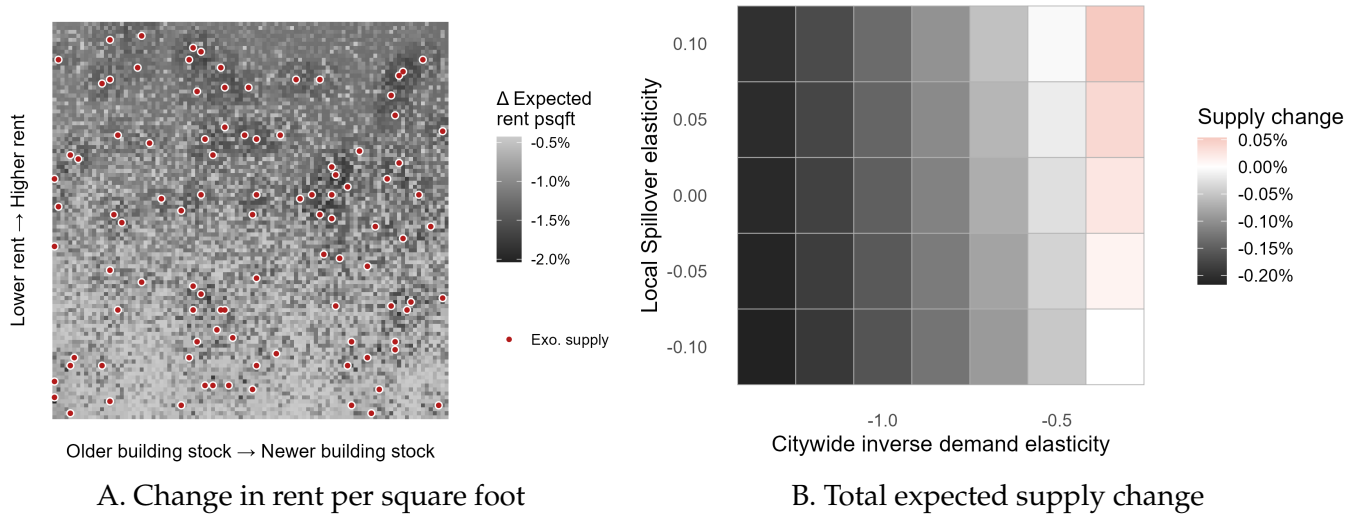
F. α elasticity of construction — η_α



L. Volatility — σ

Notes: These figures show the effects of changing the one input parameter on the change in the property value (left panels) and the change in the probability of development starting within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and a potential rent of 50. All other parameters and elasticities are shown in Table 1.

Figure B5: Parcel and Aggregates Responses to an Exogenous Supply Change of 0.91%



Notes: These figures illustrate the spatial propagation of an exogenous housing supply shock in a stylized city. In the policy experiment, 1% of parcels (shown as red dots in Panel A) are forced to redevelop ($Prob_i = 1$) at a density 100% higher than the current limit, generating an exogenous increase in housing supply equivalent to 0.91% of the current supply. Potential rents R_i are drawn such that they increase from South to North, and building age becomes older (more obsolete) from West to East. All other parameters are similar across parcels ($F_B = 3$, $\bar{F} = 6$, cap rate = 4%, total return = 8%, $\sigma = 15\%$, and $\alpha_B = 0$). All other elasticities and parameters correspond to the baseline values reported in Table 1. For illustrative purposes, we set $\zeta_L = -0.05$, so that local supply decreases local rents. Panel B shows the aggregate change in total housing supply (excluding the exogenously redeveloped parcels) as a function of the citywide inverse demand elasticity ζ_C , and the local spillover elasticity ζ_L .

building stock becomes older and more obsolete from West to East.

Panel A of Figure B5 shows how rents per square foot change across the city. The aggregate exogenous increase in supply lowers expected rents and weakens redevelopment incentives across parcels. The decline is largest for parcels with high initial demand (R_i) and for newer buildings. These parcels sit farther from their redevelopment thresholds. At the same time, construction becomes even less likely near parcels that are forced to redevelop, because of the negative spillover parameter ζ_L . The effect is strongest when several nearby parcels are forced to redevelop. The opposite occurs with $\zeta_L < 0$, in which we observe redevelopment clustering in space.

The aggregate response of all other parcels remains small, however. In a built-out city, redevelopment is expensive. Demolition costs are high. Construction takes time. Panel B of Figure B5 shows how the aggregate supply response varies with the inverse long-run demand elasticity, ζ_C , and the strength and sign of local spillovers, ζ_L . For empirically plausible values of ζ_C , the change in aggregate supply is small. It is typically less than a 0.20% decline in response to the 0.91% initial stock increase, even though redevelopment shifts across locations. Positive local spillovers raise local construction incentives, but they do not overturn the dominant role of the citywide inverse demand elasticity, for plausible calibration.

C Data, Calibration, and Baseline Outcomes

This Appendix documents the empirical foundations of the calibrated model. We describe the construction of the parcel-level dataset and the estimation of the model’s key parameters, including local rent spillovers, rental depreciation, rent levels, and expected returns. Finally, we describe the aggregation procedures used to construct the neighborhood- and citywide objects required for the model calibration. Together, these estimates and construction procedures establish the empirical inputs underlying the calibrated spatial equilibrium and the policy counterfactuals presented in the main text.

Table C1: Prior Land Use of Permitted Parcels

	Built			BBL Not Found [†]	Vacant	Total Permits	Share Built (%)
	Residential Only	Other Built [‡]	Total Built				
<i>Manhattan</i>	7	79	86	1	25	112	76.8
<i>Bronx</i>	18	310	328	16	154	498	65.9
<i>Brooklyn</i>	300	194	494	12	246	752	65.7
<i>Queens</i>	417	166	583	30	290	903	64.6
<i>Staten Island</i>	245	36	281	59	478	818	34.3
<i>Citywide</i>	987	785	1,772	118	1,193	3,083	57.5

Notes: This table reports the prior land use of parcels receiving private for-profit residential building permits filed between 2023 and 2025, based on a match between permit filings and historical PLUTO records. Vacant parcels are those recorded as vacant in every pre-filing PLUTO vintage. [†]BBL Not Found parcels have no matched pre-filing PLUTO record, most plausibly because the tax lot was created after subdivision following demolition of a prior structure; because this cannot be confirmed, these parcels are excluded from the numerator of Share Built but retained in the denominator, so the reported shares are conservative lower bounds. [‡]Other Built combines parcels of unknown, commercial, or mixed use prior to filing.

C.1 Rental Obsolescence due to Aging

The motivating evidence in Section 1 shows that older buildings redevelop more frequently. We provide here the empirical foundation for this observation by estimating the age elasticity of rents. This elasticity, in turn, determines how depreciation lowers the option value of waiting in the structural model. Following [Rosenthal \(2014\)](#), we estimate the age elasticity independently using both transaction prices and reported rental incomes.

We compute the age elasticity of real estate prices using multifamily transactions from CoStar between 2000 and 2023, and with the rental income for buildings with more than 11 units from the NOPV data (see [C.2.1](#) for details on data cleaning). The estimated price discount ϵ_z using transaction data can be applied directly to rents because the year and location fixed effects account for variations in capitalization rates. We estimate a hedonic pricing regression of the form

$$\log(Y_{ijt}) = \beta_j + \beta_t + \epsilon_z \log(\text{Age}_{it} + 1) + X_i' \beta + \varepsilon_{ijt},$$

where $\log(Y_{ijt})$ is the log building price (or rental income) in year t and location j , β_j and β_t denote spatial and time fixed effects, Age_{it} is the difference between the sales (assessment) year and the

building's construction year, and X_i includes covariates such as log building size, log number of floors, a retail-use indicator, and distance to the nearest MTA subway station. The log-log specification allows for faster depreciation at earlier ages, consistent with [Lopez and Yoshida \(2022\)](#).

Table C2: Estimation of the Age Elasticity

	Dependent variable:					
	log(SalePrice)			log(RentalIncome)		
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(\text{Age}_i + 1)$	-0.096*** (0.014)	-0.093*** (0.013)	-0.089*** (0.011)	-0.084*** (0.012)	-0.084*** (0.012)	-0.088*** (0.010)
Controls	X	X	X	X	X	X
Community District FE	X			X		
Year FE	X			X		
Community district x year FE		X			X	
Neighborhoods x year FE			X			X
Observations	35,383	35,383	35,383	165,644	165,644	165,644
R ²	0.814	0.834	0.852	0.945	0.945	0.949
Adjusted R ²	0.813	0.827	0.836	0.945	0.945	0.948

Notes: This table reports the estimates of $\log(Y)_{ijt} = \beta_j + \beta_t + \epsilon_z \log(\text{Age}_i + 1) + X_i \beta + \varepsilon_{ijt}$, where $\log(Y)$ is the log of the building price transacted in year t in location j in the first three columns and the log of the reported rental income in the last three. β_j and β_t are spatial and time fixed effects, respectively, Age_{it} is the age of the building at the time of transaction or reporting, X_i are covariates including the log size of the building in square foot, the log number of floors, an indicator for whether the property has a retail component, and the distance to the closest MTA metro station. In the first three columns, the sample consists of all building transactions from 2000 to 2023 in New York City, collected from Costar. In the last three columns, it is the reported rental income for properties with more than 11 units from the NOPV data from 2016 to 2024. Robust standard errors clustered at the community district and year are presented in parentheses. Estimates followed by ***, **, and * are statistically significant at the 1%, 5%, and 10% levels, respectively.

Table C2 reports results across several fixed-effect specifications. Estimates of ϵ_z in Columns (1-3), which capture the decline in prices due to aging, range from -0.096 to -0.084, implying an annual rental depreciation pattern that is steeper in the early years (averaging approximately 2% over the first ten years). Remarkably, the transaction-based and rental-income estimates are highly consistent despite relying on different data sources and time periods. Although rent-regulated and rent-stabilized buildings cannot be identified with certainty, we use the "exemption" field in the NOPV data to test whether the age gradient differs between buildings identified as non-market and market-rate buildings. We find no significant difference.

We therefore set $\epsilon_z = -0.088$, reflecting the average of the preferred specifications in Columns (3) and (6). Because the two estimates are obtained from independent datasets measuring different dimensions of housing value, their close agreement suggests that the estimated age elasticity reflects a stable economic relationship rather than idiosyncrasies of either dataset. We then compute $z_i = \text{Age}_i^{\epsilon_z}$ to obtain parcel-level rental depreciation. Spatial variation in z_i is shown in Panel B of Figure C6.

C.2 Estimation of local spillovers parameter ζ_L

C.2.1 Data

We define ζ_L as the *local* supply elasticity of rents: the extent to which local rents change in response to a nearby increase in supply. To estimate it for New York, we combine data on building rental incomes with building completion data.

Building completion. We collected the date, location, and size of new residential completions in New York City from the Department of City Planning Housing Database ([New York City Department of City Planning, 2026a](#)). We restrict the sample to new building construction with at least 50 residential units so that each event constitutes a credible change in neighboring amenities and competition. We date each completion by the year of its first certificate of occupancy, the point at which the units become rented. Redevelopment projects that have been filed, approved, or permitted but not yet completed help us to construct our primary control group because these sites are selected on development potential in the same way completed sites are, but their units have not yet reached the market. In total, we identify 964 completions of 50 units or more between 2016 and 2024, delivering 155,164 net units, of which 198 (20.5%) are in Manhattan.

Rents data from NOPV. We construct a building-level panel of rents from the New York City Notice of Property Value (NOPV) covering income years 2016 to 2024.²² The NOPV data records the gross income of every property that the Department of Finance (DOF) values with the income approach. For properties with at least eleven residential units, the DOF requires owners to report gross income themselves through the Real Property Income and Expense (RPIE) filing. To approximate market rents, we therefore restrict to properties with at least eleven units, which excludes condominiums (whose income is imputed). We further exclude public housing operated by the New York City Housing Authority, and buildings that devote 10% or more of their floorspace to commercial use. The resulting sample contains 211,988 building-year observations on 24,651 buildings. The median building has 32 units and 27,392 square feet, was built in 1927, and rents for \$21.44 per square foot. Ninety-one percent of buildings are observed every year, and 3.0% of income observations are interpolated.²³

C.2.2 Methodology

We estimate the effect of exposure to nearby new construction with the staggered difference-in-differences estimator of [Callaway and Sant’Anna \(2021\)](#). Let Y_{it} denote the log of rent per square foot in building i and year t , and let G_i denote the first year in which building i is exposed to a building completion within its 500-foot radius. Writing $Y_{it}(g)$ for the potential outcome under first exposure in year g and $Y_{it}(0)$ for the never-exposed potential outcome, we estimate the *group-time average treatment effect* as

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g], \quad (\text{C.1})$$

the average effect at date t among buildings first exposed in year g . Unlike the conventional two-way fixed effects (TWFE) event-study regression, this methodology compares the treated cohort

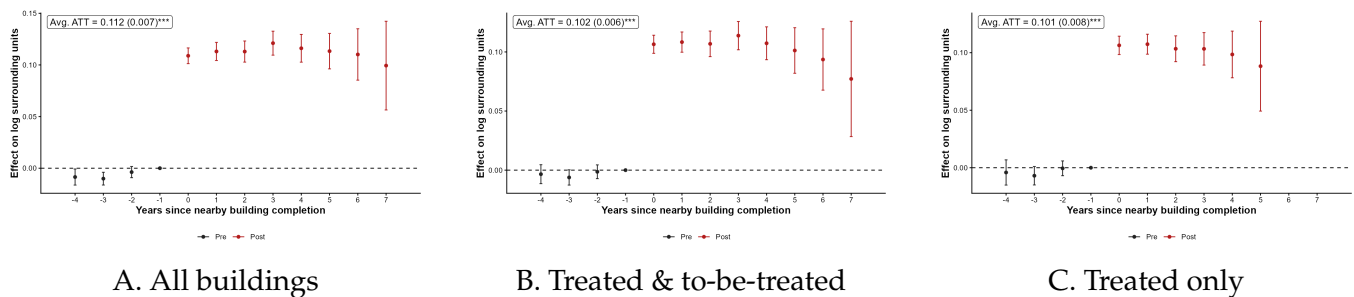
²²We thank the NYU Furman Center for sharing the data with us.

²³We construct annual rent per square foot as the gross income of the property divided by its floorspace. For interior years in which income is not reported, we interpolate linearly between the surrounding observed years.

only against cohorts that are *never treated* or *not yet treated*. With treatment effect heterogeneity (such as length-of-exposure heterogeneity here), this approach voids negative weights in the aggregation of an ATT (De Chaisemartin and d’Haultfoeuille, 2020; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024). Identification of (C.1) rests on the parallel trend assumption, which requires that, absent exposure, the rents of buildings exposed in year g would have evolved in parallel with those of the comparison group. We plot the event-study (dynamic) ATT aggregated by length of exposure and trace the path of the rent response before and after exposure.

Defining treatment and control groups. We define a building as treated if the centroid of its lot lies within 500 feet of the centroid of a lot on which a new building was completed. Treatment timing is based on the completion year of the neighboring building. Construction events on the building’s own lot are excluded, and buildings exposed to a second nearby completion during the period are dropped. We report results under three progressively more restrictive comparison groups. The first uses all buildings of at least eleven units observed in the NOPV data. These are already a restricted and broadly comparable set of residential buildings. The second and preferred sample retains only buildings that are treated or *will become* treated. The latter group is identified by buildings near sites where we observe a permit filed but not yet completed. Locations in this sample are plausibly more similar because all these sites are areas where development is being pursued. The third and most restricted sample retains treated buildings only, so that identification comes purely from variation in the timing of completion among places that all receive supply.²⁴

Figure C1: Effect of nearby new constructions on supply



Notes: Each panel plots Callaway and Sant’Anna (2021) event-study estimates of the effect of a nearby residential completion on the housing stock within 500 feet of the building. The dependent variable is the log number of residential units in the ring, measured as the PLUTO 2013 count rolled forward with net completions from the DCP Housing Database. Event time zero is the year the nearby building received its certificate of occupancy. The sample is NOPV buildings of at least eleven units, excluding public housing and buildings with a commercial floor-area share above 10%, the building’s own lot, and buildings exposed to a second nearby completion during the period. Panel A uses all such buildings, Panel B restricts the sample to buildings that are treated or that have a filed-but-uncompleted project nearby, and Panel C restricts it to treated buildings. Bars denote 95% confidence bands computed from a multiplier bootstrap clustered at the building level.

Treatment effects on neighboring supply. Before turning to rents, we verify that the treatment corresponds to a substantial increase in the surrounding housing stock. We estimate the event

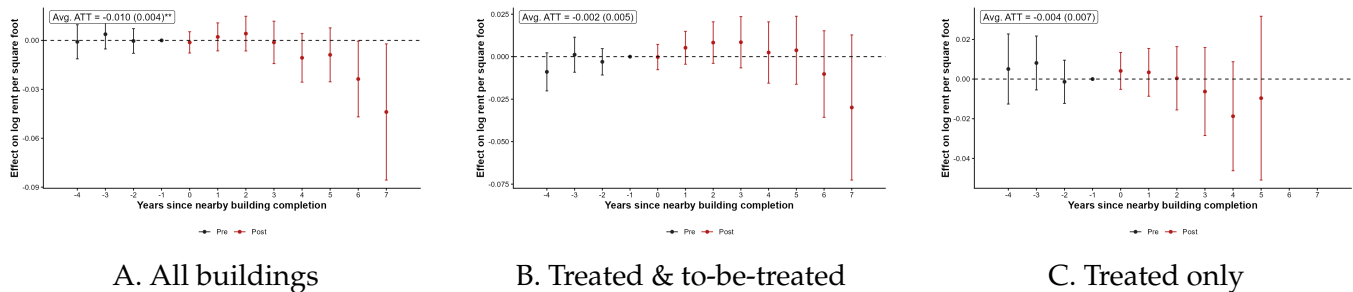
²⁴With this sample, the effects of the latest-treated groups are not identified, as no not-yet-treated units remain against which to compare them. Note that for all three samples, we allow the not-yet-treated to be used for the calculation of the ATT.

study using the log of the number of residential units in the 500-foot rings around NOPV buildings. The ring stock is the PLUTO count of neighboring units in 2013, rolled forward with the net residential completions recorded in the DCP Housing Database. Figure C1 reports the dynamic effects for the three comparison groups. Pre-treatment coefficients are flat and centered on zero. After the treatment, the surrounding unit stock rises sharply at completion. Within 500 feet, housing supply increases by about 10.0% and remains at that level thereafter. This result is consistent with a one-time, permanent addition to the neighboring stock.

C.2.3 Results

Figure C2 reports the event-study estimates for the three comparison samples. The pre-treatment coefficients are flat and statistically indistinguishable from zero in all three panels. Rents appear not to respond to nearby supply, or if they do, only marginally. The coefficients at and immediately after completion are essentially zero. Estimates drift mildly negative at longer horizons, but the effects remain small and, in the samples with the cleanest comparison groups, statistically non-significant. The average post-treatment effect ranges from -0.002 (Sample 2) to -0.010 (Sample 1) and is statistically significant at the 5% level only in Panel A, which uses the broadest, all-buildings comparison group.

Figure C2: Effect of nearby new constructions on rents



Notes: Each panel plots [Callaway and Sant'Anna \(2021\)](#) event-study estimates of the effect of a nearby residential completion on the rent of the building. The dependent variable is the log rent per square foot, gross income divided by floor area. Event time zero is the year the nearby building received its certificate of occupancy. The sample is NOPV buildings of at least eleven units, excluding public housing, buildings with a commercial floor-area share above 10%, the building's own lot, and buildings exposed to a second nearby completion during the period. Panel A uses all such buildings. Panel B restricts the sample to buildings that are treated or that have a filed-but-uncompleted project nearby, and panel C restricts it to treated buildings, so that identification comes from differences in the timing of completion. Bars denote 95% confidence bands computed from a multiplier bootstrap clustered at the building level.

These results are smaller in magnitude (and non-significant) compared to recent estimates in the literature, which report significant negative effects of local supply on rents. For instance, [Li \(2022\)](#) finds a local elasticity of rents with respect to new supply of -0.10 , and [Rollet \(2025\)](#) finds -0.42 at a comparable spatial scale. Our results do not, however, imply that rents are unresponsive to new supply, because the citywide supply effect is operative regardless of the local effect. Rather, once building-specific and time trends are absorbed by the event-study design, we find no significant additional local effect of nearby completions. This result implies that the net effect of two opposing forces is zero. The effect of the increased local housing competition, which lowers rents, and the effect of the neighborhood amenity improvements, which raise them, have the same magnitude: $\text{LocalEffect} = \text{CompetitionEffect} + \text{AmenityEffect} \approx 0$.

Calibration of ζ_L . To translate the binary treatment effect into the elasticity ζ_L , we compare the ATT on rents with the ATT on nearby supply, both computed as the average across length of exposure. Table C3 reports the event-study coefficient five years after completion, the overall dynamic ATT, and the same statistics for the supply outcome. Specifically, we define the local elasticity as the ratio of the rent and supply responses, $\zeta_L = \text{ATT}_{\text{rent}}/\text{ATT}_{\text{supply}}$. It equals -0.09 , -0.01 , and -0.04 across the three samples. We adopt $\zeta_L = -0.01$, consistent with the results with our preferred sample, and we assess robustness to $\zeta_L \in \{-0.10, -0.05, 0, +0.05, +0.10\}$, spanning positive (amenity-dominated) and negative (competition-dominated) local effects. We discuss the implications of this calibration for gentrification in Section 6.2.3.

Table C3: Effect of nearby new construction on rents – Main results and robustness

Specification	N	Rents (log rent per sqft)		Supply (log units)	Implied ζ_L
		$ATT(+5)$	Overall ATT	Overall ATT	
<i>Panel A: Main results</i>					
Sample 1: all buildings	22,204	−0.009 (0.006)	−0.010** (0.004)	0.112*** (0.007)	−0.09 [†]
Sample 2: treated + to-be-treated	4,750	0.004 (0.007)	−0.002 (0.005)	0.102*** (0.007)	−0.01
Sample 3: treated only	3,495	−0.010 (0.015)	−0.004 (0.007)	0.101*** (0.007)	−0.04
<i>Panel B. Robustness</i>					
Unit cutoff = 30	5,950	0.009* (0.005)	0.002 (0.004)	0.078*** (0.005)	0.02
Unit cutoff = 100	2,321	−0.027** (0.011)	−0.021** (0.009)	0.169*** (0.017)	−0.12 [†]
1000 ft radius	6,683	0.001 (0.005)	−0.005 (0.004)	0.033*** (0.003)	−0.16
Remove interpolated years	4,750	0.000 (0.008)	−0.001 (0.006)	0.102*** (0.007)	−0.01
At least 2 pre-periods	4,373	0.009 (0.008)	0.006 (0.004)	0.109*** (0.005)	0.05
Commercial < 5%	4,128	0.005 (0.007)	0.003 (0.005)	0.101*** (0.007)	0.03
Only use never-treated controls	4,750	0.005 (0.008)	−0.000 (0.005)	0.100*** (0.007)	−0.00

Notes: This table shows the ATT event-study estimates of the effect of a nearby residential completion on the rent of the building, and its nearby supply in units using the estimating procedure of Callaway and Sant’Anna (2021). N is the number of buildings used for the estimations, ATT(+5) is the event-study coefficient five years after the nearby completion, “Overall ATT” is the average dynamic effect over event times 0 to 7. ζ_L is the ratio of the rent and supply overall ATTs. Panel B provides robustness checks by varying one key assumption listed in the rows. Standard errors computed from a multiplier bootstrap clustered at the building level are presented in parentheses. Estimates followed by ***, **, and * are statistically significant at the 1%, 5%, and 10% levels, respectively. [†] indicates implied ζ_L for which both overall ATT are significant at least at the 10% level.

Robustness. We re-estimate the preferred specification under a battery of alternatives to provide robustness evidence of our null results. Panel B of Table C3 reports the rent and supply effects for each, together with the implied local elasticity ζ_L . The effect of treatment on surrounding units is large, positive, and significant at the one-percent level in every specification, ranging from 0.03 to 0.17. The treatment thus continues to capture a substantial and well-measured supply shock throughout. The rent effect, by contrast, is small and statistically non-significant in six of the seven specifications. Its magnitude is modest, and the sign is not systematic. The one notable exception is when we use larger building completions (those adding at least 100 units), which lower nearby rents by about 2% while raising nearby supply by roughly 17% (an implied ζ_L of -0.12), consistent with the intuition that only unusually large local shocks meaningfully depress surrounding rents. Taken together, the implied local elasticity is confined to a narrow band around zero, $\zeta_L \in [-0.16, 0.05]$, across radii, samples, sample periods, and every specification.

C.3 Parcel-level Data

We begin with the 2023 PLUTO dataset, which contains 857,944 tax lots. We apply a sequence of restrictions to isolate parcels relevant for the counterfactual analysis that we highlight in Table C4. Particularly important:

- **Residential parcels.** We retain lots that are zoned or partially zoned for residential use. To retain parcels eligible for residential use, we keep those with an “R-x” zoning classification in the first through fourth zoning categories. We also retain parcels with observed residential square footage regardless of zoning classification. We remove parcels that we identify in the NOPV to be managed by the NYC Housing Authority (i.e., public housing).
- **Complete and consistent information.** We keep parcels that have recorded built year and FAR in addition to parcels without building. We exclude parcels with extreme lot sizes by retaining parcels larger than 1,250 square feet (half of a standard Manhattan lot) and smaller than 250,000 square feet (roughly one hundred times a standard Manhattan lot).
- **Rental and caprate data availability.** We keep parcels with (i) an ACS-based rental estimate and (ii) an observed CoStar caprate measure, leaving 765,305 parcels in the final dataset.

In total, we keep 89.2% of all parcels of New York. We use the same procedure when constructing the 2015 parcel-level data (89.2% are retained).

Table C4: Parcel-level Cleaning Steps

Filter	Parcels	Share of Total	Loss
PLUTO 2023	857,944	100.00%	
with FAR information	825,703	96.24%	-3.76%
residential or partially residential	801,469	93.42%	-2.93%
not public housing	800,821	93.34%	-0.08%
with acceptable parcel size	777,541	90.63%	-2.91%
with spatial information	776,692	90.53%	-0.11%
with age information	776,353	90.49%	-0.04%
with rental (ACS) data	766,042	89.29%	-1.33%
with return (CoStar) data	765,333	89.21%	-0.09%
with all variables	765,305	89.20%	0.00%

C.4 Rental Data Construction

The model relies on a measure of rent that reflects the potential rent per unit of space, net of the effects of height, obsolescence, and affordable-unit requirements. We use the ACS median gross rent variable (B25064_001) that reports the median contract rent at the block group or census tract level. First, we assign to each parcel either the block group level rent, $Rent_{bg}$, or the tract level rent, $Rent_{tract}$, when the former is unavailable. This procedure preserves geographic heterogeneity while maintaining close to full data coverage.

To recover a rent measure consistent with the model, we adjust the ACS data in several steps. First, because ACS rent is reported per dwelling unit, we normalize it by the median apartment size in the corresponding geography. We compute the median size of residential units using PLUTO records for buildings with reliable information on total residential square

where $R_{i(j)}$ denotes the potential rent assigned to the geography j where parcel i is located, $\tilde{F}_{B,i(j)}$, $\tilde{z}_{B,i(j)}$, and $\tilde{\alpha}_{B,i(j)}$ are the weighted median built density, rental depreciation, and affordable unit share across parcels in geography j , respectively. Weighting by built square footage $F_{B,i} \times L_i$ ensures that the representative building characteristics reflect the stock of space actually inhabited by renters, consistent with the ACS rent measure. Panel A of Figure C4 shows the geographical variation in this constructed variable.

C.5 Returns, Cap Rates, and Rental Growth

We use market-level data from CoStar to construct the return series that underpins the stochastic process for rents. New York City is composed of 37 CoStar-defined submarkets that we manually geocoded. The dataset covers the last twenty years (2004 to 2023). Returns are computed in three steps.

1. **Asset value per square foot.** For each market m and year t , we construct the implied asset value per square foot using market rents and cap rates:

$$\text{ValuePSF}_{m,t} = \frac{\text{RentPSF}_{m,t}}{\text{Caprate}_{m,t}}.$$

2. **Asset and rental returns.** We compute asset returns and rental returns as follows and report the summary statistics across market in Table C5:

$$\text{AssetReturn}_{m,t} = \frac{\text{ValuePSF}_{m,t}}{\text{ValuePSF}_{m,t-1}} - 1, \quad \text{RentalReturn}_{m,t} = \frac{\text{RentPSF}_{m,t}}{\text{ValuePSF}_{m,t-1}}.$$

3. **Total return.** The annual total return in market m is the sum of the two components:

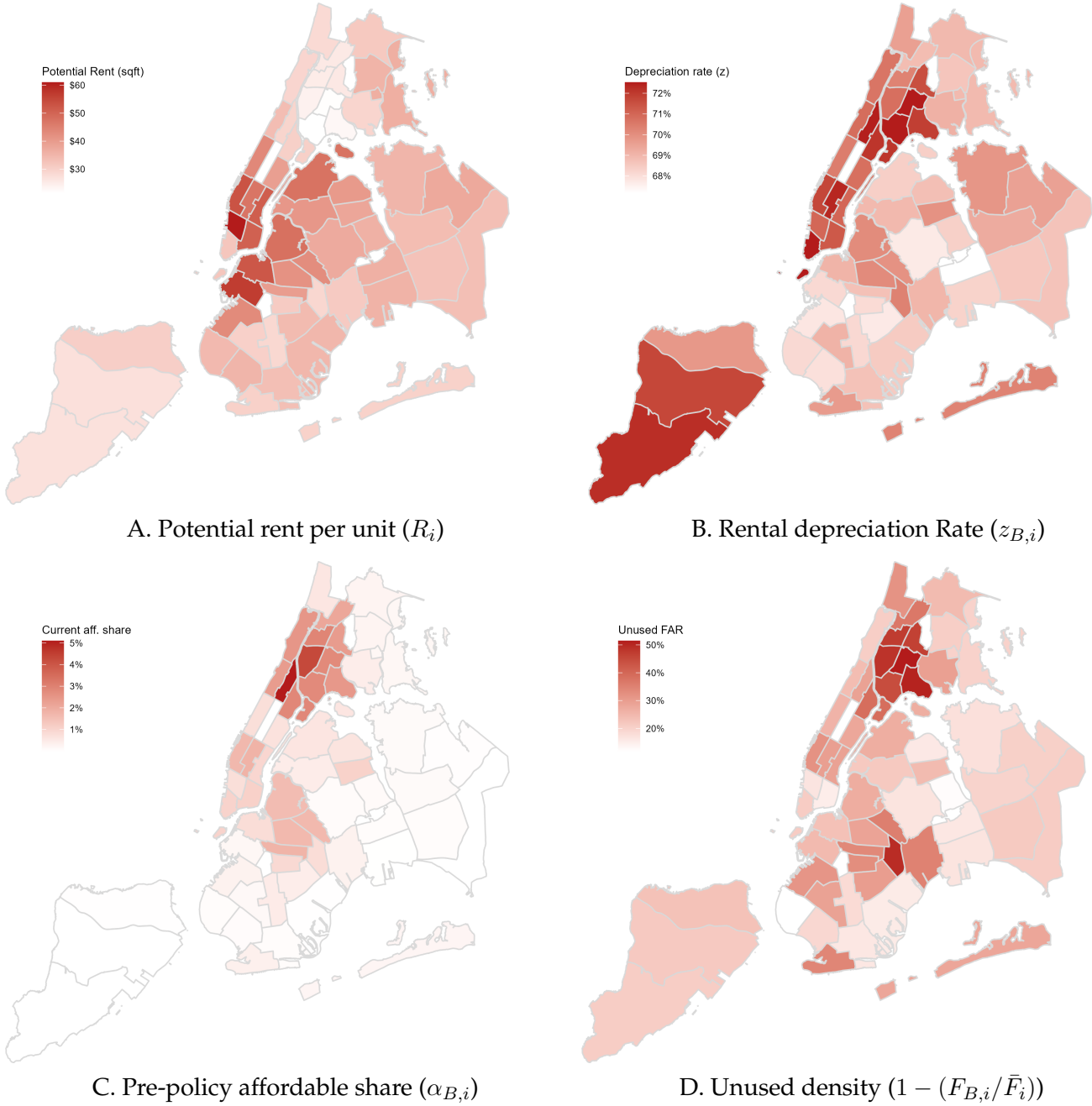
$$\text{TotalReturn}_{m,t} = \text{AssetReturn}_{m,t} + \text{RentalReturn}_{m,t}.$$

Table C5: Costar Market Level Returns

Statistic	N	Mean	St. Dev.	Min	Median	Max
Asset Return (%)	37	2.39	0.58	1.26	2.46	3.64
Rental Return (%)	37	4.85	0.59	4.05	4.85	5.98
Total Return (%)	37	7.23	0.40	6.60	7.26	8.14

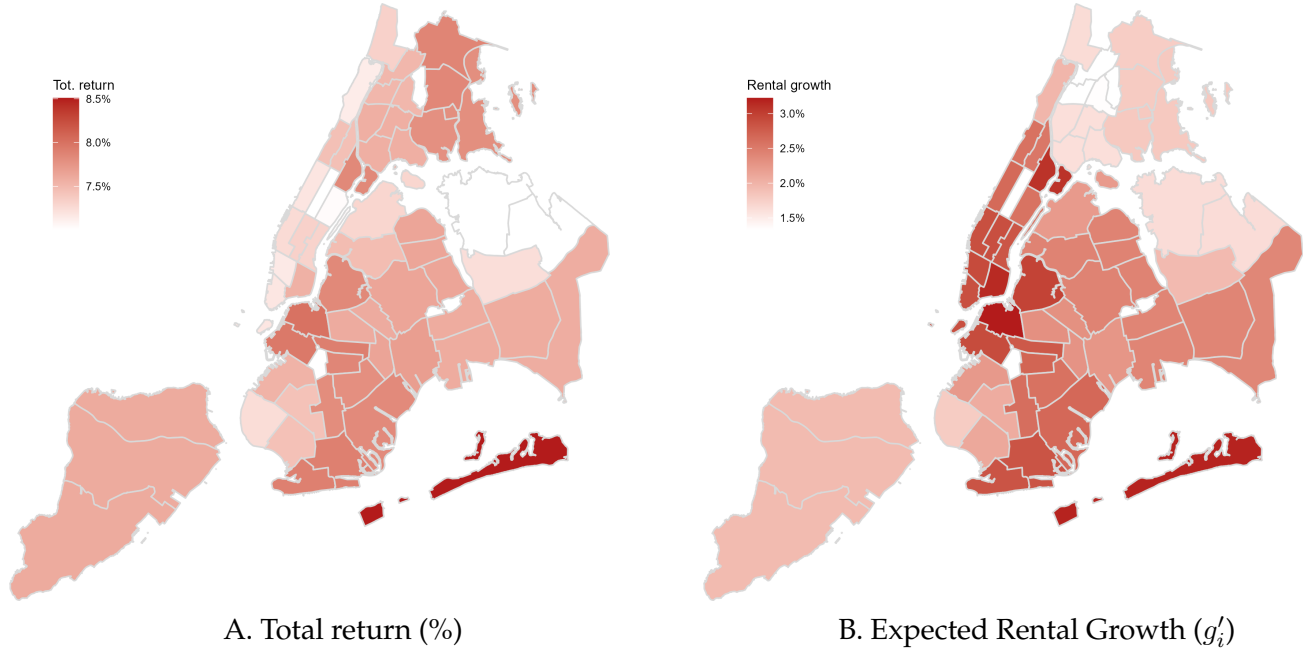
We note that the average total return across the 37 CoStar markets is 7.23%. This value falls within the estimates reported by [Amaral et al. \(2025\)](#) and [Lyons et al. \(2026\)](#) for New York. The advantage of our approach is that it preserves meaningful spatial variation in returns, rental growth, and volatility. Figure C5 shows that expected total returns decline as one moves away from the city center, a pattern consistent with the argument in [Amaral et al. \(2025\)](#) that dense urban locations tend to deliver lower total returns.

Figure C4: Spatial Variations of Key Variables



Notes: These maps show the average across all parcels of each community district of key parameters of the calibration. Panel A shows the potential rent per unit of density R_i that reflects market-rate rents for one non-obsolete unit, obtained by adjusting the annual rent for the average age and size of buildings within each neighborhood. Panel B shows the average discount due to the aging of properties $z_{B,i}$. Panel C shows the average share of buildings with affordable units. Panel D shows the unused density computed as the share of allowed FAR that is currently not built. All statistics are computed with 2023 data.

Figure C5: Total Housing Return and Expected Rent Growth



Notes: These maps show the average total return (asset and rental returns) in Panel A, and the 2023 estimated rental growth (total return minus 2023 cap rate) in Panel B computed across all parcels of the community districts. These statistics are drawn from Costar market data from 2004 to 2023.

C.6 Aggregating Parcel-Level Outcomes

We aggregate outcomes at the city level and for each neighborhood j . The two main outcomes are the expected number of net additional housing units, $\text{UnitNet}_j(\mathcal{P}, h)$, and the average expected rent per housing unit, $\text{Rent}_j(\mathcal{P}, h)$. We also compute related measures, including: the expected total supply $S_j(\mathcal{P}, h)$ in square foot, the present value of all parcels $PV(\mathcal{P})$, the present value of the property taxes $PV^{\text{Tax}}(\mathcal{P})$, and the average share of option value $\text{Option}(\mathcal{P})$. Aggregated outcomes are defined from the parcel-level outcomes as:

$$S_j(\mathcal{P}, h) = \sum_{i \in j} ((1 - \mathbf{P}_{ih}^*(\mathcal{P}))F_{B,i}L_i + \mathbf{P}_{ih}^*(\mathcal{P})F_i^*(\mathcal{P})L_i) \quad (\text{C.2})$$

$$\text{UnitConst}_j(\mathcal{P}, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i F_i^*(\mathcal{P})}{\text{UnitSize}_i} \quad (\text{C.3})$$

$$\text{UnitNet}_j(\mathcal{P}, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i (F_i^*(\mathcal{P}) - F_{B,i})}{\text{UnitSize}_i} \quad (\text{C.4})$$

$$\text{AffNet}_j(\mathcal{P}, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i (F_i^*(\mathcal{P}) - F_{B,i}) \cdot \max(\alpha_{B,i}, \alpha)}{\text{UnitSize}_i} \quad (\text{C.5})$$

$$\text{Exp.Unit}_j(\mathcal{P}, h) = \sum_{i \in j} \left(\mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i (F_i^*(\mathcal{P}))}{\text{UnitSize}_i} + (1 - \mathbf{P}_{ih}^*(\mathcal{P})) \frac{L_i F_{B,i}}{\text{UnitSize}_i} \right) \quad (\text{C.6})$$

$$\text{Rent}_j(\mathcal{P}, h) = \frac{\sum_{i \in j} \text{RentInc}_{ih}(\mathcal{P})}{\text{Exp.Unit}_j(\mathcal{P}, h)} \quad (\text{C.7})$$

$$\text{RentPerSqft}_j(\mathcal{P}, h) = \frac{\sum_{i \in j} \text{RentInc}_{ih}(\mathcal{P})}{S_j(\mathcal{P}, h)} \quad (\text{C.8})$$

$$PV_j(\mathcal{P}) = \sum_{i \in j} \pi_i(\mathcal{P}; \Theta) \quad (\text{C.9})$$

$$\text{Option}_j(\mathcal{P}) = \frac{\sum_{i \in j} \pi_i^{\text{option}}(\mathcal{P})}{PV_j(\mathcal{P})} \quad (\text{C.10})$$

$$PV_j^{\text{Tax}}(\mathcal{P}) = \sum_{i \in j} \pi_i^{\text{Tax}}(\mathcal{P}). \quad (\text{C.11})$$

where $\pi_i^{\text{option}}(\mathcal{P})$ is the redevelopment option value computed by the difference between the property value π_i and its counterpart when no building can be rebuilt (i.e., $k_c = \infty$). The present value of the taxes for a parcel is computed as the present value of the tax components on the existing and new buildings, accounting for possible tax exemption:

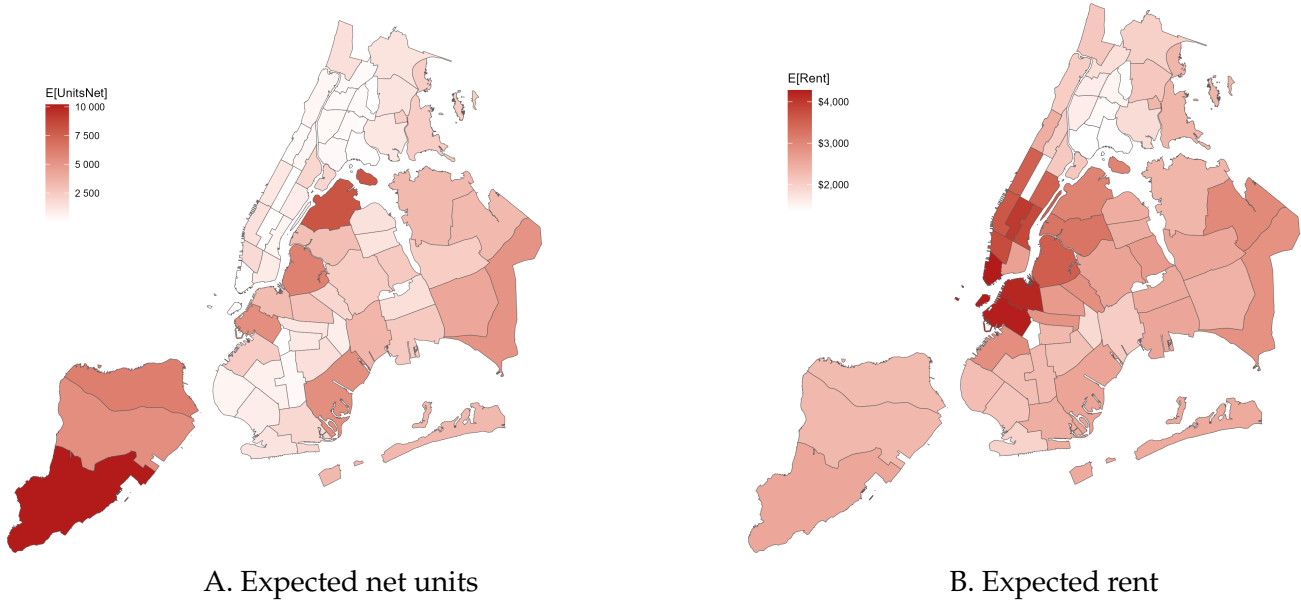
$$\begin{aligned} \pi_i^{\text{Tax}}(\mathcal{P}) = & \underbrace{\Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i) \frac{\psi}{r - \delta_i} \left[R_i - R_i^{e*} \left(\frac{R_i^{e*}}{R_i} \right)^\xi \right]}_{\text{Tax on existing property}} \\ & + \underbrace{\psi \Gamma(F_i^*, 1, \alpha, L_i) \frac{R_i^{e*}}{r - \delta_i} e^{-(r - \delta_i)(d + \tau)} \left(\frac{R_i^{e*}}{R_i} \right)^\xi}_{\text{Property taxes after } \tau \text{ years of exemption}} \end{aligned}$$

Table C6: Partial Correlations Between Inputs and Outcomes of Baseline Model

	<i>Baseline Outcomes:</i>				
	Prob	F opt	log(PV)	Option	PV/F
	<i>All parcels</i>				<i>Built parcels</i>
	(1)	(2)	(3)	(4)	(5)
Rent potential (R_i)	0.284	-0.015	0.462	0.147	0.788
Rent growth (g_i)	0.055	0.020	0.036	0.028	0.096
Rental depreciation index ($z_{B,i}$)	-0.056	-0.017	0.270	-0.015	0.311
Current density ($F_{B,i}$)	-0.139	0.823	0.654	-0.076	0.084
Non-built parcel ($\mathbf{1}_{F_{B,i}=0}$)	0.689	0.001	-0.376	0.972	

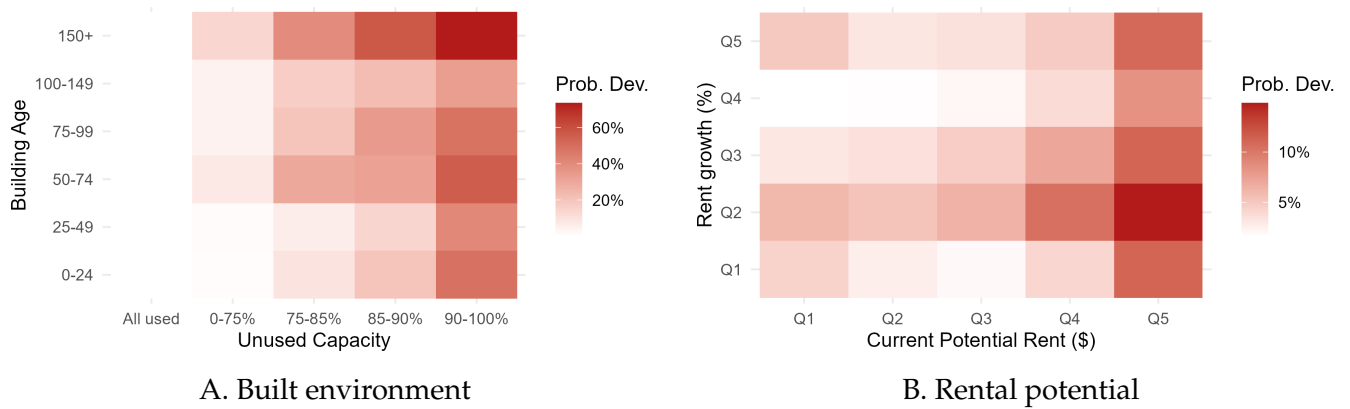
Notes: This table reports the partial correlations between key inputs (rows) and outcomes (columns) of our baseline model results. The partial correlations are conditional on community districts and all other inputs listed in the rows. It uses model outcomes over a 10-year horizon for the 765,305 parcels under the baseline calibration described in Tables 2 and 1. In Column (1), the outcome is the estimated probability of redevelopment decision $\mathbf{P}_i(\mathcal{P}_0; \Theta)$. In Column (2), the optimal density chosen endogenously to maximize property value ($F^*(\mathcal{P}_0; \Theta)$). In Column (3), the log of the present value of the parcel $\pi_i(\mathcal{P}_0; \Theta)$. In Column (4), the option value share of total value $\tilde{\pi}_i(\mathcal{P}_0; \Theta)/\pi_i(\mathcal{P}_0; \Theta)$. In Column (5), the present value per square foot of built units $\pi_i(\mathcal{P}_0; \Theta)/F_i L_i$.

Figure C6: Baseline - Market-level Aggregated Outcomes



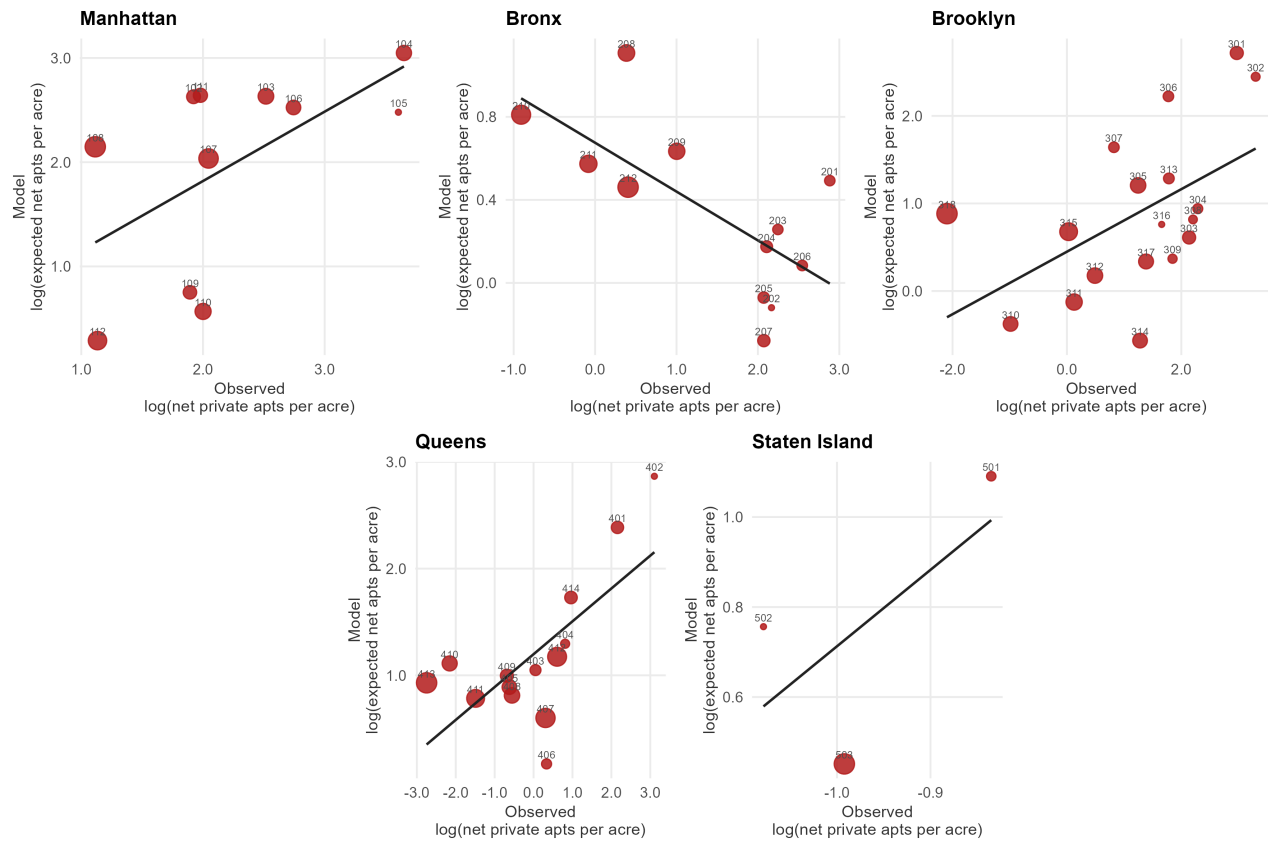
Notes: These maps show the market-level aggregated outcomes of our baseline simulation as described in Table 1. Panel A shows the expected net units addition and Panel B shows the expected rents per unit.

Figure C7: Parcel Heterogeneity and Determinants of Probability of Development



Notes: These panels show the unconditional average probability of development computed for each grid cell. Panel A splits cells into deciles based on the current density ($F_{B,i}$) and the current age of the building ($z_{B,i}$). Panel B splits cells into deciles based on the potential rent (R_i) and rental growth (g_i) using the model baseline calibration with parameters described in Table 1.

Figure C8: Expected and realized construction by borough



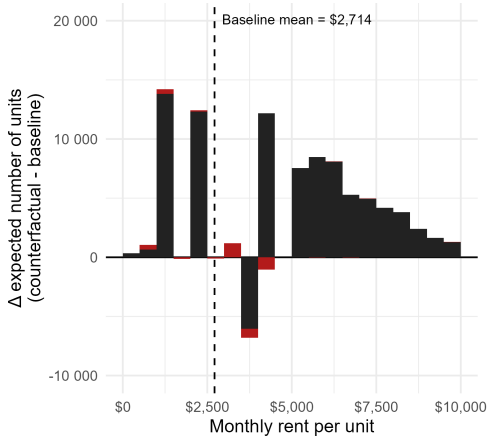
Notes: This scatter plot shows, for each borough, the log number of housing units constructed in each community district between 2016 and 2025, scaled by community district residential acreage, against the corresponding model-implied construction outcomes based on the 2015 calibration.

D Counterfactual Results – Additional Tables & Figures

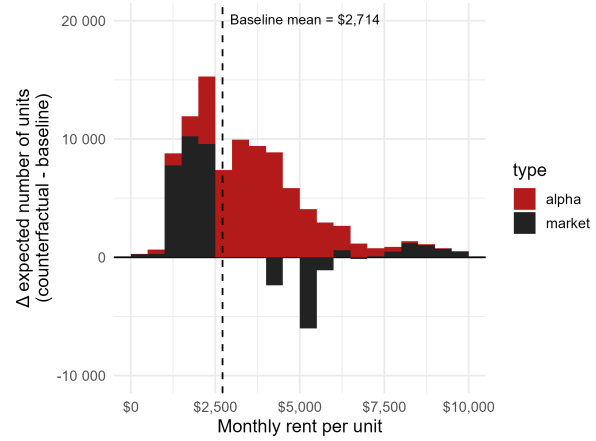
In this Appendix, we provide additional detail and robustness for the policy counterfactuals presented in Sections 6 and 7. We begin by examining the full distribution of rent changes generated by each policy, showing directly how redevelopment replaces lower-rent housing with higher-rent housing. We then examine the robustness of these counterfactuals to alternative elasticity assumptions, voluntary policy adoption, targeted policy interventions, and changes in average housing-unit size.

D.1 Distribution of rent

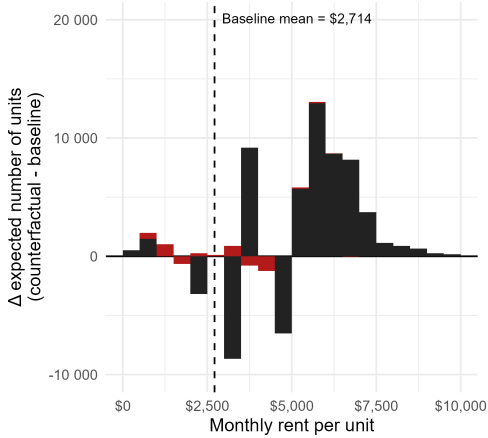
Figure D1: Counterfactuals — Change in Distribution of Rent



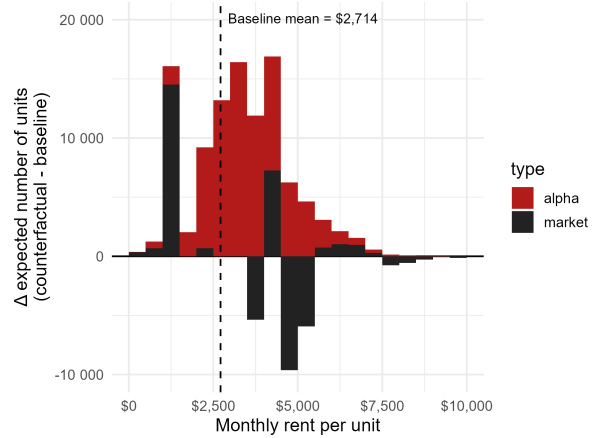
A. Upzoning ($\phi = .50$)



B. Density incentive zoning ($\alpha = 0.3$ & $\phi = .50$)



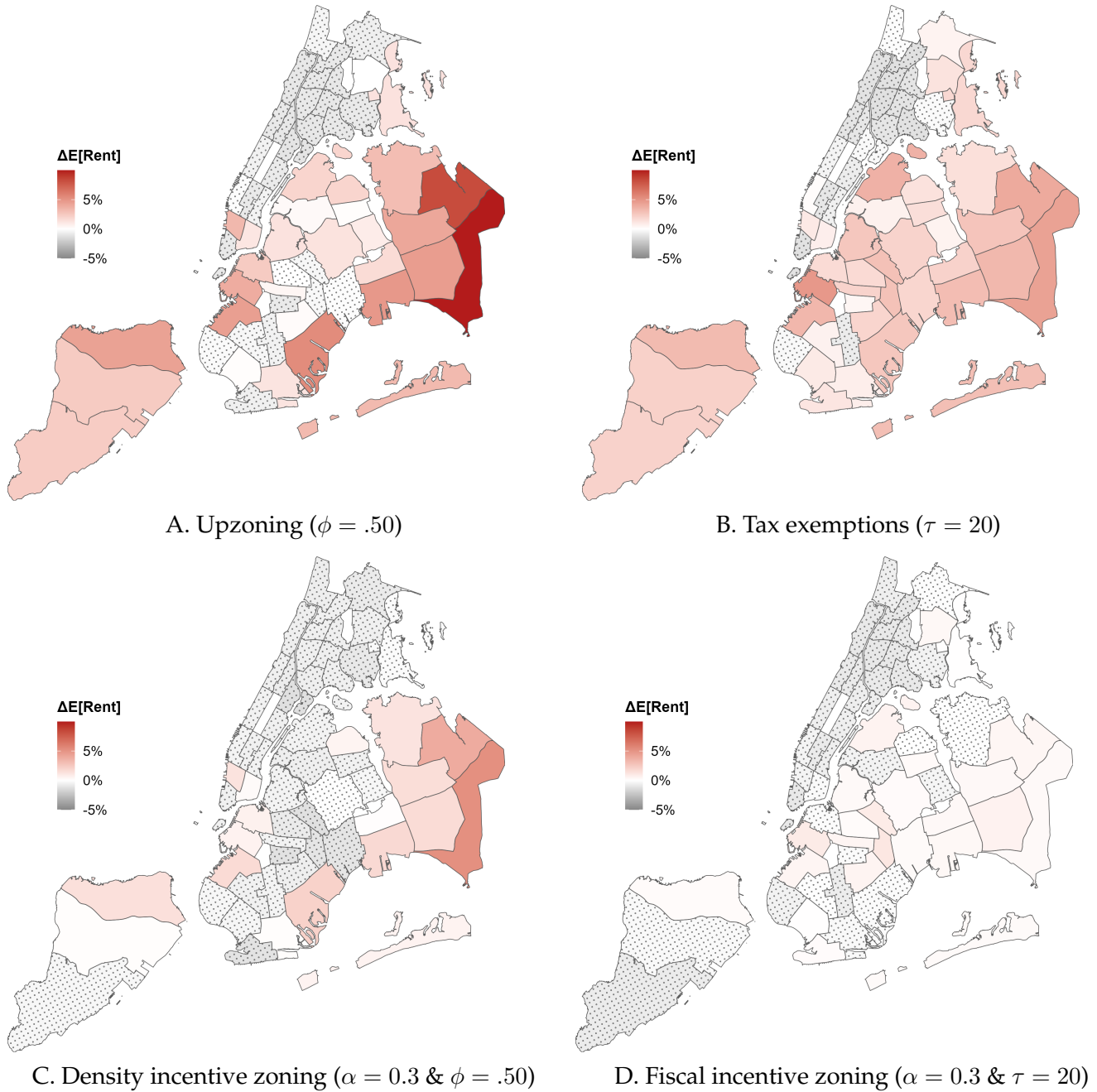
C. Tax exemption ($\tau = 20$)



D. Fiscal incentive zoning ($\alpha = 0.30$ & $\tau = 20$)

Notes: The histograms report the change in the expected number of apartments in each \$500 rent bin under the counterfactual policy relative to the baseline, evaluated at a 10-year horizon. Bars above (below) zero indicate rent bins in which the expected number of units increases (decreases). Changes are decomposed into market-rate units (black bars) and units subject to affordability mandates (red bars).

Figure D2: Counterfactuals — Change in Expected Average Rent

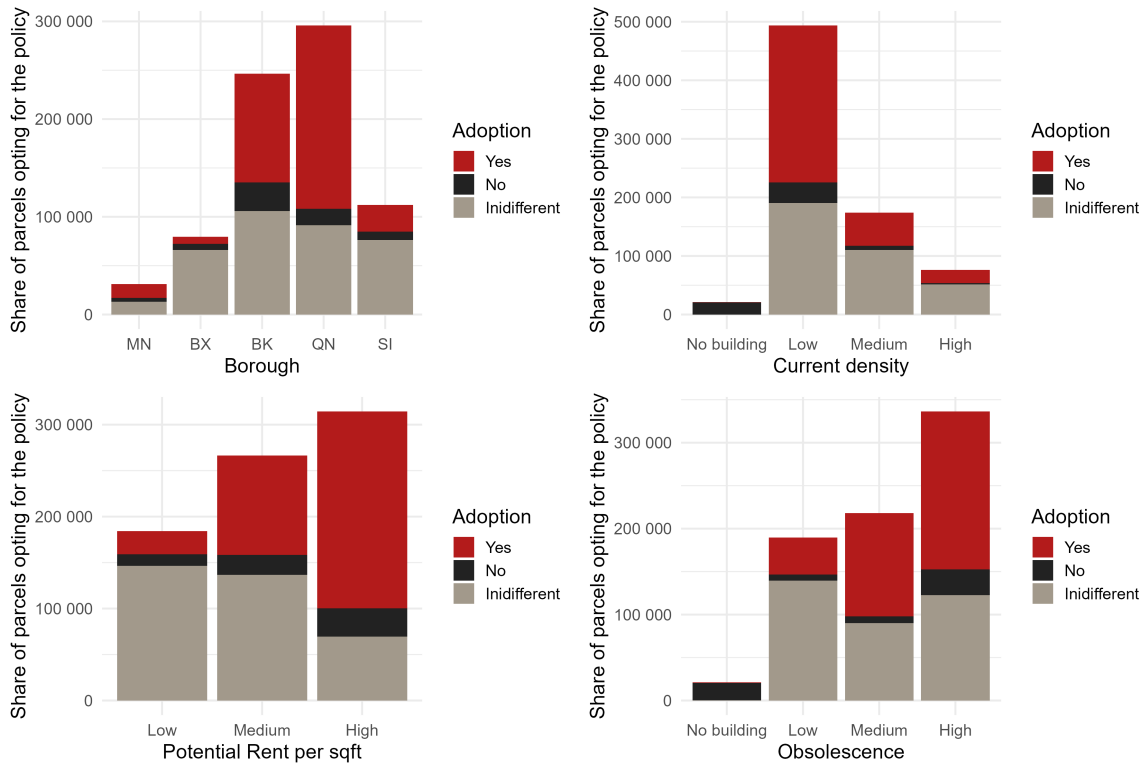


Notes: These maps show the percentage change in average expected rent per unit for different counterfactual policy analyses. It shows for each community district the change compared to the baseline outcomes with horizon 10-year.

D.2 Voluntary Adoption of Policies

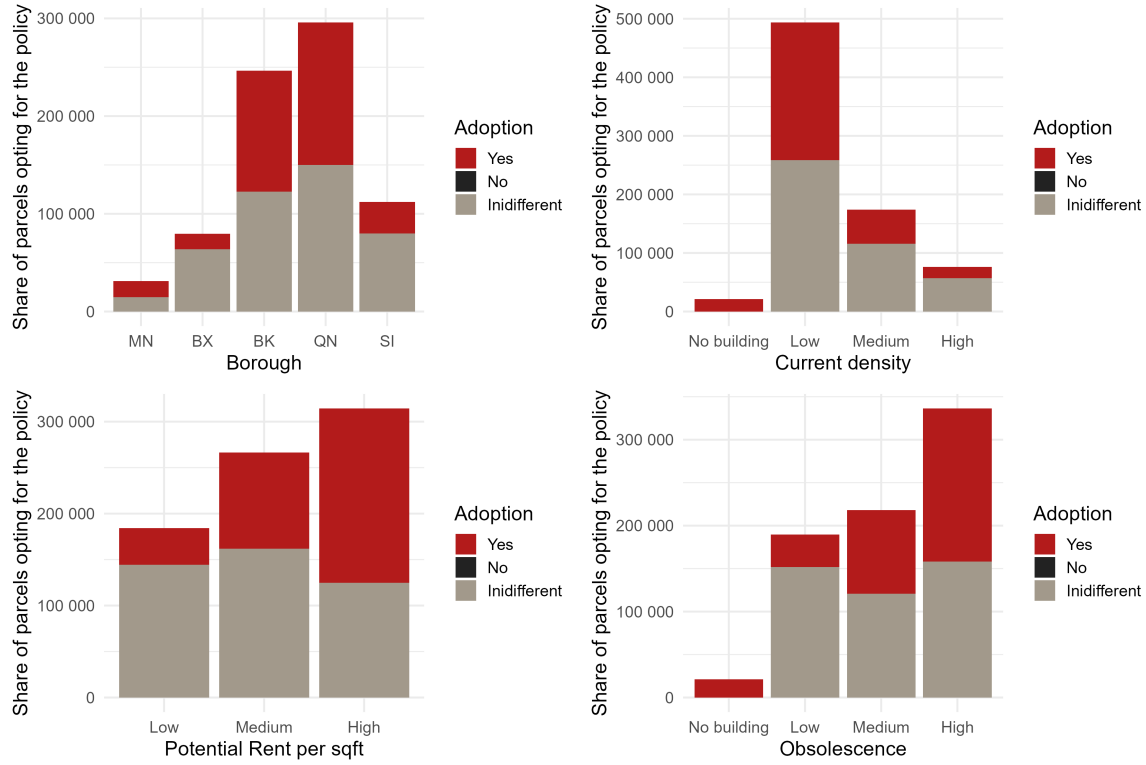
Table D1 reports the share of parcels that would opt into each policy. The model predicts that approximately 45% of parcels adopt either the voluntary Density Incentive Zoning or the Fiscal Incentive Zoning. About 12% of adopters differ across these two bundles, underscoring the importance of parcel-level heterogeneity in the adoption decision. When density and fiscal incentives are combined, adoption rises sharply to 72.1%, with 26.7% of parcels switching from non-adoption to adoption once the fiscal incentive is added to the density incentive zoning. Figures D3 and D4 decompose adoption rates by location and building characteristics. Participation is systematically higher among parcels with low current density, high potential rent, and those located in Queens and Brooklyn.

Figure D3: Voluntary Adoption of Density Incentive Policy



Notes: These bar graphs show the share of parcels that opt-in ($\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$), opt-out of the policy $\mathcal{P}$, or are indifferent under the policy counterfactual that combine density bonuses ($\phi = 0.50$), and an affordable mandate ($\alpha = .30$), compared to the status-quo policies.

Figure D4: Voluntary Adoption of Fiscal Incentive Policy



Notes: These bar graphs show the share of parcels that opt-in ($\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$), opt-out of the policy $\mathcal{P}$, or are indifferent under the policy counterfactual that combine tax exemptions ($\tau = 20$), and an affordable mandate ($\alpha = .30$), compared to the status-quo policies.

Table D1: Share of Parcels that Opt-in Voluntary Policy Interventions

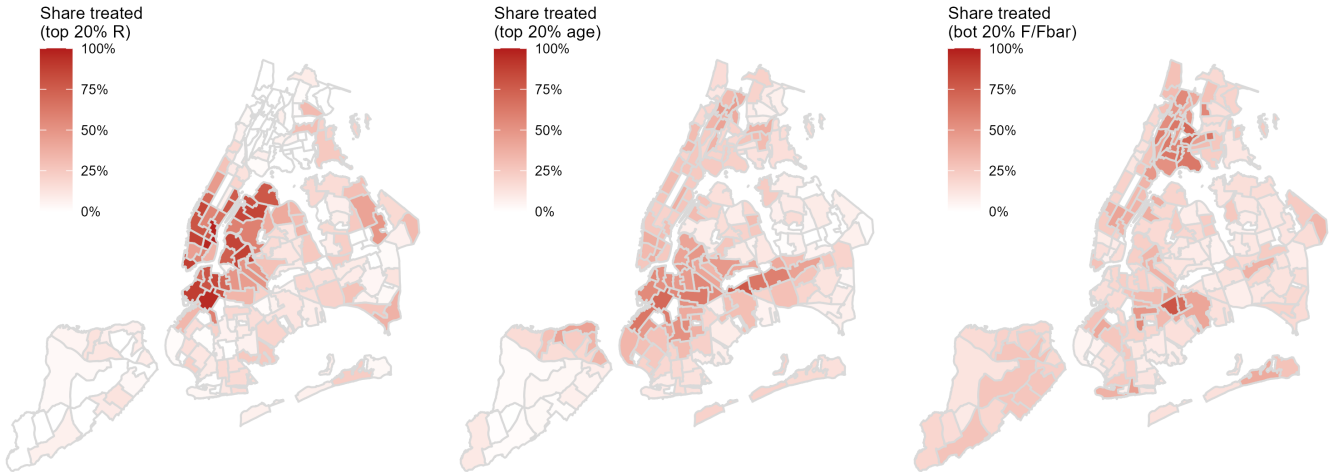
	Number of Parcels	Share (%)
All Parcels	765,305	100
(7) Density Incentive Zoning ($\alpha = 0.30$ & $\phi = 0.50$)	347,845	45.5
but not in fiscal IZ	99,299	13.0
(8) Fiscal Incentive Zoning ($\alpha = 0.30$ & $\tau = 20$)	334,315	43.7
but not in density IZ	85,769	11.2
(9) All Policies	551,956	72.1
but not in density IZ (i.e., switchers with τ)	204,111	26.7
but not in fiscal IZ (i.e., switchers with ϕ)	219,215	28.6

Notes: This table reports the number and the share of parcels that would opt-in the policy bundle $\mathcal{P}$ under voluntary adoptions. These parcels are defined as those that are better off under the new policy as in the baseline policy ($\pi_i(\mathcal{P}) \geq \pi_i(\mathcal{P}_0)$).

D.3 Targeted Counterfactuals

In this appendix, we document whether housing policies become more effective when targeted toward parcels with different characteristics. Because the model predicts that different policy instruments affect redevelopment through different economic margins, they should be most effective when applied to parcels where those margins are most relevant. The targeted counterfactuals therefore provide a mechanism check on the model by testing whether these predicted comparative statics are consistent in the policy simulations. We examine these predictions by targeting policies to three groups of parcels: (1) parcels with high underlying demand (the highest 20% R_i), (2) parcels with obsolete buildings (the lowest 20% $z_{B,i}$), and (3) parcels with substantial unused development capacity (the lowest $F_i/\bar{F}$). Figure D5 shows the location of those targeted parcels. Those parcels are not the same. When focusing on high-rent potential parcels, we target mostly parcels in Manhattan, western Brooklyn, and Queens. Most obsolete buildings are located in Brooklyn, and high-unused-capacity parcels are located mostly in the Bronx. The lowest overlap is between the first and last groups, since it is rare for parcels to have unused capacity in high-demand areas.

Figure D5: Targeted Parcels in Targeted Counterfactuals



We reconduct our two unilateral policy counterfactuals on targeted parcels and present the results in Table D2. Targeting the 20% most valuable parcels delivers 63% of the citywide residential floor-space production under upzoning but only 44% under the tax exemption policy. The pattern reverses for the least-utilized parcels. Targeting those yields 79% of citywide production under tax exemption but only 53% under upzoning. This asymmetry reflects the mechanics of each instrument. Density bonuses are most effective where land values are high enough to make additional floor area profitable, whereas tax exemptions operate on the margin of development timing and are therefore more powerful on underbuilt parcels. Targeting obsolete parcels, by contrast, produces limited outcomes across all three policies. Although targeting substantially changes housing production and aggregate property value, it has comparatively little effect on the breadth of policy incidence, as the share of parcels experiencing property-value losses remains remarkably stable across targeting strategies.

Under all counterfactuals targeting high-value or obsolete parcels, expected citywide rents

Table D2: Targeted Counterfactual Results

	Built Environment			Rent outcomes		Costs of Interventions		
	$\Delta S(\mathcal{P}, h)$ (% chg.)	$\Delta \text{NetUnit}(\mathcal{P})$ (# of units)	$\Delta \text{Aff. Unit}(\mathcal{P})$ (# of units)	$\Delta \text{Rent}(\mathcal{P})$ (% chg.)	$\Delta \text{Shr. Q1}$ (pp)	$\Delta PV^{\text{Tax}}(\mathcal{P})$ (% chg.)	$\Delta PV(\mathcal{P})$ (% chg.)	Shr. PV-losers (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(0)	Baseline under $\mathcal{P}_0$ (Numbers in level)			Baseline Outcomes				
	3.70 Bn.	126, 215	449	\$2, 714.18	20.00%	401.69 Bn.	1, 557.34 Bn.	-
(1)	Upzoning policy — $\phi = 50\%$							
(1a)	1.37	55, 008	61	1.31	0.51	-1.04	-1.74	98.12
(1b)	0.87	38, 159	82	1.19	0.29	-0.42	-1.08	98.08
(1c)	0.38	16, 918	15	0.59	0.12	-0.29	-0.49	98.85
	0.72	27, 490	-12	0.23	0.28	-0.47	-0.91	98.10
(2)	Tax exemption — $\tau = 20$							
(2a)	1.92	77, 877	484	1.21	0.81	-4.55	-1.84	92.78
(2b)	0.86	40, 341	506	1.28	0.27	-2.48	-0.69	95.82
(2c)	0.52	22, 501	16	0.66	0.23	-1.10	-0.55	97.33
	1.51	60, 827	443	0.28	0.63	-3.82	-1.37	93.29

Notes: Row (0) reports baseline outcomes under the no-policy environment $\mathcal{P}_0$ in levels. Rows (1)–(2) report changes relative to the baseline. Each row reports outcomes under alternative regulatory environments $\mathcal{P}$, all measured over a 10-year horizon. Within each policy panel, All parcels applies the policy citywide; the remaining rows restrict the mandate and incentives to the top 20% most expensive parcels (by rent potential), the top 20% most obsolete parcels (by age), and the bottom 20% least utilized parcels (by $F_0/\bar{F}$), respectively, with no policy applied elsewhere. Column (1) reports the percentage change in total expected supply in square feet. Column (2) reports the change in expected housing units constructed net of demolitions. Column (3) shows the change in affordable units out of this net change. Column (4) reports the percentage change in average expected citywide rent per housing unit. Column (5) depicts the change in percentage points in the share of units falling within the first quintile of the baseline rent distribution. Columns (6) and (7) report the percentage change in the present value of property tax revenues and total residential asset values, respectively. Column (8) shows the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$).

rise, consistent with the redevelopment margin effect dominating the supply effect. Targeting the least-utilized parcels, however, produces the lowest rent rise. This provides perhaps one of the strongest internal validations of the paper’s replacement effect. Because least-utilized parcels include many vacant or near-vacant lots, redevelopment there involves little demolition of existing housing. Development therefore primarily expands rather than replaces the housing stock, substantially attenuating the replacement effect. In dense urban environments where new construction requires tearing down existing structures, supply expansions exert an inherent upward pressure on rents that is largely absent when development occurs on empty land.

D.4 Size of Units and Main Results

In this Appendix, we examine whether the paper’s central replacement-effect result depends on expressing housing supply in housing units rather than floor area. Because redevelopment decisions, construction costs, and spatial spillovers are all modeled per square foot, housing units enter only as a reporting convention. Nevertheless, expressing outcomes in housing units could potentially affect the interpretation of policy counterfactuals if unit sizes systematically change over time.

Robustness to the square-footage basis. We first address the normalization concern directly by replicating the rent decomposition exercise in square-foot terms rather than housing units. Figure D6 shows that the same pattern emerges, if anything more strongly. The replacement effect dominates the supply effect across all six policy scenarios. This confirms that the dominance of the replacement effect is not an artifact of the unit-size reporting assumption but reflects the underlying mechanism linking redevelopment decisions to rent outcomes.

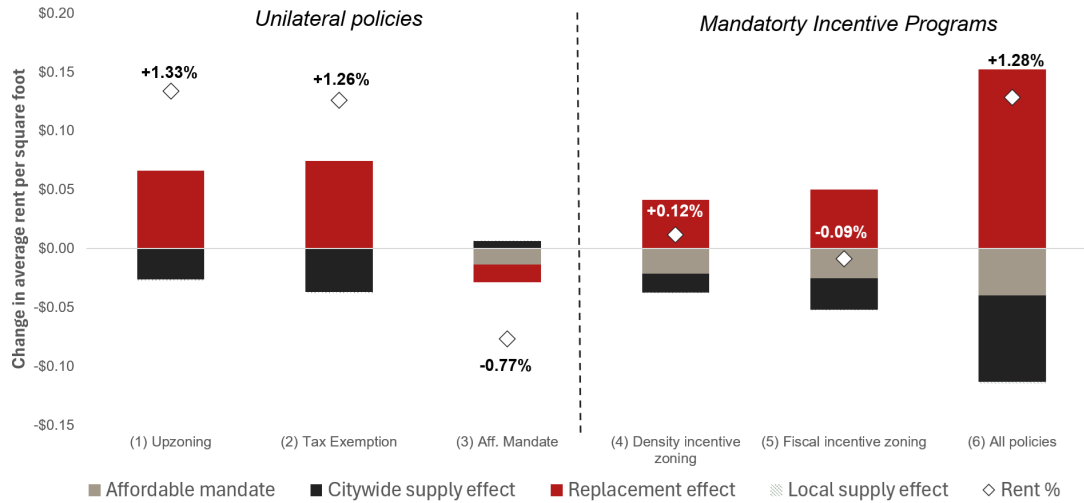
Result Sensitivity to change in unit size. To provide context, we first plot in Figure D7 the rolling weighted average of residential unit size by year of construction, using PLUTO data. After several decades of growth, unit sizes have declined substantially since the early 2000s. Over the last decade, newly built units were on average 7.0% smaller than the existing stock. Because our baseline results hold unit size fixed, this structural shift implies that our housing production figures may be biased downward if smaller units continue to be built over the horizon h .

To assess whether this historical decline could materially affect our conclusions, we introduce a unit-size-adjustment parameter β , defined as the ratio of new unit size to the existing stock’s unit size. Although Figure D7 documents an approximately 7% decline in average unit size over the past decade, we examine a substantially wider range of values, $\beta \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$, to test whether our conclusions remain robust under changes well beyond those observed historically. With this adjustment, the expected number of units on parcel i under policy $\mathcal{P}$ becomes

$$\text{Unit}_i(\mathcal{P}, h; \beta) = \underbrace{\frac{(1 - \mathbf{P}_{ih}^*(\mathcal{P}))F_{B,i}L_i}{\text{UnitSize}_i}}_{\text{Exp. units with no development}} + \underbrace{\frac{\mathbf{P}_{ih}^*(\mathcal{P})F_i^*(\mathcal{P})L_i}{\beta \text{UnitSize}_i}}_{\text{Exp. units with development}}.$$

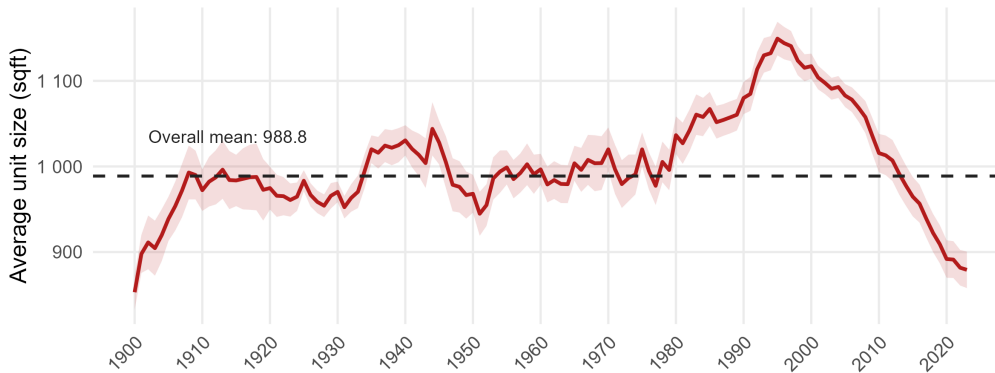
so that $\beta < 1$ implies smaller new units and hence more units per square foot of new construction. We also adjust the supply feedback so that rent growth responds to changes in the number of housing units rather than floor area. Because rental income is unaffected by the normalization, rent per unit mechanically decreases with β to the extent that there is positive housing production

Figure D6: Counterfactual Results — Expected Rents in Square Footage Basis



Notes: The points report the average expected rent per square foot change under each counterfactual policy relative to the baseline at a 10-year horizon. The stacked bars decompose this change into four components. The Affordable Mandate component isolates the direct effect of the affordable share α on rental income, holding expected construction activity fixed. The Supply effect captures the change in expected rents attributable to additional citywide housing supply. The Amenity effect is computed analogously and reflects the impact of local supply increases on neighborhood rents. The Replacement effect is defined as the residual component and captures rent changes associated with replacing older, depreciated structures with new construction, including changes in redevelopment probabilities and endogenous updates to expected rental growth. Policies (1)–(3) correspond to unilateral interventions—upzoning, tax exemption, and affordability mandate—while policies (4)–(6) combine these instruments.

Figure D7: Weighted Average Size of Constructed Unit

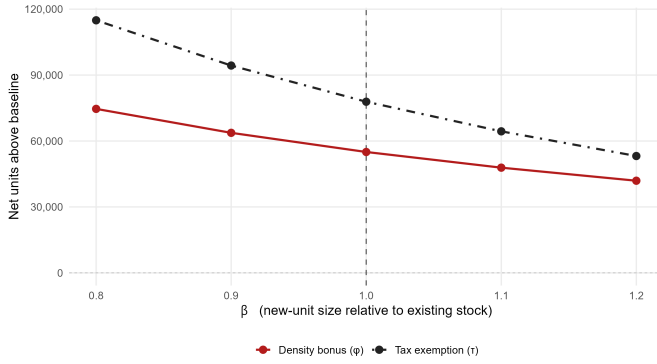


Notes: This figure plots the rolling average of residential unit size (in square feet) by year of construction using information from PLUTO. The rolling averages series uses a ± 5 -year window. The shaded ribbons represent 95% confidence intervals around the rolling mean.

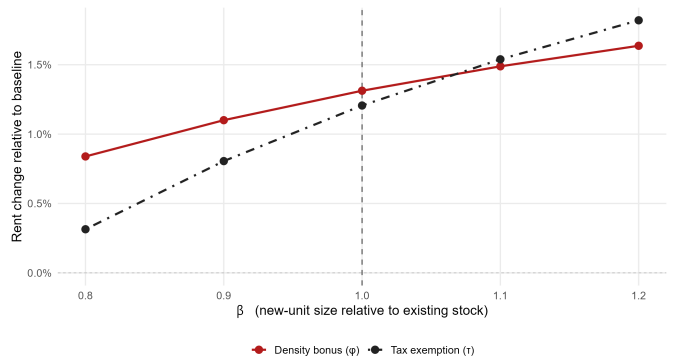
($F_i^* P_i^* > 0$). Figure D8 reports how net unit production (Panel A) and average expected rent per unit (Panel B) vary with β for upzoning ($\phi > 0$) and tax exemption ($\tau > 0$) counterfactuals.

Rents rise under both policies for all values of β considered. Thus, only a drastic and sustained reduction in unit size would be sufficient to generate rent decreases alongside increases in supply because smaller units spread the rental value created by redevelopment across a larger

Figure D8: Effect of new apartment unit size on aggregate outcomes



A. Change in housing unit



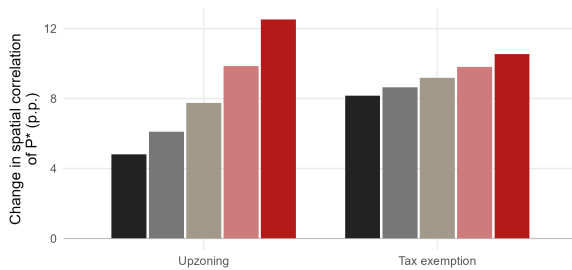
B. Change in expected average rent

Notes: These figures show the sensitivity of the main policy results to the assumption governing the size of newly built units relative to the existing stock, captured by the parameter β . Panel A reports the change in expected net units constructed relative to the no-policy baseline, for the density bonus (ϕ , solid red line) and the tax exemption (τ , dash-dot black line), as a function of β . Panel B reports the corresponding change in expected rent per unit relative to the no-policy baseline.

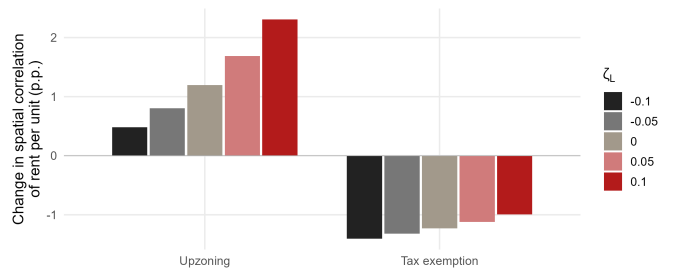
number of housing units, diluting the replacement premium measured on a per-unit basis. Even the substantially more extreme reductions in unit size considered here are insufficient to reverse the paper's central finding that increased construction raises average rents.

D.5 Other Supporting Figures and Tables

Figure D9: Local Spillover Strength and the Spatial Correlation of Constructions and Rents



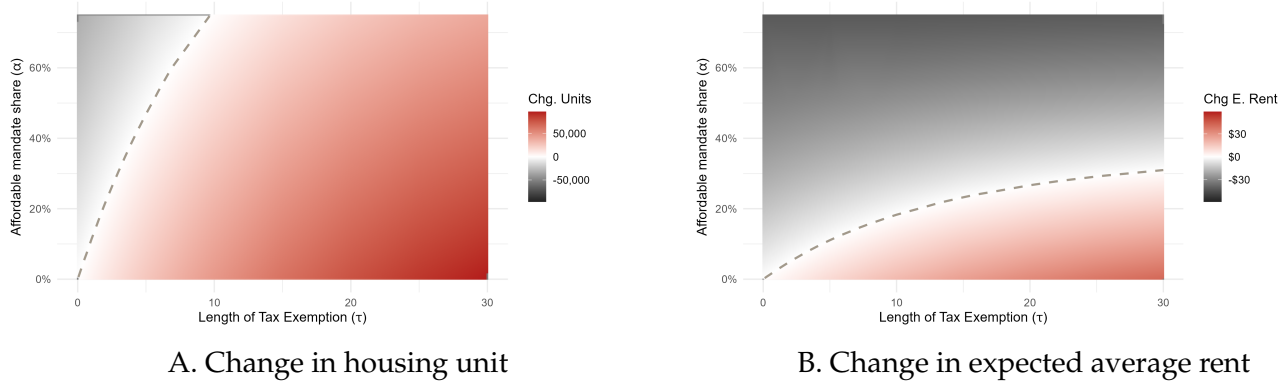
A. Probability of Construction



B. Rents per Units

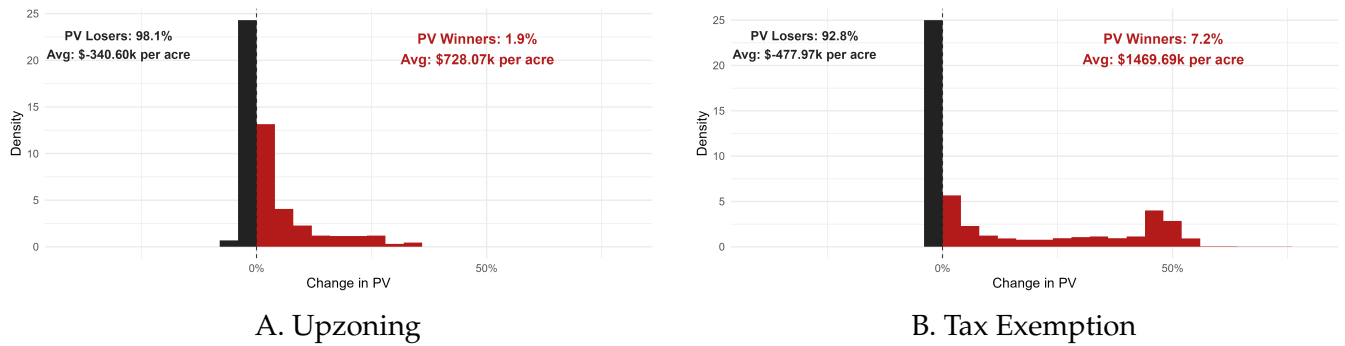
Notes: The bars report changes in the spatial correlation of construction probabilities (Panel A) and rents per unit (Panel B) by varying the local elasticity of rents with respect to new supply (ζ_L). Spatial correlations are measured using Moran's I , computed with inverse-distance weights over all parcels within 500 feet of each target parcel. All values are expressed as deviations from the baseline spatial correlation at the 10-year horizon.

Figure D10: Continuous Policy Outcomes - Fiscal Incentive Zoning



Notes: These heatmaps show the outcomes of a Fiscal Incentive Zoning program under continuous values of τ (x-axis) and α (y-axis). Changes are computed over the outcomes of the status-quo under policy $\mathcal{P}_0$. Panel A shows the net change in the number of housing units $E[\text{AptNet}|\mathcal{P}]$. Panel B shows the change in average expected rents per unit. The dotted lines show outcomes that equate the value under the status-quo policy $\mathcal{P}_0$.

Figure D11: Distribution of change of present values



Notes: These bar graphs show the distribution of the percentage change in parcel present value $\Delta\pi_i(\mathcal{P})$ under the upzoning policy (Panel A) and the tax exemption policy (Panel B), relative to the baseline $\mathcal{P}_0$. Each bar represents the density of parcels experiencing a given percentage change in present value. Red bars denote parcels whose present value increases under the policy ($\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$), while black bars denote parcels whose present value decreases ($\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$). Average changes are expressed per acre of land.

Table D3: Counterfactual Results

	Built Environment			Rent outcomes		Costs of Interventions		
	$\Delta S(h; \mathcal{P})$ (% chg.)	$\Delta \text{NetUnit}(\mathcal{P})$ (# of units)	$\Delta \text{Aff. Unit}(\mathcal{P})$ (# of units)	$\Delta \text{Rent}(\mathcal{P})$ (% chg.)	$\Delta \text{Shr. Q1}$ (pp)	$\Delta PV^{\text{Tax}}(\mathcal{P})$ (% chg.)	$\Delta PV(\mathcal{P})$ (% chg.)	Shr. PV-losers (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(0)	3.70 Bn.	126, 215	<u>Baseline Outcomes</u> 449	\$2, 714.18	20.00%	401.69 Bn.	1, 557.34 Bn.	-
			<u>Unilateral Policy</u>					
(1)	1.37	55, 008	61	1.31	0.51	-1.04	-1.74	98.12
(2)	1.92	77, 877	484	1.21	0.81	-4.55	-1.84	92.78
(3)	-0.33	-13, 441	42, 286	-0.76	-0.10	-0.07	0.34	5.46
			<u>Mandatory Incentive Programs</u>					
(4)	0.83	33, 283	61, 992	0.11	0.33	-1.04	-1.17	98.78
(5)	1.39	56, 846	79, 613	-0.13	0.62	-3.98	-1.41	93.74
(6)	3.80	152, 897	121, 681	1.22	1.45	-6.69	-4.23	93.89
			<u>Voluntary Incentive Programs</u>					
(7)	0.89	35, 728	30, 739	0.68	0.12	-0.34	-0.68	98.12
(8)	1.46	59, 561	81, 057	0.16	0.41	-3.54	-0.95	93.09
(9)	4.11	165, 277	127, 819	2.03	1.01	-5.72	-3.23	93.09
			<u>Mandatory Carrot & Stick Zoning with no supply effect</u>					
(10)	1.06	42, 562	66, 159	0.91	-0.16	0.27	-0.02	12.20
(11)	1.67	68, 283	85, 711	1.07	-0.18	-2.13	0.52	1.56
(12)	5.43	218, 673	154, 624	5.34	-0.76	-1.68	0.96	0.08

Notes: Row (0) reports baseline outcomes under the no-policy environment $\mathcal{P}_0$ in levels. Rows (1)–(12) report changes relative to the baseline. Each row reports outcomes under alternative regulatory environments $\mathcal{P}$, all measured over a 10-year horizon. Rows (1)–(6) correspond to mandatory policies; rows (7)–(9) to the voluntary counterparts of policies (4)–(6); rows (10)–(12) to mandatory policies shutting down the supply feedback channel. Column (1) reports the percentage change in total expected supply in square feet. Column (2) reports the change in expected housing units constructed net of demolitions. Column (3) shows the change in affordable units out of this net change. Column (4) reports the percentage change in average expected citywide rent per housing unit. Column (5) depicts the change in percentage points in the share of units falling within the first quintile of the baseline rent distribution. Columns (6) and (7) report the percentage change in the present value of property tax revenues and total residential asset values, respectively. Column (8) shows the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$).

E Model Extension: Allowing for a Renovation Option

In this Appendix, we examine whether the paper’s central results depend specifically on redevelopment through replacement, or whether they continue to hold when developers can instead renovate—incrementally upgrading the existing housing stock rather than replacing it. We extend the model by introducing renovation as an alternative to redevelopment. Renovation restores the quality of an existing structure, $z_{B,i} \rightarrow 1$, while leaving density unchanged. This isolates a quality-improving margin while holding housing supply fixed.

E.1 Model and Calibration Assumptions

Developers choose between two irreversible actions, $m \in \{\text{renov}, \text{rebuild}\}$, in addition to maintaining the existing structure. Parcel value satisfies

$$\pi(R_i, F_{B,i}, z_{B,i}, \alpha_{B,i}; \mathcal{P}, \Theta) = \max \left\{ \pi_i^{\text{renov}}(\dots; \mathcal{P}, \Theta), \pi_i^{\text{rebuild}}(\dots; \mathcal{P}, \Theta) \right\}.$$

Each intervention m has its own optimal trigger $R_i^{e*,m}$ and construction duration d^m . The developer chooses the option that yields the highest parcel value at time $t = 0$. If the rebuild dominates the renovation options in value at $t = 0$, the developer selects rebuild, and the relevant trigger is $R_i^{e*,\text{rebuild}}$, regardless of whether $R_i^{e*,\text{renov}}$ is lower or higher.

The renovation extension keeps the redevelopment structure and introduces four modifications that isolate a quality-improving, non-supply-expanding margin:

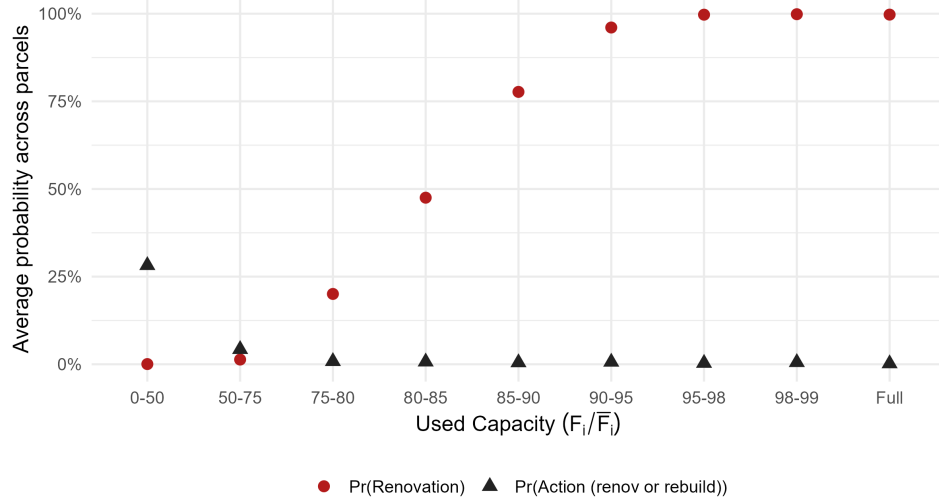
1. **No density gain:** Renovation does not increase building density, so $F_i^{*,\text{renov}} = F_{B,i}$.
2. **No demolition cost:** Demolition costs vanish under renovation, so $\gamma(F_{B,i}, L_i) = 0$.
3. **Shorter duration:** Renovation completes more quickly than redevelopment, with $d^{\text{renov}} < d$.
4. **Policies do not apply to renovation:** Policies do not apply to renovation, so $\mathcal{P}^{\text{renov}} = \mathcal{P}_0$.

Developers who renovate neither receive tax incentives nor face affordability mandates.

These features imply that renovation improves building quality without expanding housing supply. The renovation margin leaves the supply channel, captured by ζ_C , unchanged and raises rents through quality improvements and amenity spillovers captured by ζ_L . Because the policies do not apply to renovated buildings, developers can avoid the policy bundle $\mathcal{P}$ by selecting renovation. This introduces an opt-out margin that attenuates policy effects. This distinction matters for policy design because, in this context, redevelopment determines housing supply, while renovation changes only the quality of the existing stock.

The renovation option introduces a single new parameter: the renovation duration d^{renov} . We calibrate d^{renov} from permit-level data in the [New York City Department of City Planning Housing Database Project-Level Files](#), computing the average time between permit issuance and project completion for “Alteration” permits. This procedure yields $d^{\text{renov}} = 2.21$ years, which is 36% shorter than the redevelopment duration d , so renovation reduces exposure to foregone rental income.

Figure E1: Renovation Decision and Probability of Action by Used Capacity



Notes: Each point represents the mean probability within a bin of the density used ratio $F_i / \bar{F}_i$, where F_i is the current floor-area ratio (FAR) and $\bar{F}_i$ is the zoning limit. Red circles show the probability of choosing the renovation option over redevelopment. Black triangles show the probability of any replacement action occurring within the 10-year horizon.

E.2 Results

E.2.1 Baseline results and diagnosis

Developers choose renovation only on parcels that lie close to their zoning limit, where redevelopment no longer increases density. Figure E1 plots, across bins of used capacity $F_i / \bar{F}_i$, the probability of renovation (red), and the probability of any action (renovation or redevelopment) within the 10-year horizon (black). When substantial capacity remains unused, renovation is negligible. Below 75% of capacity, virtually no parcel prefers renovation over redevelopment, consistent with evidence that developers build to their maximum allowed density (Brueckner et al., 2024; Büchler and Lutz, 2024). The renovation option becomes relevant only above 85% of capacity, and dominates above 95%. When $F_i \approx \bar{F}_i$, redevelopment yields no density gain, so renovation becomes the optimal intervention by model construction.

Renovation is the preferred option on many parcels, yet has limited aggregate implications for housing supply because it arises mainly on parcels that are unlikely to redevelop. The probability of any action is highest ($\approx 30\%$) for low-density parcels with substantial redevelopment upside, and then declines sharply as parcels approach their zoning limit. For built-up parcels, the foregone rental income during renovation rarely offsets the associated quality gains, even with the shorter renovation period. Allowing for renovation, therefore, leaves aggregate development activity largely unchanged, including in regions where renovation is the dominant intervention.

The first row of Table E1 reports baseline outcomes with the renovation option. Aggregate housing supply remains essentially unchanged. Total expected production equals 170,356 units, and 126,166 units net of demolitions. These figures are nearly identical to the baseline without the renovation option, which delivers 167,758 total units and 126,215 net. The larger difference between the two numbers indicates that, under this model, a greater share of the housing stock consists of renovated units. This result confirms that the aggregate housing supply implications

Table E1: Results with Renovation Option

	Built Environment			Rent outcomes		Costs of Interventions			
	$\Delta \text{Shr. Renov}$ (%)	$\Delta S(h; \mathcal{P})$ (% chg.)	$\Delta \text{NetUnit}$ # of Apt.	ΔAffNet # of Apt.	ΔRent (%)	Shr. Q1 (pp)	ΔPV^{Tax} (%)	ΔPV (%)	Shr. PV-losers (%)
(0)	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(0) Baseline under $\mathcal{P}_0$ (numbers in levels)	52.8%	3.70 Bn.	126, 167	<u>Baseline Outcomes</u>	\$2, 715.01	20.00	\$401.7 Bn.	\$1, 557.34 Bn.	–
				456					
				<u>Unilateral Policy</u>					
(1) Upzoning ($\phi = 50\%$)	–52.51	1.55	62, 407	81	1.92	0.16	–0.10	–0.93	96.89
(2) Tax exemption ($\tau = 20$)	–49.30	2.03	82, 451	512	1.66	0.54	–3.88	–1.14	91.87
(3) Affordability mandate ($\alpha = 30\%$)	8.18	–0.30	–11, 978	44, 216	–0.52	–0.24	0.33	0.72	4.64
				<u>Mandatory Carrot & Stick Zoning</u>					
(4) Density Incentive Zoning	–52.38	0.97	39, 005	64, 519	0.57	0.02	–0.23	–0.46	97.30
(5) Fiscal Incentive Zoning	–18.01	1.49	60, 641	81, 940	0.25	0.36	–3.37	–0.77	92.79
(6) Combined Incentive Zoning	–52.66	4.14	166, 559	128, 425	2.08	0.97	–5.63	–3.13	93.00

Notes: Row (0) reports baseline outcomes when the renovation option is available alongside redevelopment under the no-policy environment $\mathcal{P}_0$. Rows (1)–(6) report changes relative to this baseline under alternative regulatory environments $\mathcal{P}$, all measured over a 10-year horizon. Each row corresponds to a counterfactual regulatory environment $\mathcal{P}$, and all reported quantities reflect deviations from the baseline calibration $\mathcal{P}_0$ over a 10-year horizon. Column (0) reports the share of the parcels that opt-in the renovation option. Column (1) reports the percentage change in total expected supply in square feet. Column (2) reports the change in expected housing units constructed net of demolitions. Column (3) shows the change in affordable units out of this net change. Column (4) reports the percentage change in average expected citywide rent per housing unit. Column (5) depicts the change in percentage points in the share of units falling within the first quintile of the baseline rent distribution. Columns (6) and (7) report the percentage change in the present value of property tax revenues and total residential asset values, respectively. Column (8) shows the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$).

of renovation are, however, modest. Expected citywide rent is marginally higher with renovation (\$2,615 vs. \$2,614), which reflects quality improvements in the existing stock rather than a reduction of housing supply.

E.2.2 Counterfactuals results

Renovation creates an escape margin from policy. Developers who choose renovation avoid the policy bundle $\mathcal{P}$, since $\mathcal{P}^{\text{renov}} = \mathcal{P}_0$. Table E1 reports the counterfactual results with the renovation option and shows that this escape margin is quantitatively important once policies include affordability mandates.²⁵ Under the mandate alone, the share of parcels that prefer renovation increases by 8.2 percentage points, which reflects the incentive to avoid the α requirement. In contrast, policies that offer incentives tilt choices toward redevelopment, reduce the reliance on renovation, and increase gross housing production.

Expected average rent rises more across all counterfactuals when the renovation option is available. The share of units in the bottom quintile of the baseline rent distribution changes little relative to the model without renovation. This pattern reflects substitution toward renovation for low-rent parcels, which preserves the composition of the lower tail of the rent distribution. Renovation raises rents through quality improvements while leaving the supply channel ζ_C unchanged. The supply mechanism that would otherwise dampen rent growth, therefore, weakens, which makes it more difficult for policies to reduce rents through replacement of the existing stock.

Renovation also dampens policy incidence. Present value losses borne by landowners fall less when the renovation option is available, and the share of PV-losers is also slightly reduced across all counterfactuals. The expanded set of developer choices drives this pattern. Renovation offers a lower-cost alternative to redevelopment that preserves parcel value on otherwise constrained parcels. Without renovation, parcels close to their zoning limit have no margin to respond to adverse policies other than waiting. Renovation introduces an adjustment margin that partially hedges this exposure.

E.3 Summary

Allowing for renovation does not alter the main aggregate results. Development activity continues to concentrate on parcels with unused capacity, where redevelopment dominates renovation. As a result, renovation has negligible effects on aggregate housing supply. At the margin, renovation improves quality and generates amenity spillovers without expanding the housing stock. When renovation is exempt from α and τ , it creates an escape margin from policy that weakens the effects of affordability mandates. These findings reinforce the central mechanism of the paper. Redevelopment decisions that replace the existing stock shape housing outcomes, whereas renovation operates mainly through incremental upgrading of that stock.

²⁵To solve the fixed point, developers need to make predictions about which developers renovate and which prefer the rebuild option. Because this is a binary decision that can create an unstable equilibrium, we smooth these outcomes by transforming the discrete choice into a probability of choosing one option over the other, following the same procedure as the adoption algorithm described in Appendix A.8.

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